OVERVIEW

The Eight Project Performance Domains

Chapter 1 named the eight performance domains; this deck opens each one. A domain is a group of related activities critical to delivering project outcomes, and all eight run at the same time and interact. Each section defines the domain, gives its key terms and practices, and ends with a "checking outcomes" table describing what success looks like: stakeholders (who is affected and how to engage them), team (management vs. leadership, culture, high performance), development approach and life cycle (predictive, hybrid, adaptive; cadence), planning (estimates, schedules, budget and reserves), project work (processes, constraints, procurement, knowledge), delivery (value, quality), measurement (metrics, KPIs, dashboards) and uncertainty (ambiguity, complexity, volatility, risk responses).

Concurrent, not sequential

The domains are interactive, interrelated and interdependent areas of focus that run throughout the project, however value is delivered.

Centralized vs. decentralized

Centralized: one accountable PM, more control and standardization, cheaper, less creative. Decentralized: authority near the work, faster and more creative, more expensive.

Approach spectrum

Predictive (stable, known requirements) ↔ hybrid ↔ adaptive (uncertain, volatile requirements; iterative and incremental).

Risk responses

Threats: avoid, escalate, transfer, mitigate, accept. Opportunities: exploit, escalate, share, enhance, accept.

HOW THE LECTURE IS ORGANIZED

Slides 1–5 What a performance domain is; 1 Stakeholders

Slides 6–18 2 Team: management and leadership, culture, high performance, emotional intelligence

Slides 19–34 3 Development approach and life cycle: cadence, predictive, hybrid, adaptive

Slides 35–52 4 Planning: estimates, schedules, dependencies, budget and reserves, procurement, metrics

Slides 53–65 5 Project work: processes, constraints, procurement and bids, knowledge

Slides 66–72 6 Delivery: value, quality, cost of quality

Slides 73–87 7 Measurement: metrics, KPIs, presenting information

Slides 88–107 8 Uncertainty: ambiguity, complexity, volatility, risk and reserves

EXAM FOCUS

Midterm 251 asked about the nature of performance domains, the purpose of cadence and when an iterative approach fits; its case study asked you to choose Agile, Waterfall or Hybrid. The instructor's 2026 midterm asked which opportunity strategy applies when a team acts to make sure an early finish happens (exploit). Finals drew on estimation, scheduling, risk responses, constraints, the scope statement and burn charts from this deck.

HOW TO READ THIS PAGE

Every original slide is shown unchanged. Yellow marks on a slide point to a numbered note: hover or tap a mark to jump to its note, and hover a note to see its marks. Questions tagged **Past exam** appeared on a real paper; **Prediction** questions are likely angles, not confirmed.

 Slide 1



Project Performance Domains

SE423: Software Project Management

1

NOTES

1. Chapter 1 listed the 8 domain names; this deck opens each one in turn. The number badge (1–8) on each domain's title slide keeps the order: stakeholders, team, development approach and life cycle, planning, project work, delivery, measurement, uncertainty.
2. For each domain learn its purpose, its key terms (many are examined as definitions), and one or two practices you could describe in a scenario answer.

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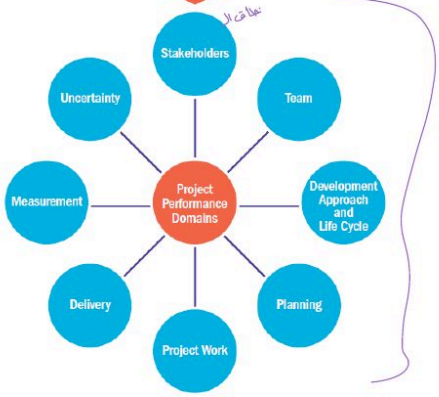
Project Performance Domains

- A project performance domain is a group of related activities that are critical for the effective delivery of project outcomes.
- Project performance domains are interactive, interrelated, and interdependent areas of focus that work in unison to achieve desired project outcomes.
- Performance domains run concurrently throughout the project, regardless of how value is delivered (frequently, periodically, or at the end of the project).

Principles of Project Management

Be a diligent, respectful, and caring steward	Create a collaborative team environment	Effectively engage with stakeholders	Focus on value
Recognize, evaluate, and respond to system interactions	Demonstrate leadership behaviors	Tailor based on context	Build quality into processes and deliverables
Navigate complexity	Optimize risk responses	Embrace adaptability and resiliency	Enable change to achieve the envisioned future state

Guide Behavior



2

NOTES

1. **Definition**: a project performance domain is a **group of related activities that are critical** for the effective delivery of project outcomes.
2. The domains are **interactive, interrelated and interdependent** areas of focus that work in unison: weak stakeholder engagement hurts planning, poor measurement hides uncertainty, and so on.
3. They **run concurrently** throughout the project, whether value is delivered frequently, periodically or only at the end. They are **not** phases done in order, which is exactly the trap in the real MCQ.

EXAM ANGLE

Past exam · Midterm 251 · V1 · MCQ 3 — Q1

Which of the following best explains the nature of project performance domains?

A. They represent distinct areas of focus that guide effective project management.
B. They are rigid processes that must be followed in order, from initiation to closure.
C. They are optional tools that project managers may use only in large projects.
D. They are specific deliverables required at the end of each project phase.

Answer: A. They represent distinct areas of focus that guide effective project management. — Performance domains are areas of focus that run concurrently; they are not sequential processes or deliverables. (Also Midterm 251 V2 MCQ 1.)

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Stakeholder Performance Domain 1

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- **Stakeholder**
  - An individual, group, or organization that may affect, be affected by, or perceive itself to be affected by a decision, activity, or outcome of a project, program, or portfolio.
- **Stakeholder Analysis** تحليل المستفيدين
  - A method of systematically gathering and analyzing quantitative and qualitative information to determine whose interests should be taken into account throughout the project.
  - includes stakeholders who are internal and external to the organization, those who are supportive of the project, and those who may not be supportive or are neutral.



3

NOTES

1. **Stakeholder**: an individual, group or organization that **may affect, be affected by, or perceive itself to be affected by** a decision, activity or outcome of a project, program or portfolio. All three count: someone who only *believes* they are affected is still a stakeholder.
2. **Stakeholder analysis**: systematically gathering and analysing quantitative and qualitative information to decide **whose interests to take into account** throughout the project.
3. It covers stakeholders **internal and external** to the organization, and those who **support, oppose or are neutral** toward the project. Ignoring opponents is how projects get blocked late.

EXAM ANGLE

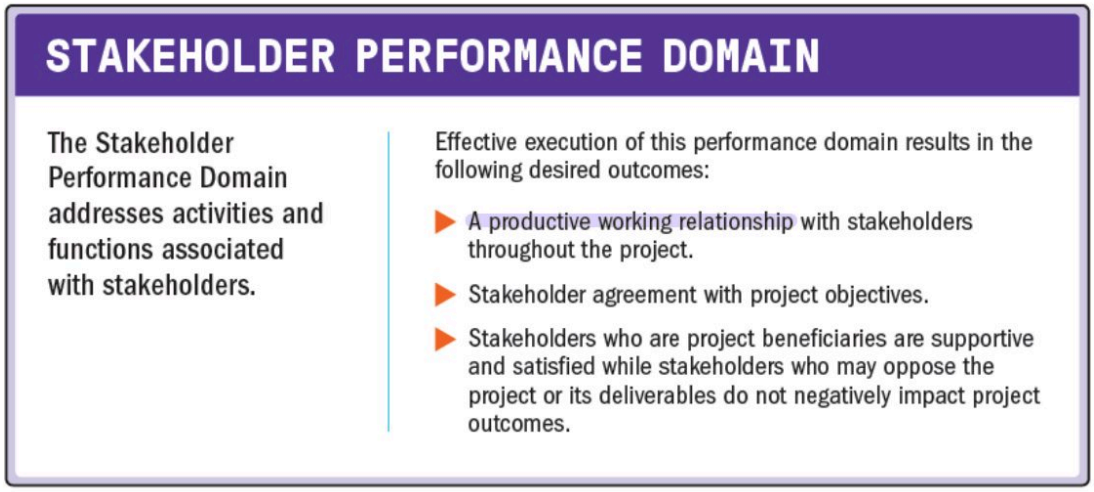
Past exam · Midterm 251 · V1 · Case Q3 Q1

Edugate scheduling system: identify three main stakeholder groups for this project, describe their roles, and explain how their input will shape the system.
Answer: Students are the end users; their input shapes usability and flexibility (0.3). The IT/Edugate team ensures integration, security and compliance with PSU IT policies (0.3). Faculty advisors and administration ensure study-plan rules are enforced and use the system to monitor students (0.4).

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Stakeholder Performance Domain

- Technical project management skills are important, but having the interpersonal and leadership skills to work effectively with stakeholders is maybe more important.



4

NOTES

1 Technical PM skills matter, but **interpersonal and leadership skills** for working with stakeholders may matter more. Most stakeholder problems (resistance, conflicting expectations) are solved by listening, negotiating and building trust, not by a better Gantt chart. The circles diagram groups stakeholders from the core team outward to customers, end users and regulators.

 Checking Outcomes: Stakeholder Performance Domain

Checking Outcomes of Stakeholder Performance Domain

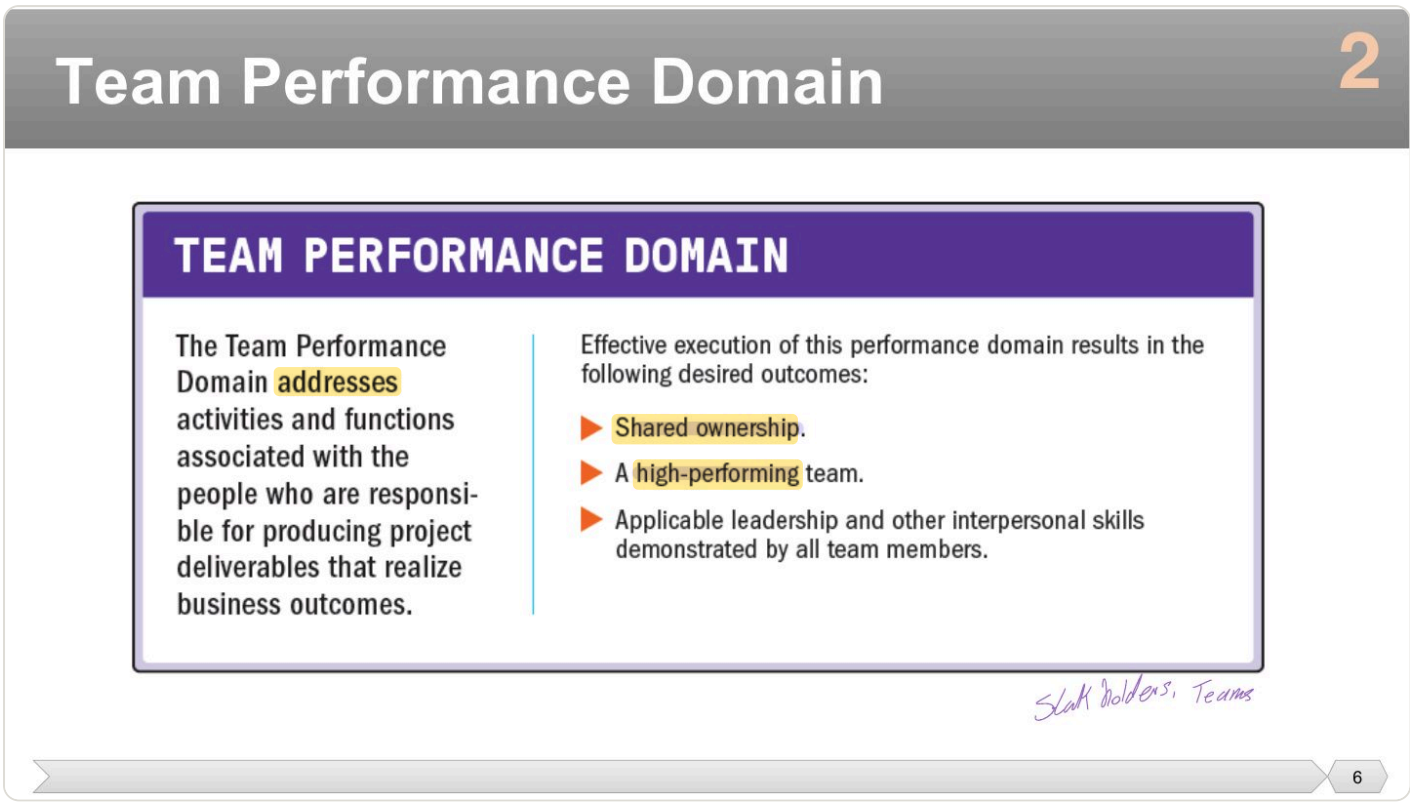
Outcome	Check
A productive working relationship with stakeholders throughout the project	Productive working relationships with stakeholders can be observed. However, the movement of stakeholders along a continuum of engagement can indicate the relative level of satisfaction with the project.
Stakeholder agreement with project objectives	A significant number of changes or modifications to the project and product requirements in addition to the scope may indicate stakeholders are not engaged or aligned with the project objectives.
Stakeholders who are project beneficiaries are supportive and satisfied; stakeholders who may oppose the project or its deliverables do not negatively impact project results	Stakeholder behavior can indicate whether project beneficiaries are satisfied and supportive of the project or whether they oppose it. Surveys, interviews, and focus groups are also effective ways to determine if stakeholders are satisfied and supportive or if they oppose the project and its deliverables. A review of the project issue register and risk register can identify challenges associated with individual stakeholders.

5

NOTES

1. The “checking outcomes” table: the left column is what success looks like for the domain, the right column is the evidence that it happened. You won’t be asked to reproduce it, but it is a good structure for a written answer about the domain. Here: **a productive working relationship** with stakeholders throughout, checked by observing it and by how stakeholders move along the engagement continuum (unaware → resistant → neutral → supportive → leading).
2. **Stakeholder agreement with project objectives**, checked by the number of changes: many scope modifications suggest stakeholders weren’t aligned.
3. **Supportive beneficiaries and harmless opponents**, checked with surveys, interviews and focus groups, and by reviewing the issue and risk registers.

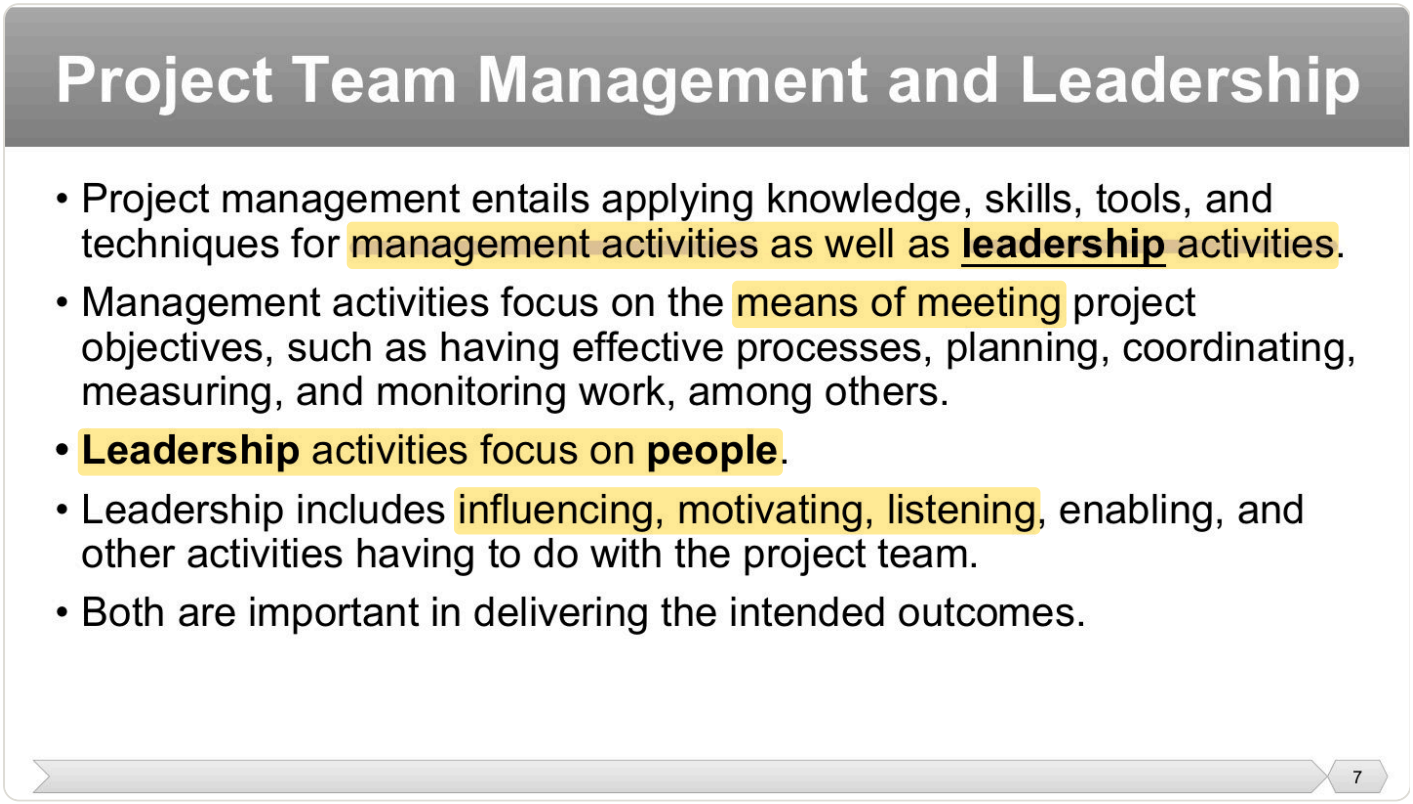
 Team Performance Domain



NOTES

1. The **team** domain covers the activities and functions of the people who produce the project deliverables.
2. Desired outcomes: **shared ownership** of the work, a **high-performing team**, and leadership and interpersonal skills shown by **all** team members, not only the PM.

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NOTES

1. Project management applies knowledge, skills, tools and techniques to **both management and leadership** activities.
2. **Management** focuses on the **means** of meeting objectives: effective processes, planning, coordinating, measuring and monitoring work. It manages things.
3. **Leadership** focuses on **people**: influencing, motivating, listening, enabling. It leads people. Both are needed to deliver the outcomes.

EXAM ANGLE

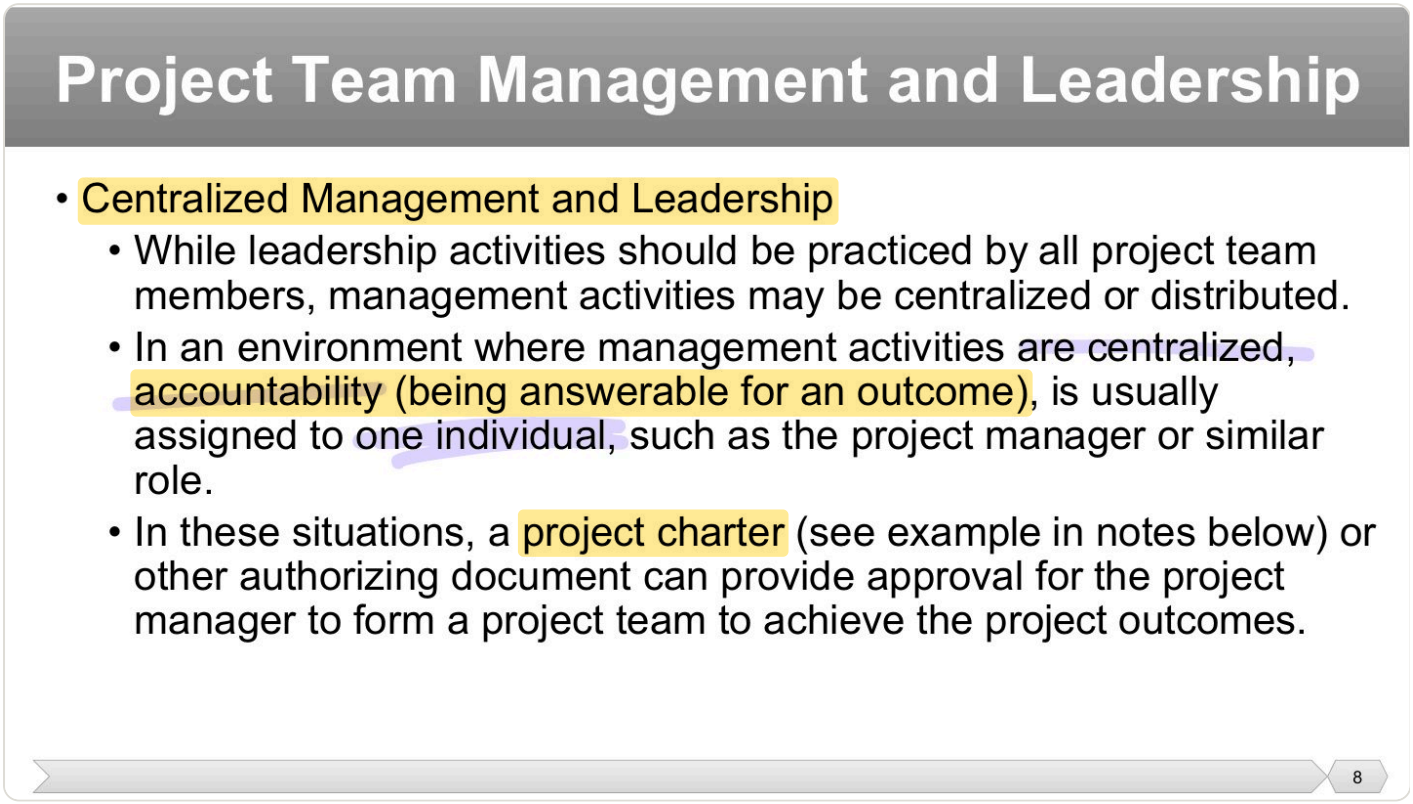
Prediction Q1

Influencing, motivating and listening to the team are examples of which kind of activity?

A. Leadership activities
B. Management activities
C. Governance activities
D. Procurement activities

Answer: A. Leadership activities

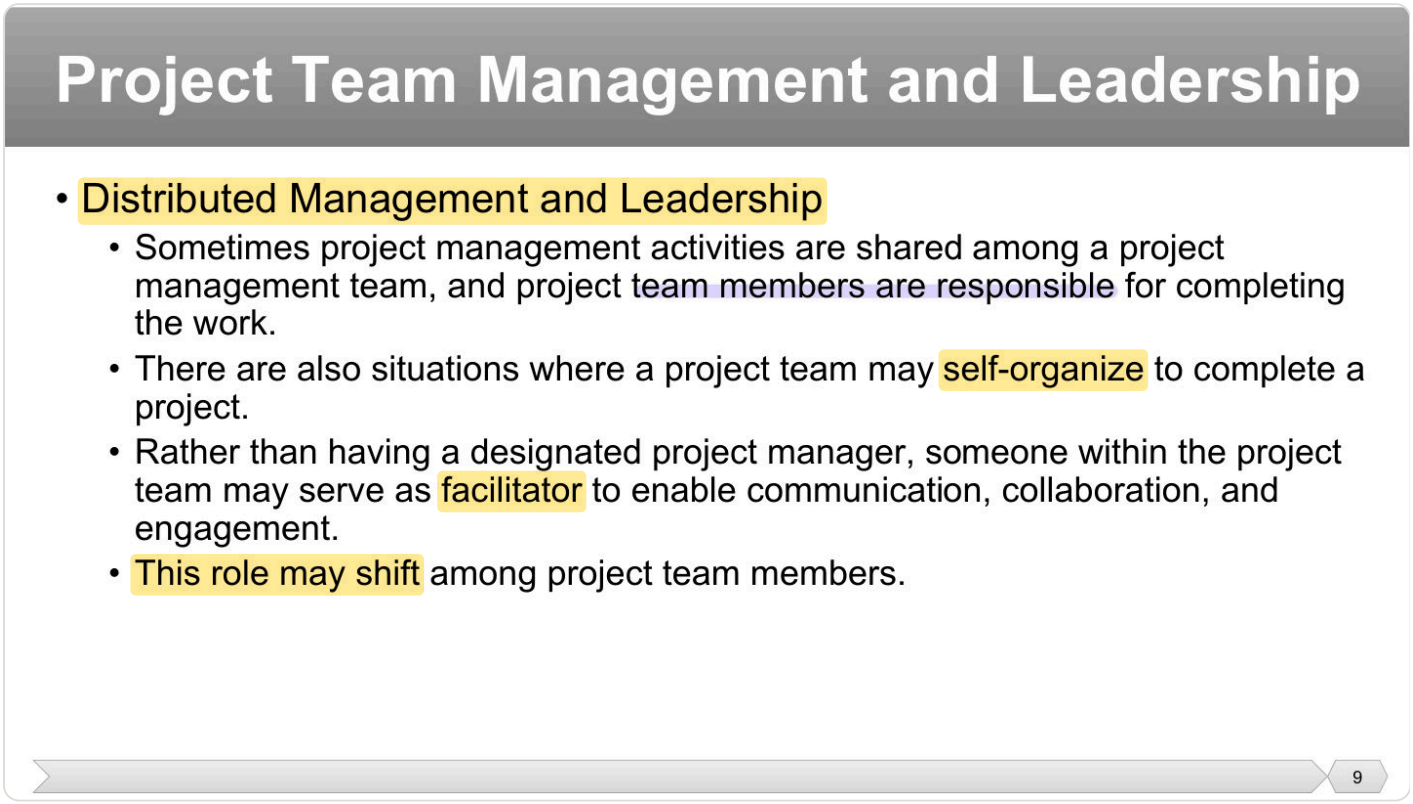
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NOTES

1. Leadership should be practised by **everyone**, but **management** activities can be centralized or distributed.
2. In a **centralized** environment, **accountability** (being answerable for an outcome) is usually assigned to **one** person, the PM or a similar role.
3. A **project charter** or other authorizing document gives that PM approval to form the team and use resources.

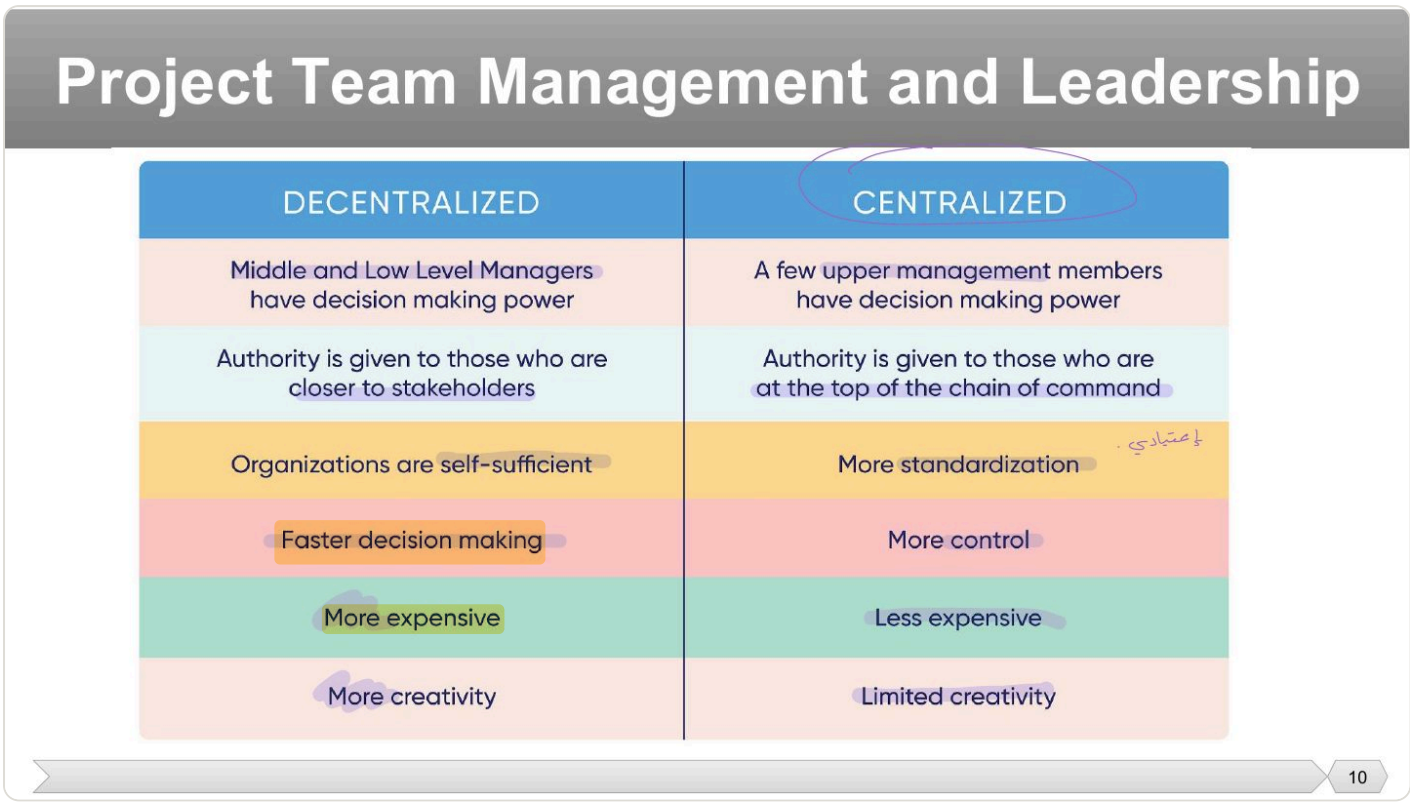
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NOTES

1. In **distributed** management, PM activities are shared among a project management team while team members do the work.
2. A team may even **self-organize** to complete the project.
3. Instead of a designated PM, someone acts as **facilitator** to enable communication, collaboration and engagement (like a Scrum master), and **the role can shift** among members.

Centralized vs. Decentralized



NOTES

1. **Decentralized**: middle and lower-level managers have decision power, and authority goes to those **closer to the stakeholders**; units are self-sufficient.
2. **Centralized**: a few **upper-management** members decide, with authority at the top of the chain of command.
3. Decentralized gives **faster decisions and more creativity** but is **more expensive** (duplicated resources and skills in each unit).
4. Centralized gives **more standardization and control** and is **less expensive**, with limited creativity. Learn it as three opposite pairs: speed vs. control, more vs. less expensive, creativity vs. standardization.

EXAM ANGLE

Past exam · Midterm 251 · V1 · MCQ 1 Q1

In decentralized project management, decision-making authority is primarily:

A. Distributed among team members to encourage shared accountability.

B. Concentrated in the hands of a single project manager.

C. Transferred to external auditors for oversight.

D. Reserved exclusively for senior executives in the organization.

Answer: A. Distributed among team members to encourage shared accountability. — (Also Midterm 251 V2 MCQ 2.)

Prediction Q2

Which is a characteristic of decentralized management, according to the table?

A. Faster decision making but more expensive

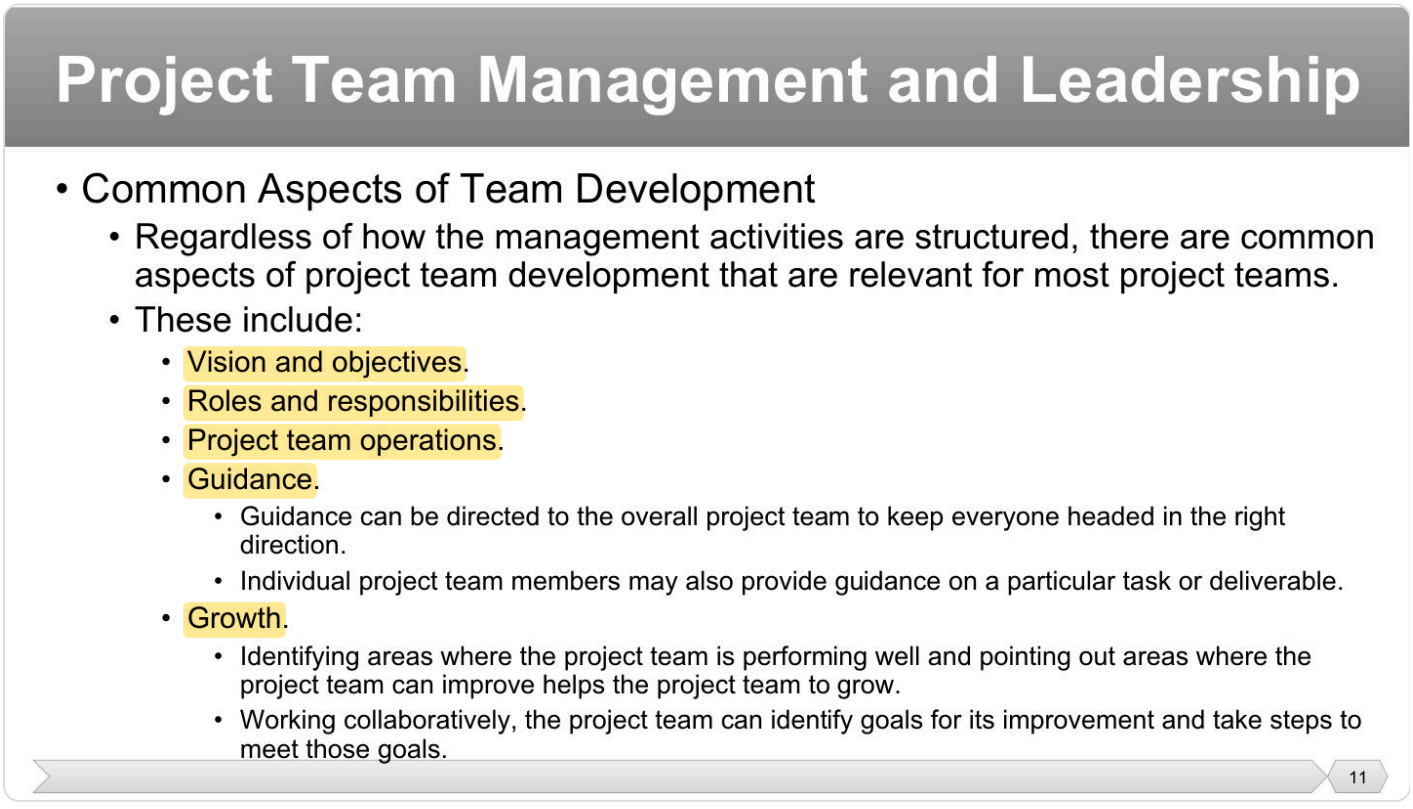
B. More standardization and control

C. Authority at the top of the chain of command

D. Less expensive with limited creativity

Answer: A. Faster decision making but more expensive

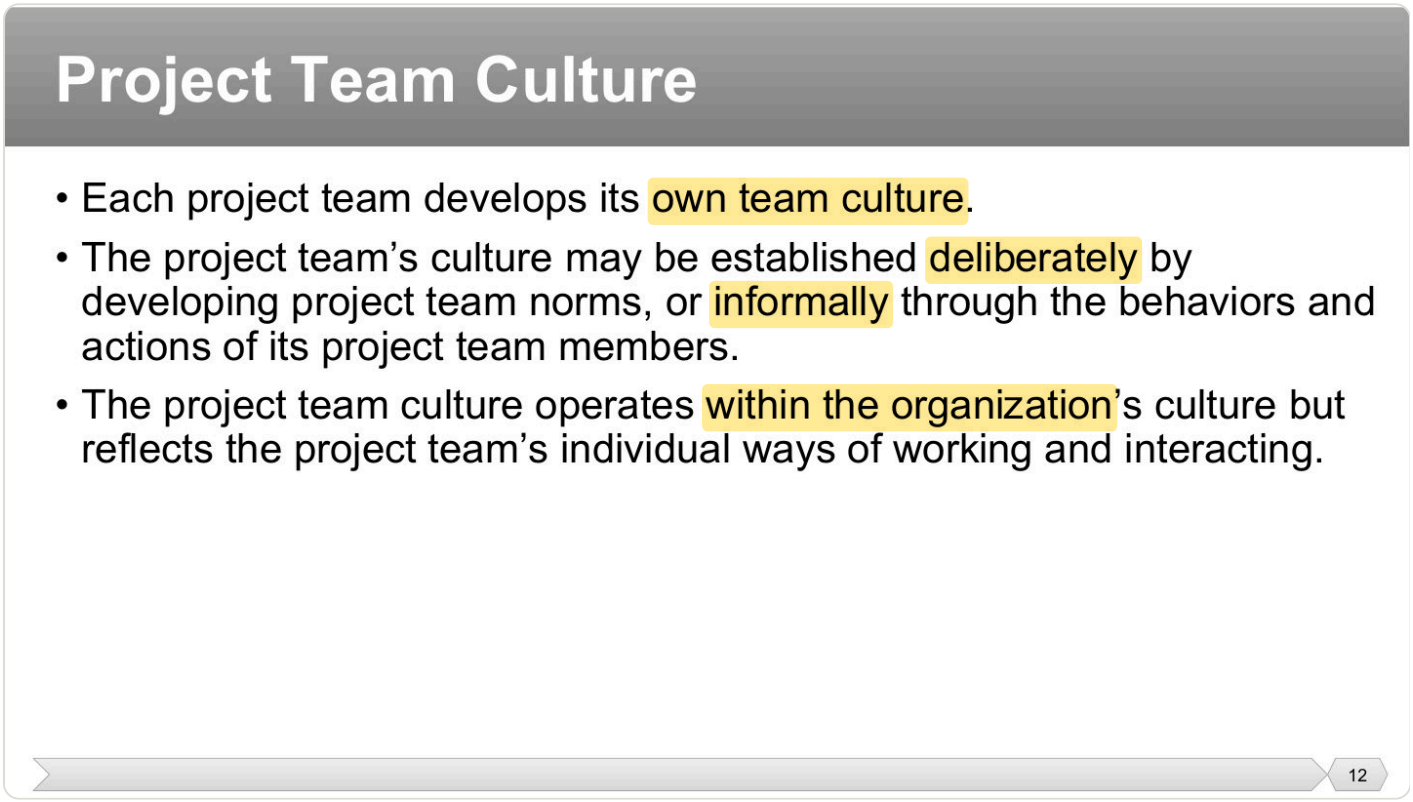
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NOTES

1. Whatever the structure, most teams need the same foundations: a shared **vision and objectives**, clear **roles and responsibilities**, and agreed **team operations** (how meetings, decisions and communication work).
2. **Guidance**: for the whole team, to keep everyone heading the same way, and from individual members on particular tasks or deliverables.
3. **Growth**: point out what the team does well and where it can improve, then set improvement goals together (what a retrospective does).

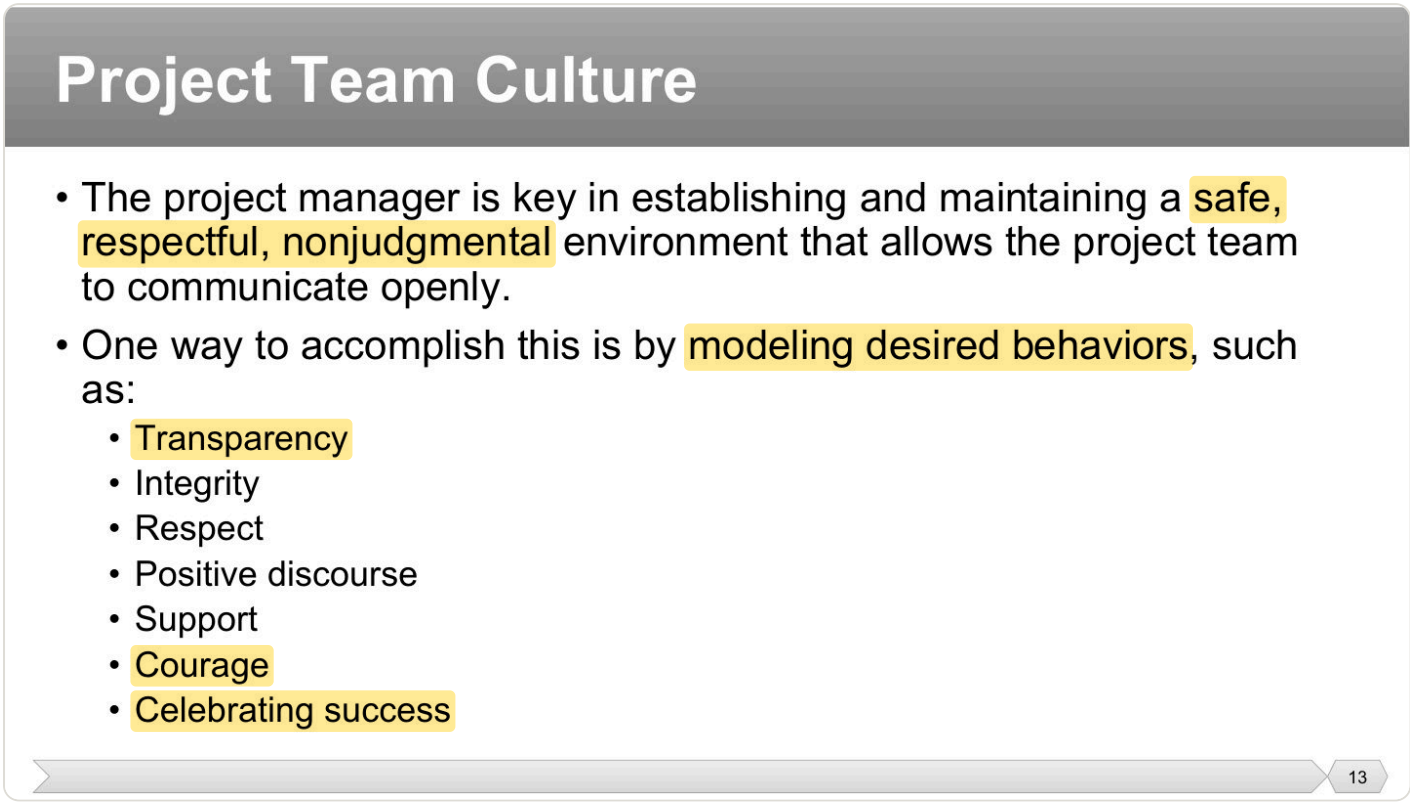
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NOTES

1. Every team develops its **own culture**.
2. It can be built **deliberately**, by agreeing team norms ("we review every pull request within a day"), or form **informally** from how members actually behave.
3. It operates **within the organization's culture** but reflects the team's own ways of working.

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NOTES

1. The PM is key to a **safe, respectful, nonjudgmental** environment where people can speak openly (the basis of trust in Chapter 3).
2. The main way is **modeling** the behaviors the PM wants: transparency, integrity, respect, positive discourse, support and **courage** (raising problems, admitting mistakes).
3. **Celebrating success** is one of them: recognizing milestones and contributions keeps morale high.

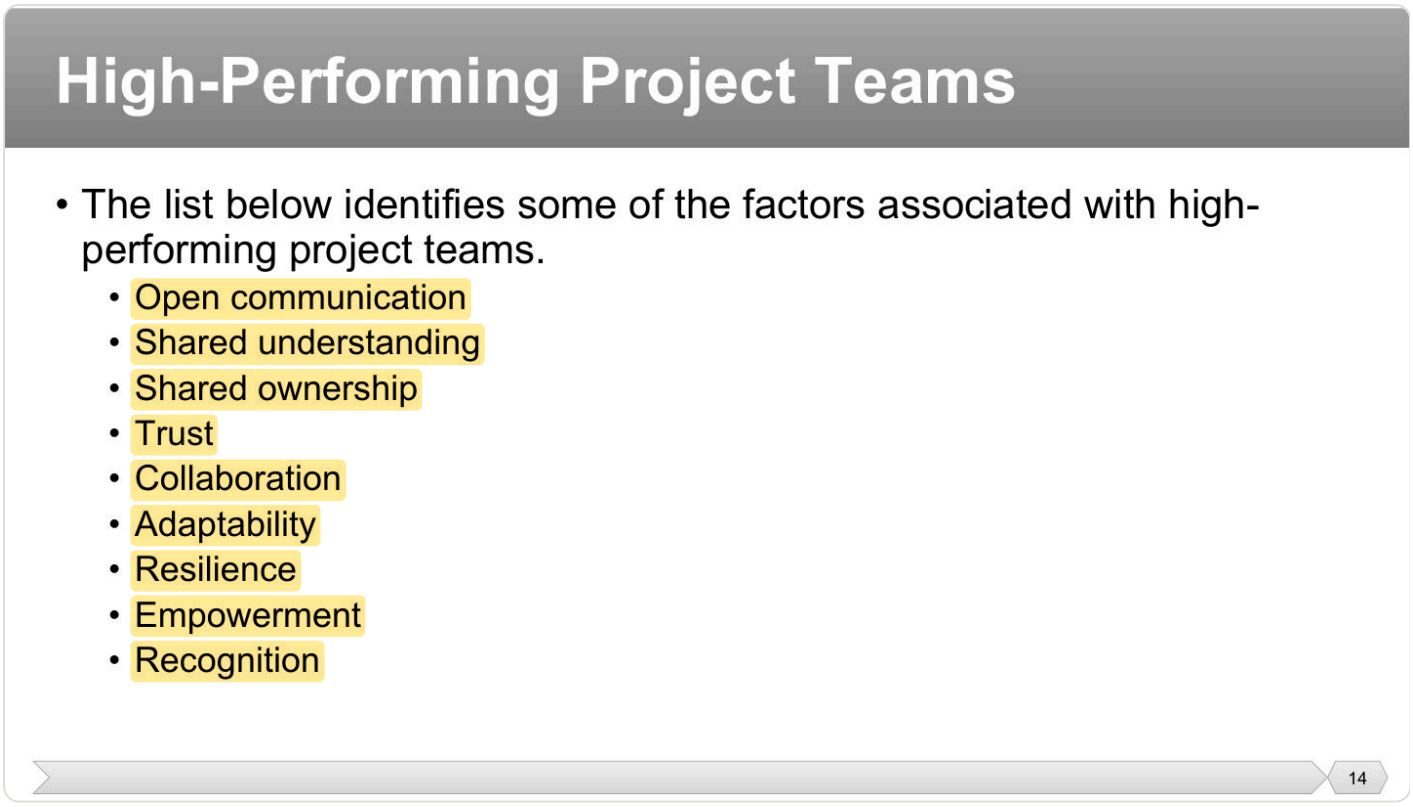
EXAM ANGLE

Past exam · Final 251 · Q5.3 Q1

Your team has just completed a major milestone in the project development process. How would you celebrate this achievement collaboratively with your team members? How would you recognize individual contributions to the team's success? What actions would you take to ensure that team morale remains high as you move forward in the project?

Answer: Celebrate as a team, e.g. a team lunch and an announcement of the milestone to the stakeholders. Recognize individuals specifically and publicly: name what each person contributed in the meeting, send thank-you notes to their managers, and give certificates or small rewards. Keep morale high with a retrospective that captures what went well, clear and achievable next goals, a balanced workload to avoid burnout, growth opportunities, and continued recognition, not only at milestones.

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NOTES

1. Factors of high-performing teams. **Open communication, shared understanding** of goals and **shared ownership** of the work: everyone knows what they are building and feels it is theirs.
2. **Trust** and **collaboration**: members rely on each other and work together instead of protecting their own part.
3. **Adaptability** and **resilience**: the team adjusts to change and recovers from setbacks.
4. **Empowerment** (the team can make decisions) and **recognition** (contributions are acknowledged).

EXAM ANGLE

Past exam · Midterm 251 · V1 · Short Q1 Q1

What are key characteristics of a good team?
Answer: A good team trusts each other enough to speak openly and admit mistakes, shares a clear goal with clear roles, debates ideas in healthy conflict, commits to decisions, holds each other accountable, and puts team results ahead of individual credit. (The slide's factors: open communication, shared understanding and ownership, trust, collaboration, adaptability, resilience, empowerment, recognition.)

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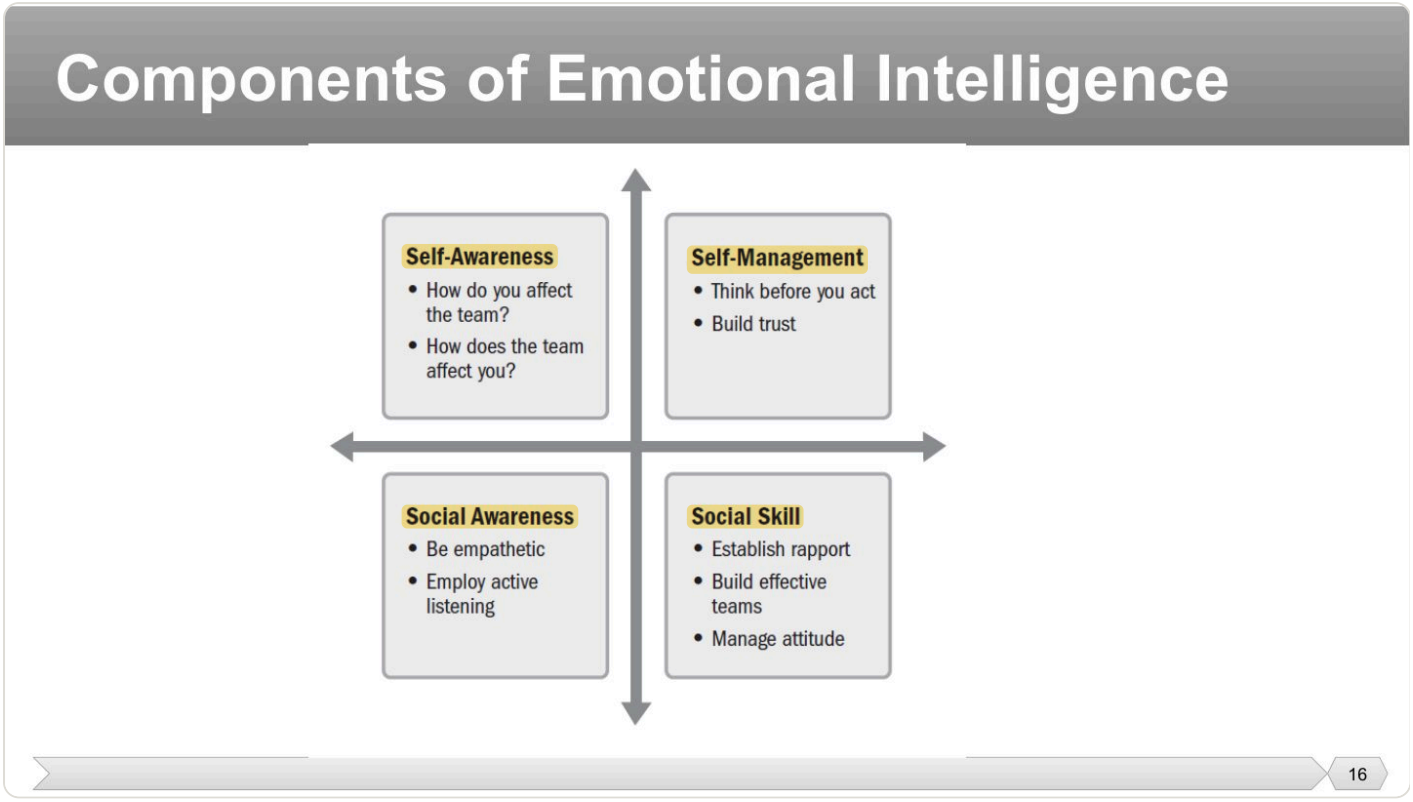
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NOTES

1. **Vision**: establish and maintain a shared picture of what the project is for, so decisions point the same way.
2. **Critical thinking**: weigh evidence, question assumptions and make sound judgments.
3. **Motivation**: understand what drives each person (intrinsic or extrinsic) and use it.
4. **Interpersonal skills**: emotional intelligence, decision-making and conflict resolution (next slide).

 Components of Emotional Intelligence



NOTES

1. Emotional intelligence (EI) has four components: two about yourself, two about others. **Self-awareness**: how do you affect the team, and how does the team affect you?
2. **Self-management**: think before you act, build trust.
3. **Social awareness**: be empathetic, use active listening.
4. **Social skill**: establish rapport, build effective teams, manage attitude.

EXAM ANGLE

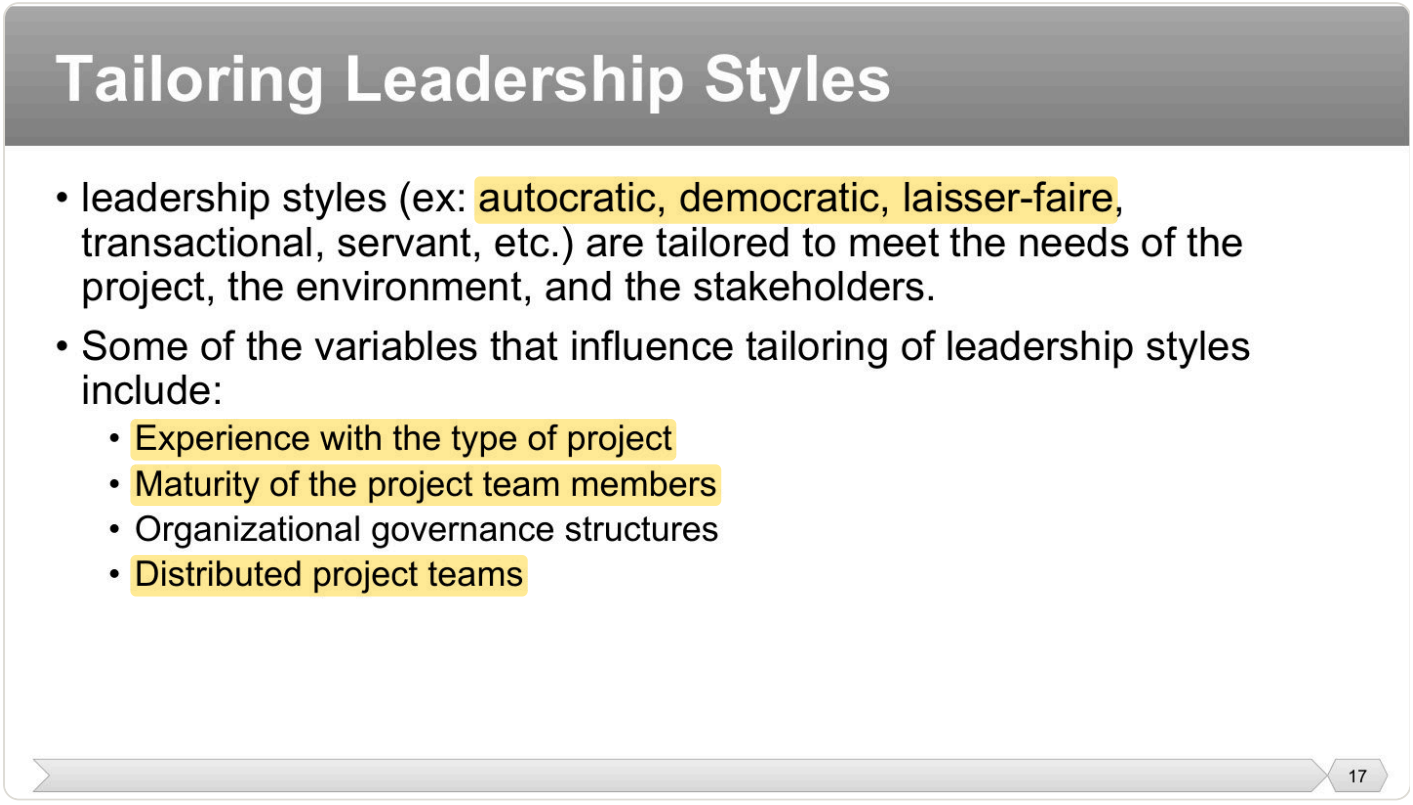
Prediction Q1

"Be empathetic" and "employ active listening" belong to which component of emotional intelligence?

A. Social awareness
B. Self-management
C. Self-awareness
D. Social skill

Answer: A. Social awareness

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NOTES

1. Leadership styles (autocratic, democratic, laissez-faire, transactional, servant…) are **tailored** to the project, environment and stakeholders; no style is always right.

2. What drives the choice: **experience** with this type of project, the **maturity** of the team members (a senior team needs less direction), the organization's **governance structure**, and whether the team is **distributed** (which needs more deliberate communication).

EXAM ANGLE

Past exam · Midterm 251 · V1 · Short Q5 — Q1

How can a project manager balance technical leadership with adaptive leadership in a software team? **Answer:** First diagnose the problem. When it can be solved by known means (a coding standard, an architecture choice the PM knows well), use technical leadership: give direction from expertise and vision. When it can't (new technology, unclear requirements), use adaptive leadership: frame the purpose, ask questions, let the team experiment and learn, and motivate through visible progress. Switching between the two as the situation changes is the balance. (Also Midterm 251 V2 Short Q5.)

 Checking Outcomes: Team Performance Domain

Checking Outcomes—Team Performance Domain

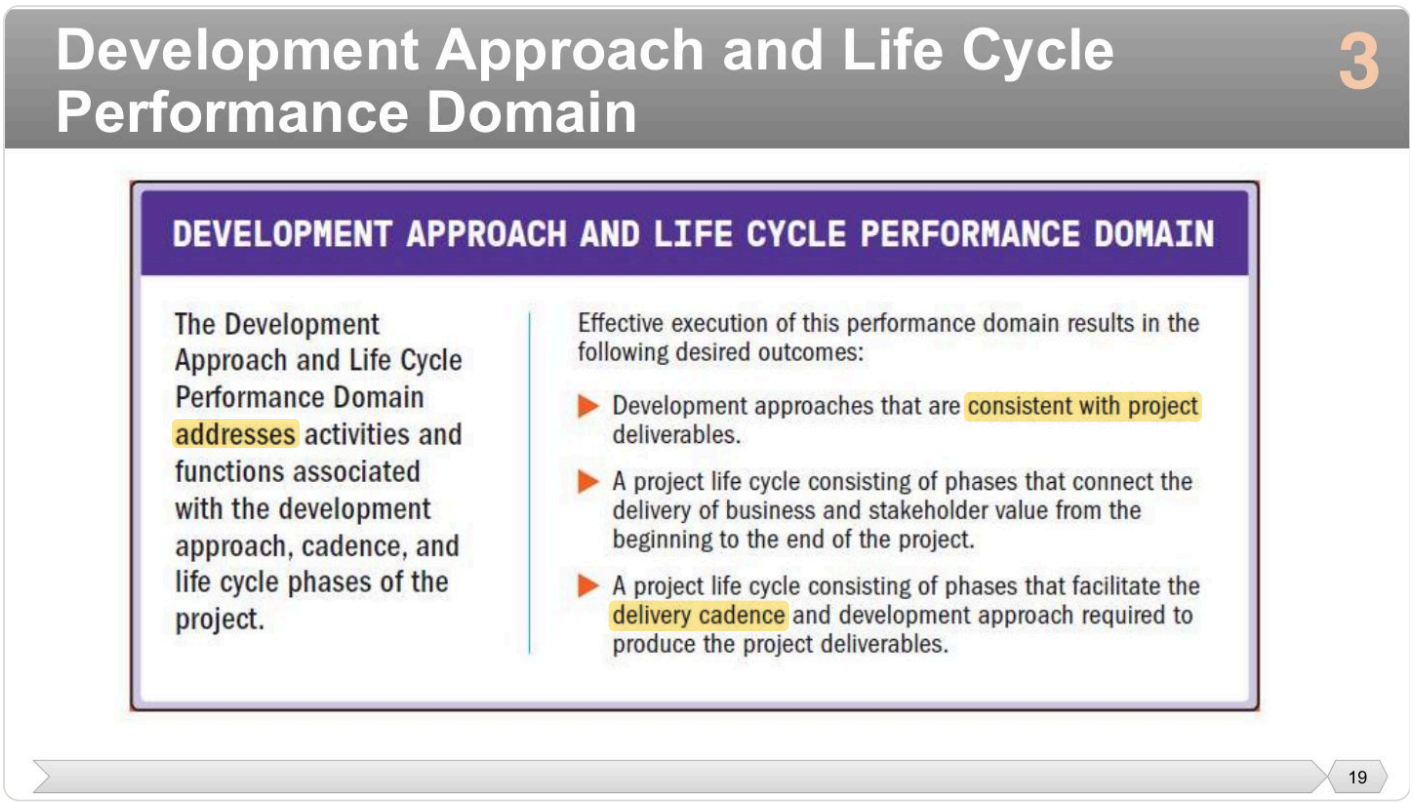
Outcome	Check
Shared ownership	All project team members know the vision and objectives. The project team owns the deliverables and outcomes of the project.
A high-performing team	The project team trusts each other and collaborates. The project team adapts to changing situations and is resilient in the face of challenges. The project team feels empowered and empowers and recognizes members of the project team.
Applicable leadership and other interpersonal skills are demonstrated by all project team members	Project team members apply critical thinking and interpersonal skills. Project team member leadership styles are appropriate to the project context and environment.

18

NOTES

1 The “checking outcomes” table: the left column is what success looks like for the domain, the right column is the evidence that it happened. You won’t be asked to reproduce it, but it is a good structure for a written answer about the domain. Team: **shared ownership** (members are engaged and take initiative), a **high-performing team** (trust, collaboration, adapting to change) and **leadership shown by all** (evidenced by members’ interpersonal skills and initiative).

 Development Approach and Life Cycle Performance Domain



NOTES

1. This domain covers the **development approach**, **cadence** and **life-cycle phases**.
2. Outcomes: approaches that are **consistent with the deliverables**, and a life cycle whose phases connect the delivery of value from start to end and support the required **delivery cadence**.

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Development Approach and Life Cycle Performance Domain

- The following definitions are relevant to the Development Approach and Life Cycle Performance Domain:
  - Deliverable
    - Any unique and verifiable product, result, or capability to perform a service that is required to be produced to complete a process, phase, or project.
  - Development Approach.
    - A method used to create and evolve the product, service, or result during the project life cycle, such as a predictive, iterative, incremental, adaptive, or hybrid method.
  - Cadence.
    - A rhythm of activities conducted throughout the project.
  - Project Phase.
    - A collection of logically related project activities that culminates in the completion of one or more deliverables.
  - Project Life Cycle.
    - The series of phases that a project passes through from its start to its completion.

NOTES

1. **Deliverable**: any unique and verifiable product, result or capability that must be produced to complete a process, phase or project.

2. **Development approach**: the method used to create and evolve the product: predictive, iterative, incremental, adaptive or hybrid.

3. **Cadence**: a **rhythm of activities** conducted throughout the project (two-week sprints, monthly releases). A predictable rhythm helps coordination and communication.

4. **Project phase**: a set of logically related activities ending in one or more deliverables. **Project life cycle**: the series of phases from start to completion.

EXAM ANGLE

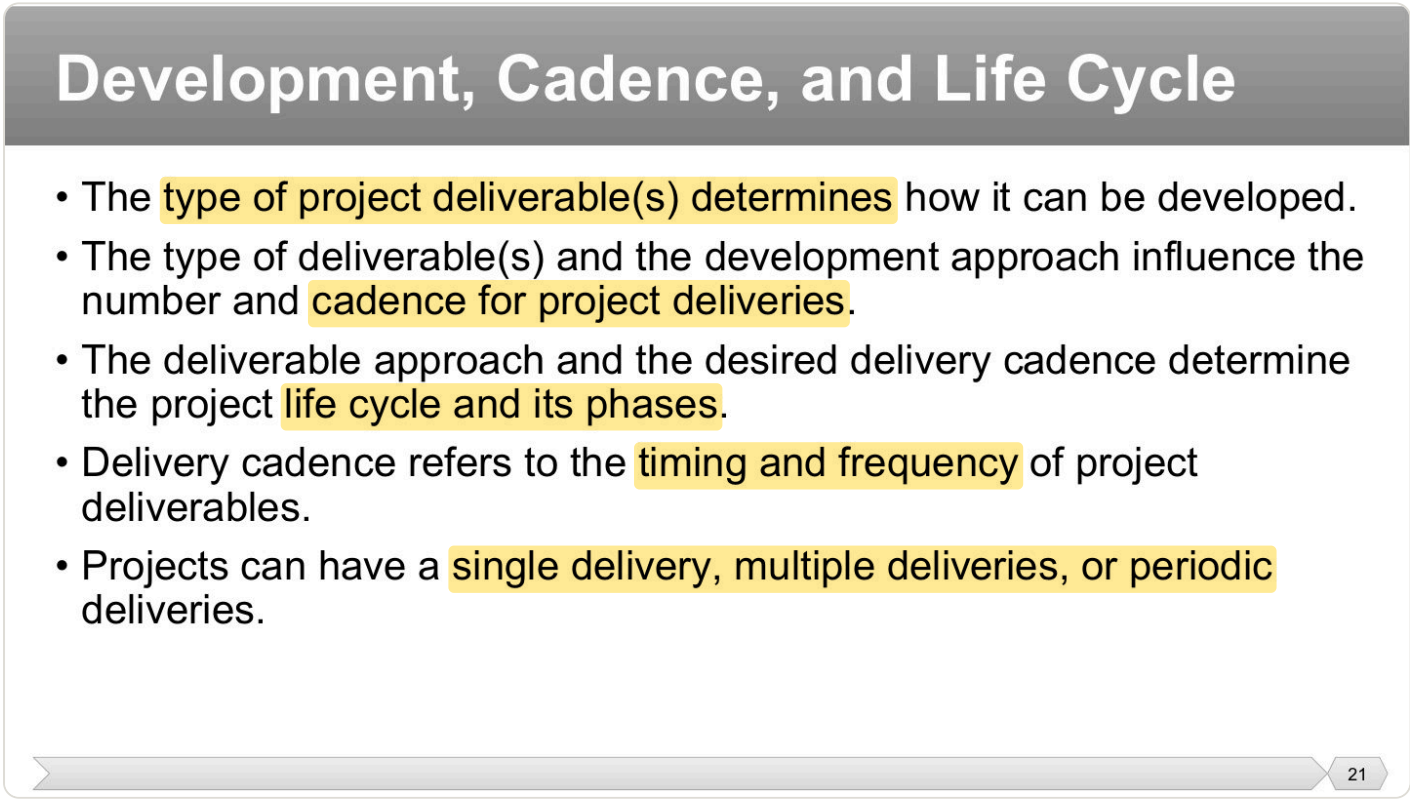
Past exam · Midterm 251 · V1 · MCQ 4 Q1

In project management, what is the primary purpose of establishing a project cadence?

A. To create a predictable rhythm that supports coordination, communication, and timely delivery.
B. To enforce strict deadlines that cannot be adjusted under any circumstances.
C. To eliminate the need for stakeholder engagement during development.
D. To ensure that all deliverables are produced only at the end of the project.

Answer: A. To create a predictable rhythm that supports coordination, communication, and timely delivery. — Cadence is the rhythm of activities (single, multiple, periodic or continuous delivery) that keeps the work coordinated. (Also Midterm 251 V2 MCQ 5.)

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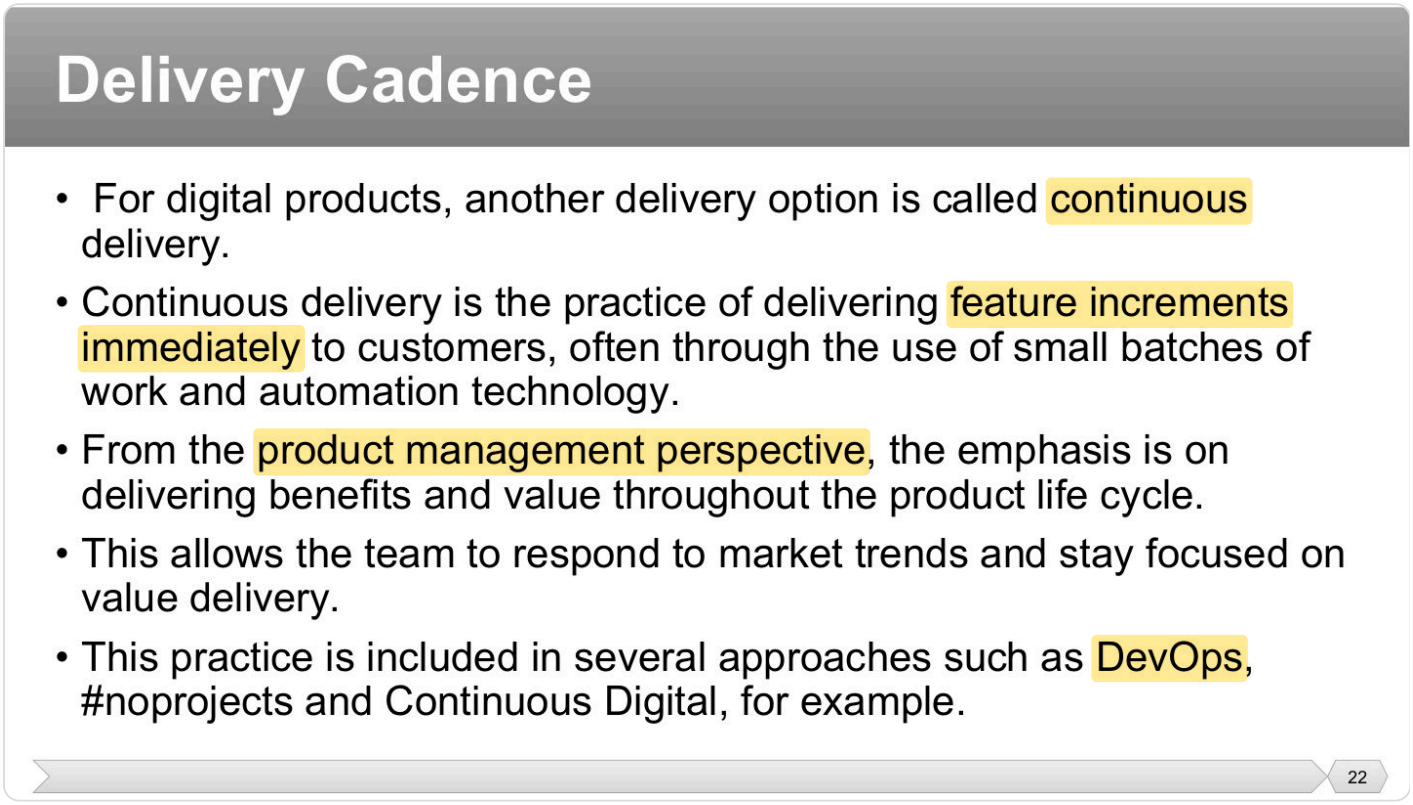
NOTES

1. The chain of decisions starts with the **type of deliverable**: it determines how the product can be developed (you can't release half a bridge, but you can release half an app).
2. The deliverable type and approach then influence the **number and cadence of deliveries**...
3. ...and the approach plus the desired cadence determine the **life cycle and its phases**.

4. **Delivery cadence** is the **timing and frequency** of deliveries.

5. Projects can have a **single** delivery (at the end), **multiple** deliveries (at different points), or **periodic** deliveries (on a fixed schedule, e.g. monthly).

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NOTES

1. For digital products there is a fourth option: **continuous delivery**.

2. It delivers **feature increments immediately** to customers, using small batches of work and automation (automated build, test and deployment).

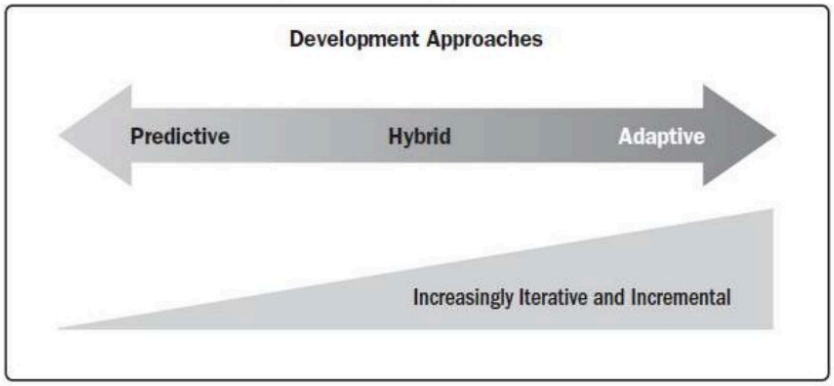
3. From the **product** perspective the emphasis is delivering value throughout the product's life, so the team can respond to market trends.

4. It is part of **DevOps**, #noprojects and Continuous Digital (Chapter 4 covers DevOps).

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Development Approaches

- A development approach is the means used to create and evolve the product, service, or result during the project life cycle.
- There are different development approaches, and different industries may use different terms to refer to development approaches.
- Three commonly used approaches are
  - predictive,
  - hybrid, and
  - adaptive.

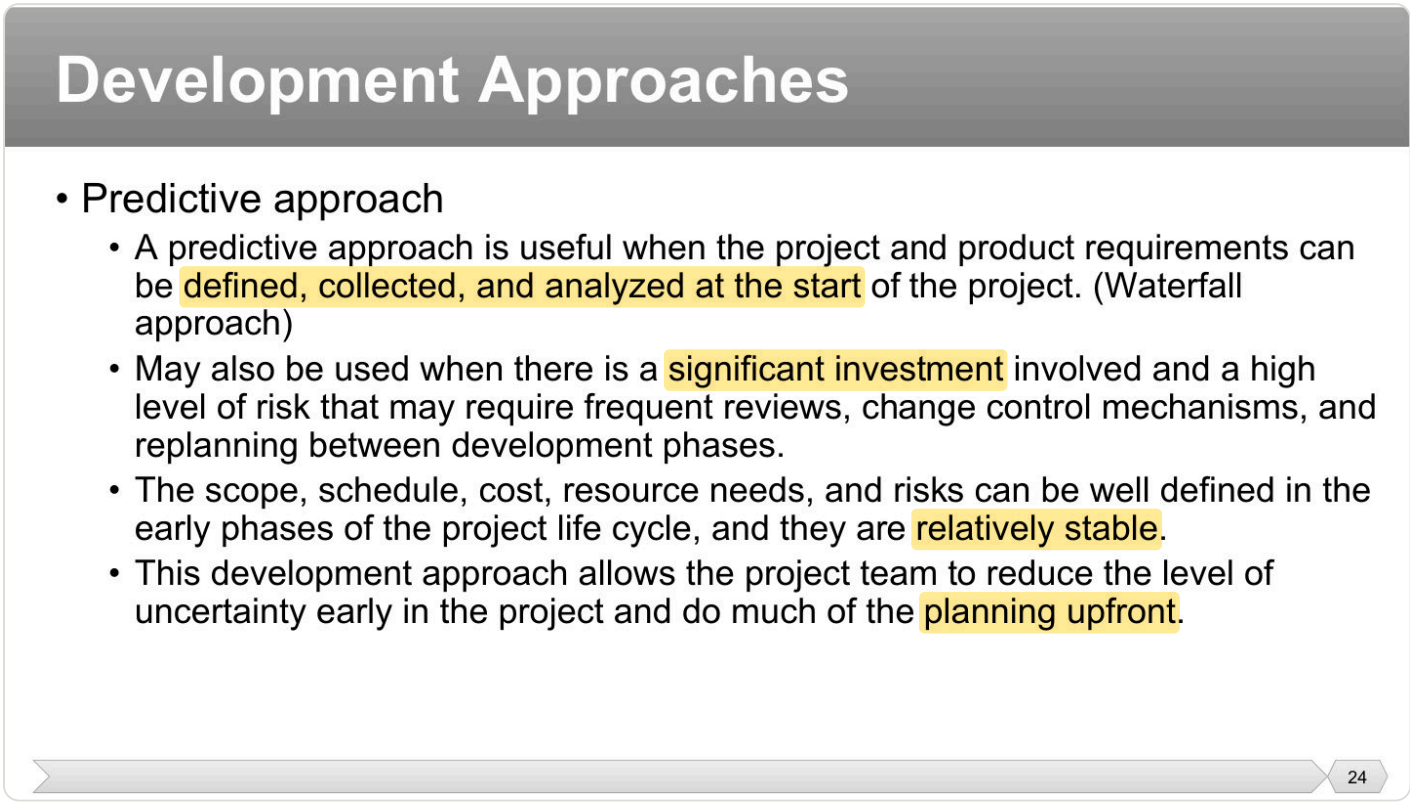


23

NOTES

1. A **development approach** is the means used to create and evolve the product during the life cycle.
2. Industries use **different names** for the same approaches.
3. The three common ones are **predictive, hybrid and adaptive**, a **spectrum** rather than three boxes: the further toward adaptive, the more iterative and incremental the work.

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NOTES

1. **Predictive** (waterfall) is useful when requirements can be **defined, collected and analysed at the start**.
2. It is also used when there is **significant investment** and high risk that call for frequent reviews, change control and replanning between phases (a new hospital information system).
3. Scope, schedule, cost, resources and risks can be defined early and are **relatively stable**.
4. It reduces uncertainty early and does most of the **planning up front**.

EXAM ANGLE

Prediction Q1

A government payroll system must implement fixed legal rules that will not change during the project. Which development approach fits, and why?

Answer: Predictive (waterfall): the requirements can be defined and analysed at the start and are stable, so scope, schedule and cost can be planned up front with phase reviews and change control.

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Development Approaches

- Hybrid approach
  - A hybrid development approach is a combination of adaptive and predictive approaches.
  - This means that some elements from a predictive approach are used and some from an adaptive approach are used.
  - This development approach is useful when there is uncertainty or risk around the requirements.
  - Hybrid is also useful when deliverables can be modularized, or when there are deliverables that can be developed by different project teams.

NOTES

1. **Hybrid** combines elements of adaptive and predictive approaches.
2. It is useful when there is **uncertainty or risk around the requirements** for part of the product.
3. It also fits when deliverables can be **modularized** or built by different teams: a fixed back-end integration planned predictively, with a user-facing app built in sprints.

EXAM ANGLE

Past exam · Midterm 251 · V1 · Case Q2 — Q1

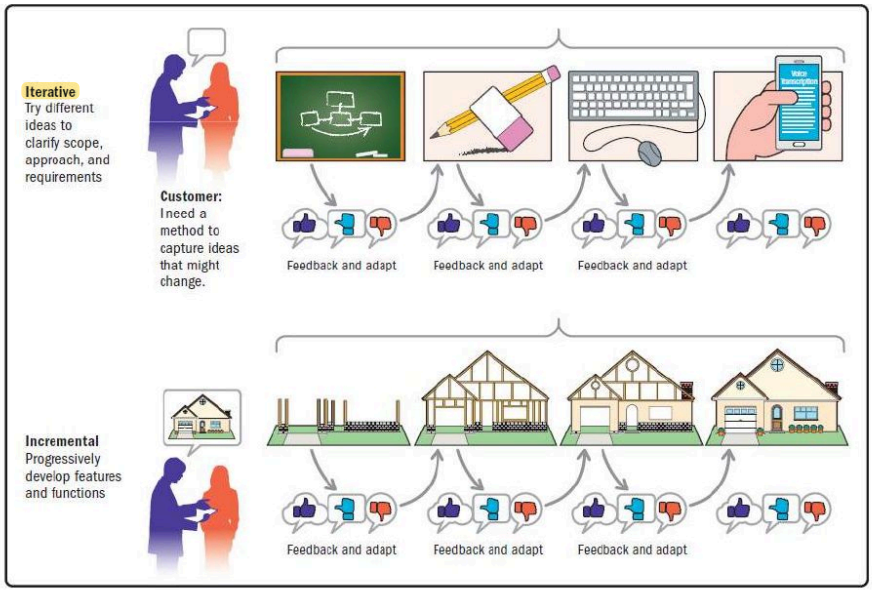
Edugate scheduling system: given the evolving requirements (student usability, advisor requests, IT integration rules), which development approach (Agile, Waterfall, or Hybrid) would you recommend, and why would it be the best fit for this project?

Answer: Agile (Scrum) (0.5), because it handles the evolving requirements from students, advisors and IT, delivers iteratively so each sprint gets feedback, and integrates with DevOps pipelines for frequent tested releases (0.5). (A hybrid answer that plans the fixed IT integration predictively and builds the student-facing features iteratively is also well argued.) The instructor's key gives Agile (Scrum).

 Iterative vs. Incremental

Development Approaches

- Hybrid approach



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NOTES

1. **Iterative** (top row): try different ideas to clarify scope, approach and requirements, refining the **same** thing after each round of feedback (sketch → mock-up → prototype → app). The customer needs a way to capture ideas that might change.
2. **Incremental** (bottom row): progressively **add new** features and functions, each a working piece (foundation → frame → walls → finished house), with feedback after each increment. Adaptive approaches use both. Iterative refines; incremental adds.

EXAM ANGLE

Prediction Q1

Adding one new working feature to the product in each cycle is best described as:

A. Incremental development
B. Iterative development
C. Predictive development
D. Waterfall development

Answer: A. Incremental development

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Development Approaches

- Adaptive approach
  - Adaptive approaches are useful when requirements are subject to a high level of uncertainty and volatility and are likely to change throughout the project.
  - A clear vision is established at the start of the project, and the initial known requirements are refined, detailed, changed, or replaced in accordance with user feedback, the environment, or unexpected events.
  - Adaptive approaches use iterative and incremental approaches. However, on the far side of the adaptive methods, the iterations tend to get shorter and the product is more likely to evolve based on stakeholder feedback.

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NOTES

1. **Adaptive** approaches suit requirements with **high uncertainty and volatility** that will change throughout the project.
2. A **clear vision** is set at the start; the known requirements are then refined, detailed, changed or replaced based on user feedback, the environment or unexpected events.
3. Adaptive approaches are **iterative and incremental**; toward the far adaptive end, iterations **get shorter** and the product evolves more from stakeholder feedback.

EXAM ANGLE

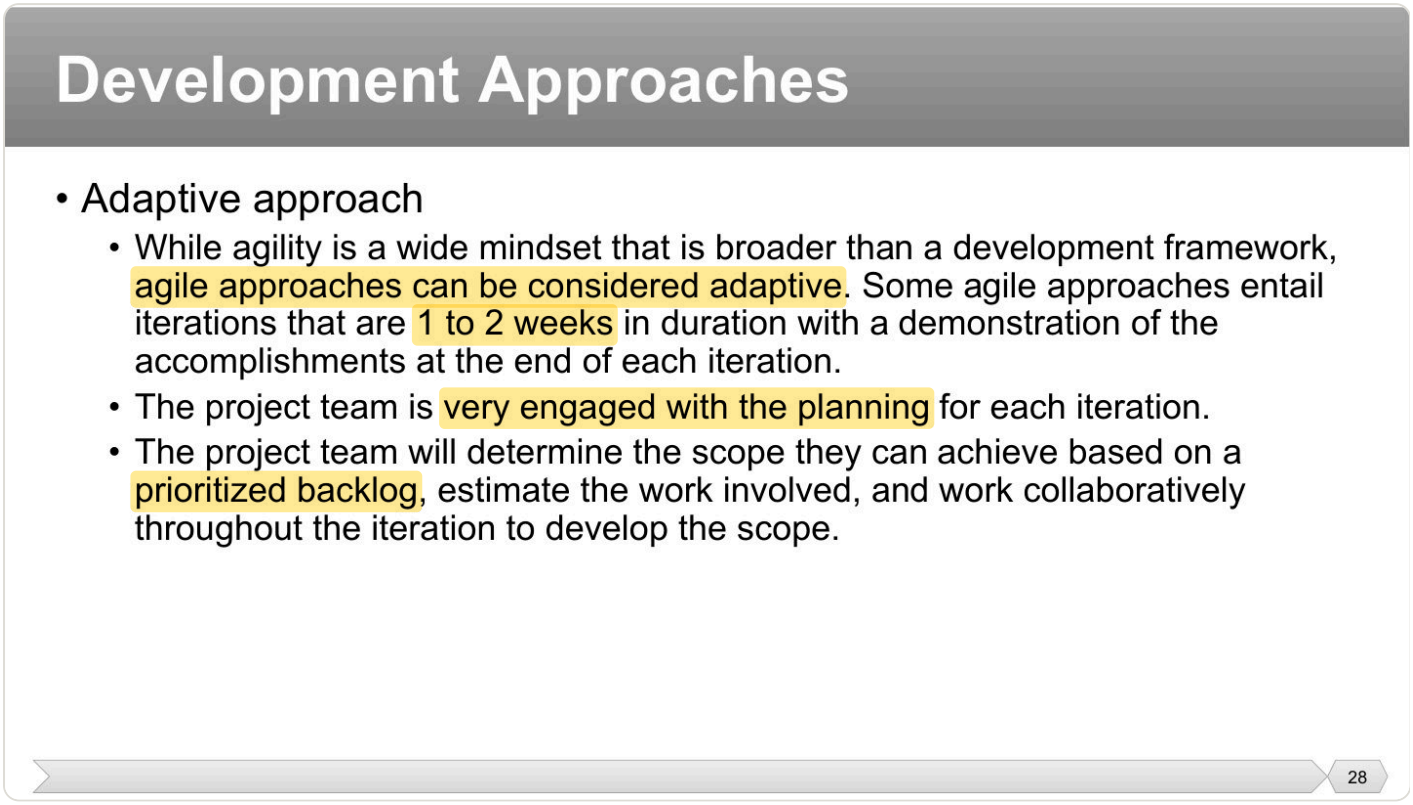
Past exam · Midterm 251 · V1 · MCQ 6 Q1

In which type of project would an iterative development approach be most beneficial?

A. A project where requirements are unclear and expected to evolve over time.
B. A project with well-defined, unchanging requirements from the start.
C. A project that must deliver the entire product only at final completion.
D. A project that cannot accommodate stakeholder feedback during execution.

Answer: A. A project where requirements are unclear and expected to evolve over time. — Iterative development refines ideas through feedback, which suits unclear or evolving requirements. (Also Midterm 251 V2 MCQ 6.)

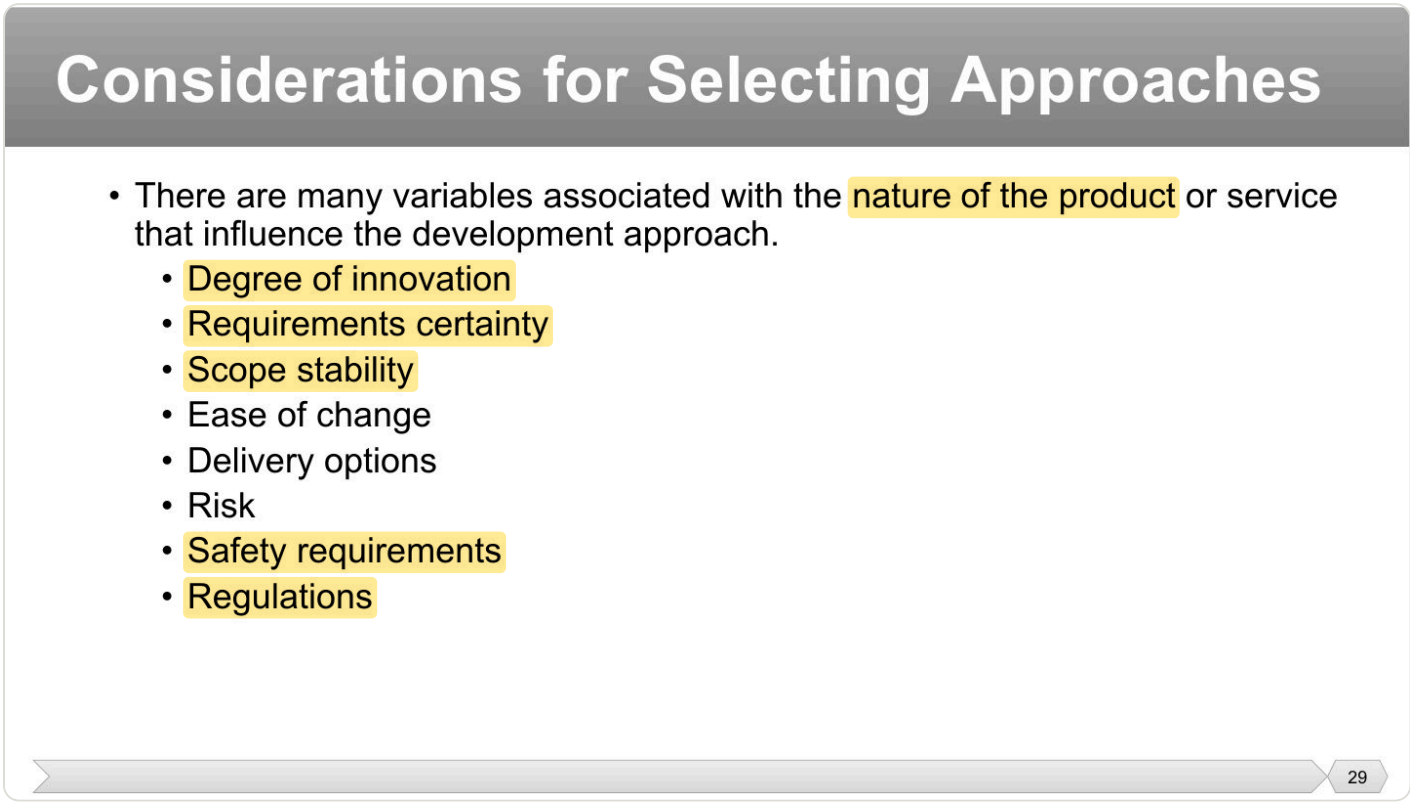
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NOTES

1. **Agility** is a mindset broader than any framework, but agile approaches count as adaptive. Some use **1–2 week iterations** that end with a demonstration of what was accomplished.
2. The team is **very engaged in planning** each iteration.
3. It decides how much scope it can achieve from a **prioritized backlog**, estimates the work itself, and works collaboratively through the iteration to build it.

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NOTES

1. Product variables that influence the approach. Rule of thumb: high **innovation**, uncertain **requirements**, unstable **scope**, easy **change** and frequent **delivery options** push toward adaptive.
2. Strict **safety requirements** and **regulations** push toward predictive (they need documented, controlled phases).

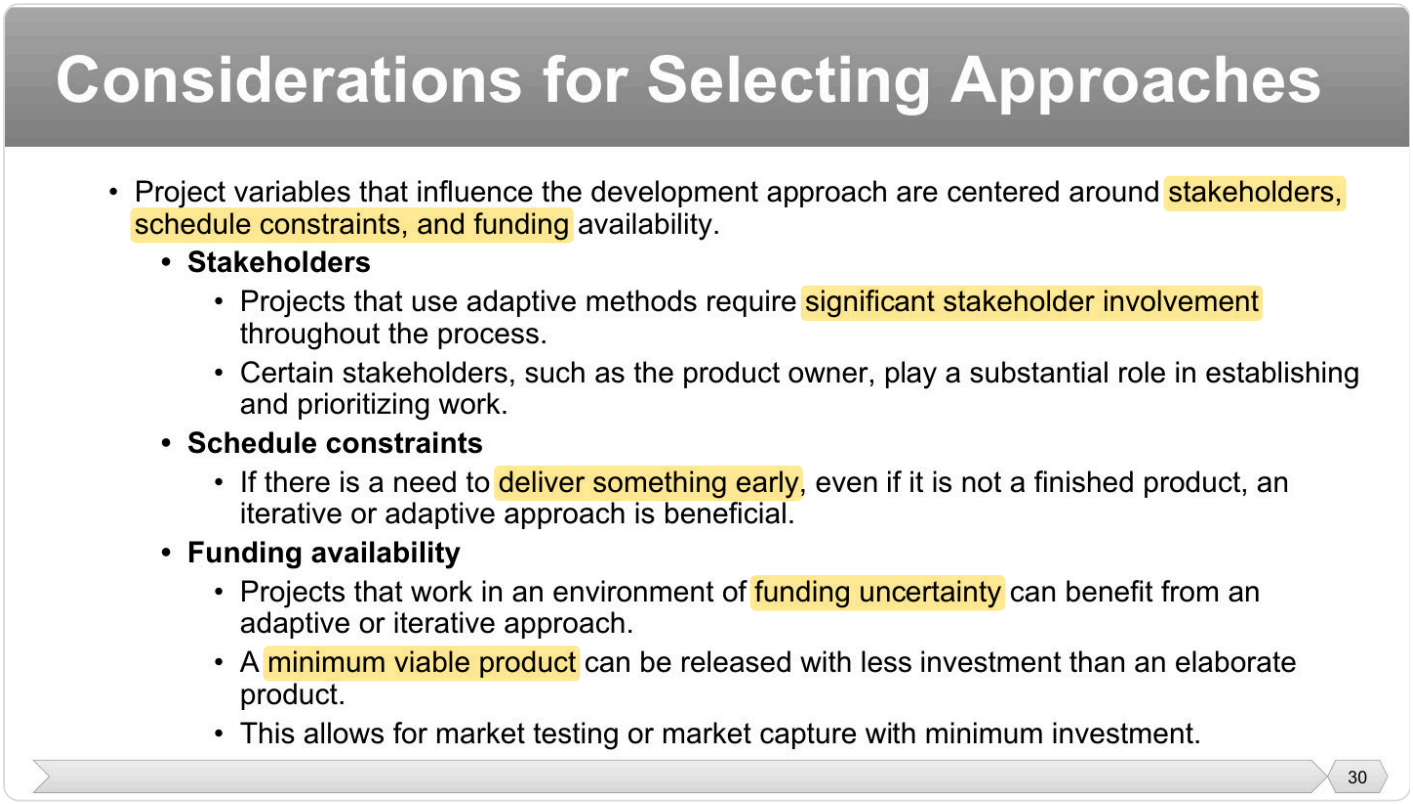
EXAM ANGLE

Prediction Q1

A medical device must meet strict safety regulations and its requirements are stable. Which way do these product variables push the development approach?

Answer: Toward predictive: strict safety requirements and regulation, plus certain and stable requirements, favour up-front planning, documented phases and phase-gate reviews.

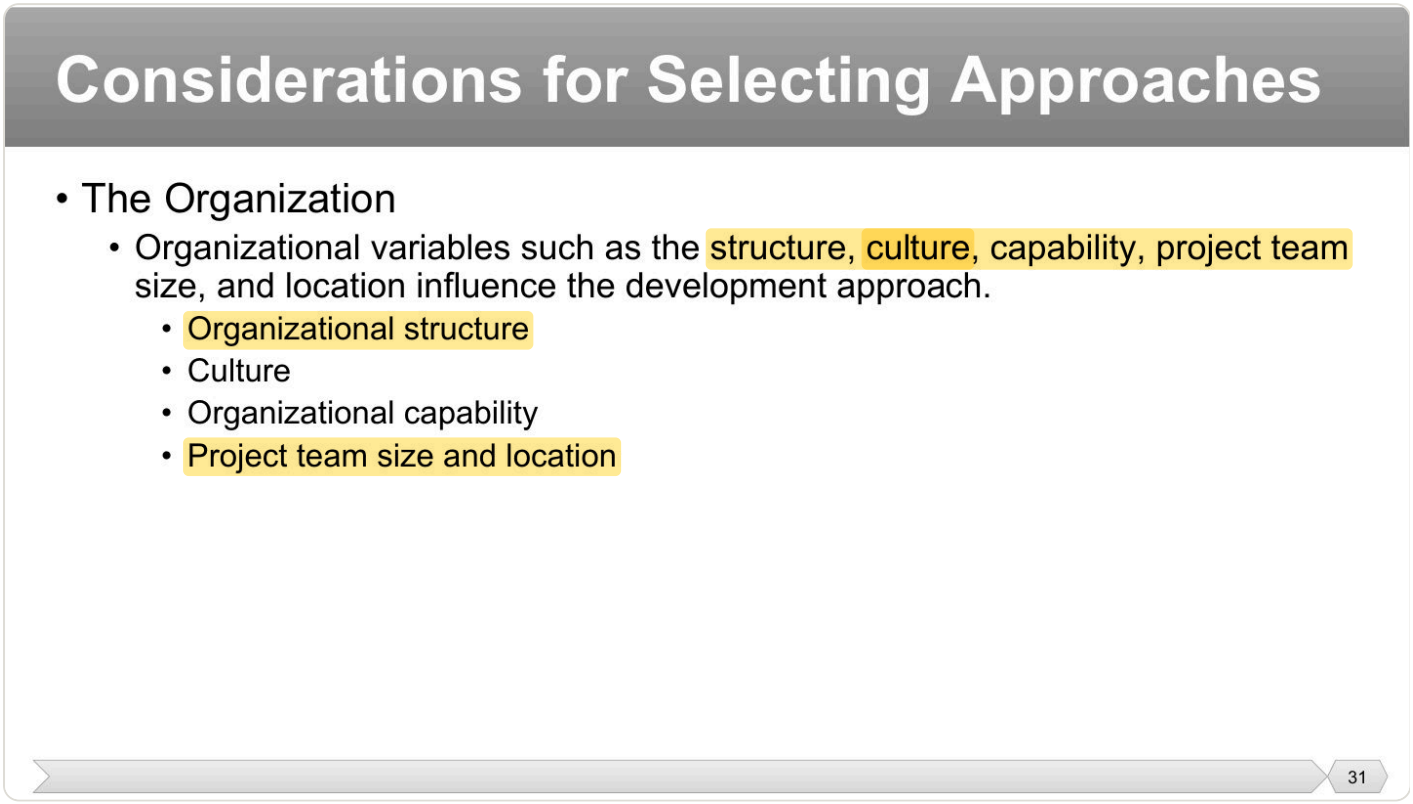
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NOTES

1. Project variables: **stakeholders, schedule constraints and funding availability**.
2. **Stakeholders**: adaptive methods need **significant stakeholder involvement** throughout; a product owner sets and prioritizes the work.
3. **Schedule**: if something must be **delivered early**, even unfinished, an iterative or adaptive approach helps.
4. **Funding**: under **funding uncertainty**, adaptive or iterative work is safer...
5. ...because a **minimum viable product (MVP)** can be released with less investment to test or capture the market.

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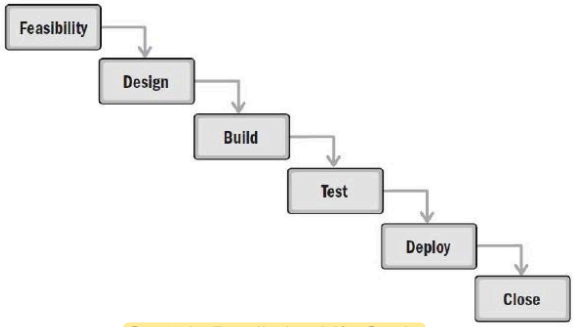
NOTES

1. Organizational variables also matter: **structure, culture, capability, team size and location**.
2. A strongly hierarchical, documentation-heavy culture struggles with adaptive work; a small co-located team adopts it easily; large or distributed teams need more structure.

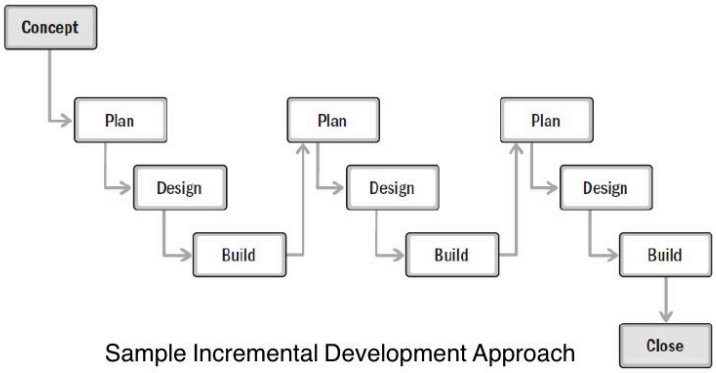
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Life Cycle and Phase Definition

- Project phases often have a phase gate review (also known as stage gate) to check that the desired outcomes or exit criteria for the phase have been achieved before proceeding to the next phase.
- Exit criteria may tie to acceptance criteria for deliverables, contractual obligations, meeting specific performance targets, or other tangible measures.



Sample Predictive Life Cycle

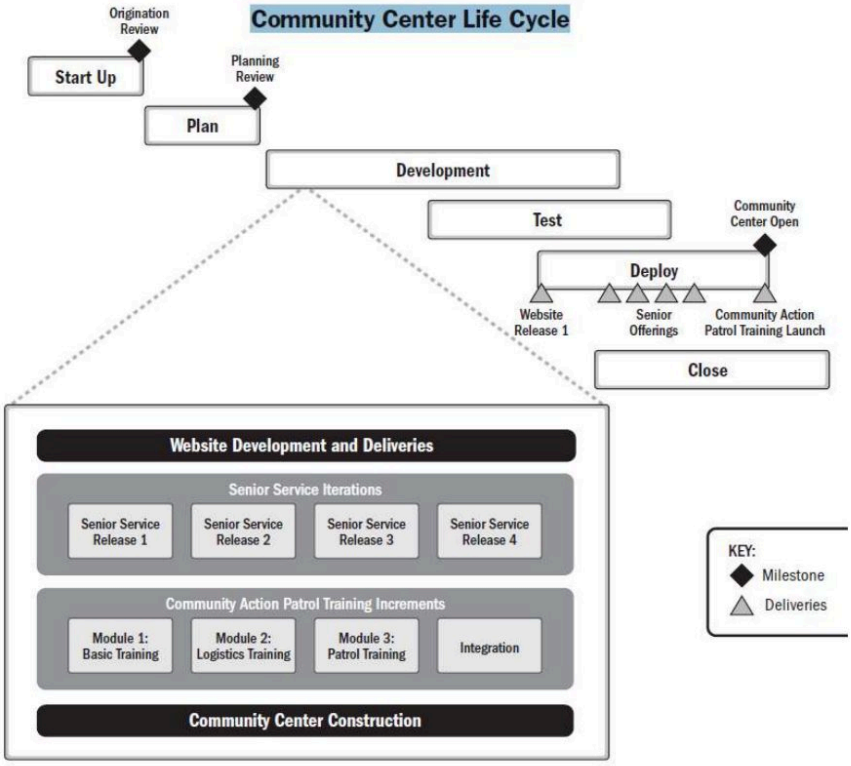


Sample Incremental Development Approach

NOTES

1. Phases often end with a **phase gate review** (stage gate): a checkpoint that the phase's desired outcomes or **exit criteria** are met before moving on.
2. Exit criteria may be tied to **acceptance criteria** for deliverables, contractual obligations or performance targets.
3. The diagrams: a **predictive** life cycle (feasibility → design → build → test → deploy → close, one phase after another) and an **incremental** one (plan → design → build repeated, each round delivering an increment).

A hybrid life cycle: the community center



NOTES

1. A real **hybrid** example. The community center building follows predictive phases with milestone reviews: start up → plan → development → test → deploy → close.

2. Inside development, the **website** is delivered in iterations (senior service releases 1–4) and the **training** (community action patrol modules 1–3 and integration) in increments, while construction proceeds predictively. One project, different approaches for different deliverables.

 Checking Outcomes: Development Approach and Life Cycle

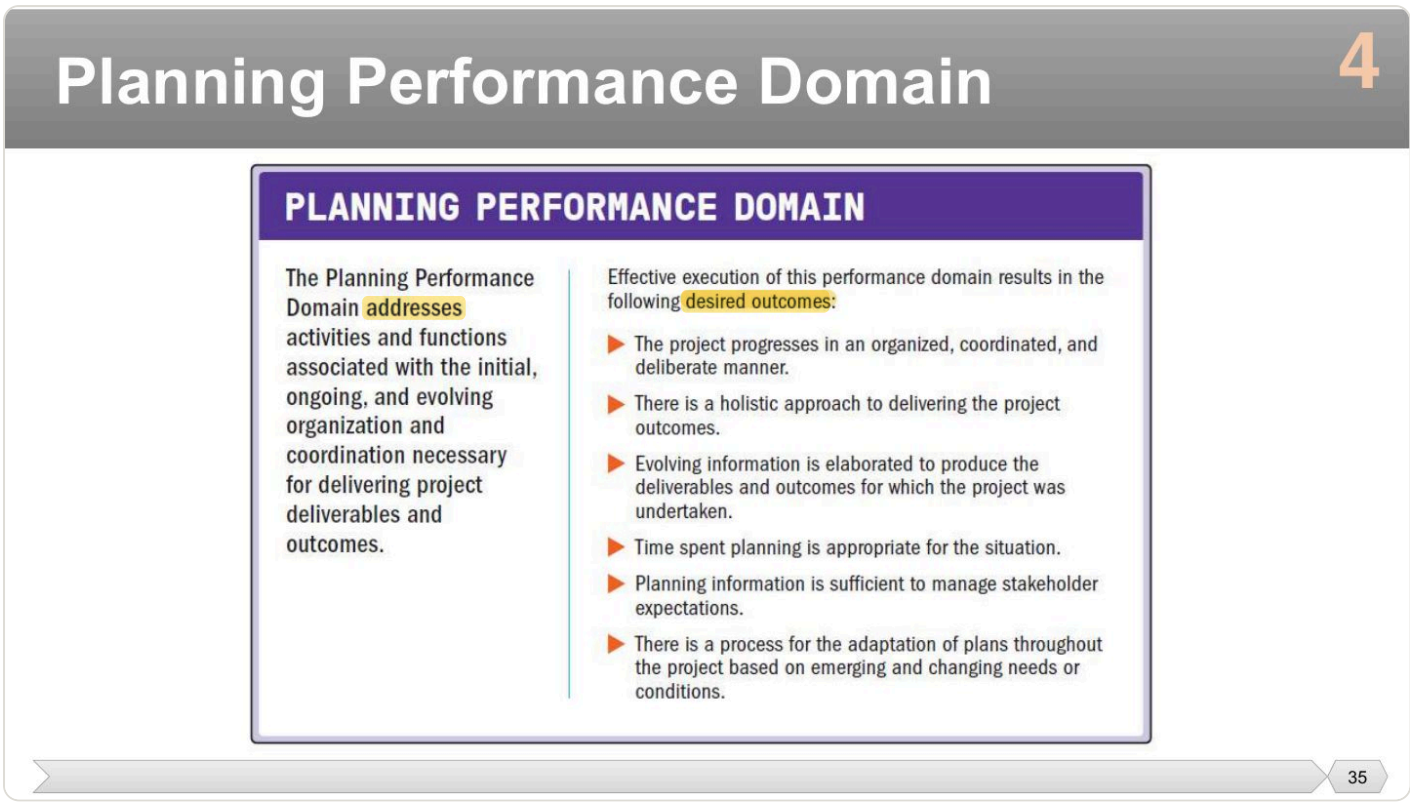
Checking Outcomes—Development Approach and Life Cycle Performance Domain

Outcome	Check
Development approaches that are consistent with project deliverables	The development approach for deliverables (predictive, hybrid, or adaptive) reflects the product variables and is appropriate given the project and organizational variables.
A project life cycle consisting of phases that connect the delivery of business and stakeholder value from the beginning to the end of the project	Project work from launch to close is represented in the project phases. Phases include appropriate exit criteria.
Project life cycle phases that facilitate the delivery cadence and development approach required to produce the project deliverables	The cadence for development, testing, and deploying is represented in the life cycle phases. Projects with multiple deliverables that have different delivery cadences and development methods are represented by overlapping phases or phase repetitions, as necessary.

NOTES

1 The “checking outcomes” table: the left column is what success looks like for the domain, the right column is the evidence that it happened. You won’t be asked to reproduce it, but it is a good structure for a written answer about the domain. Development approach and life cycle: approaches **consistent with the deliverables** (an approach that suits each product), a life cycle that **delivers value from start to end**, and phases that **support the delivery cadence**.

 Planning Performance Domain

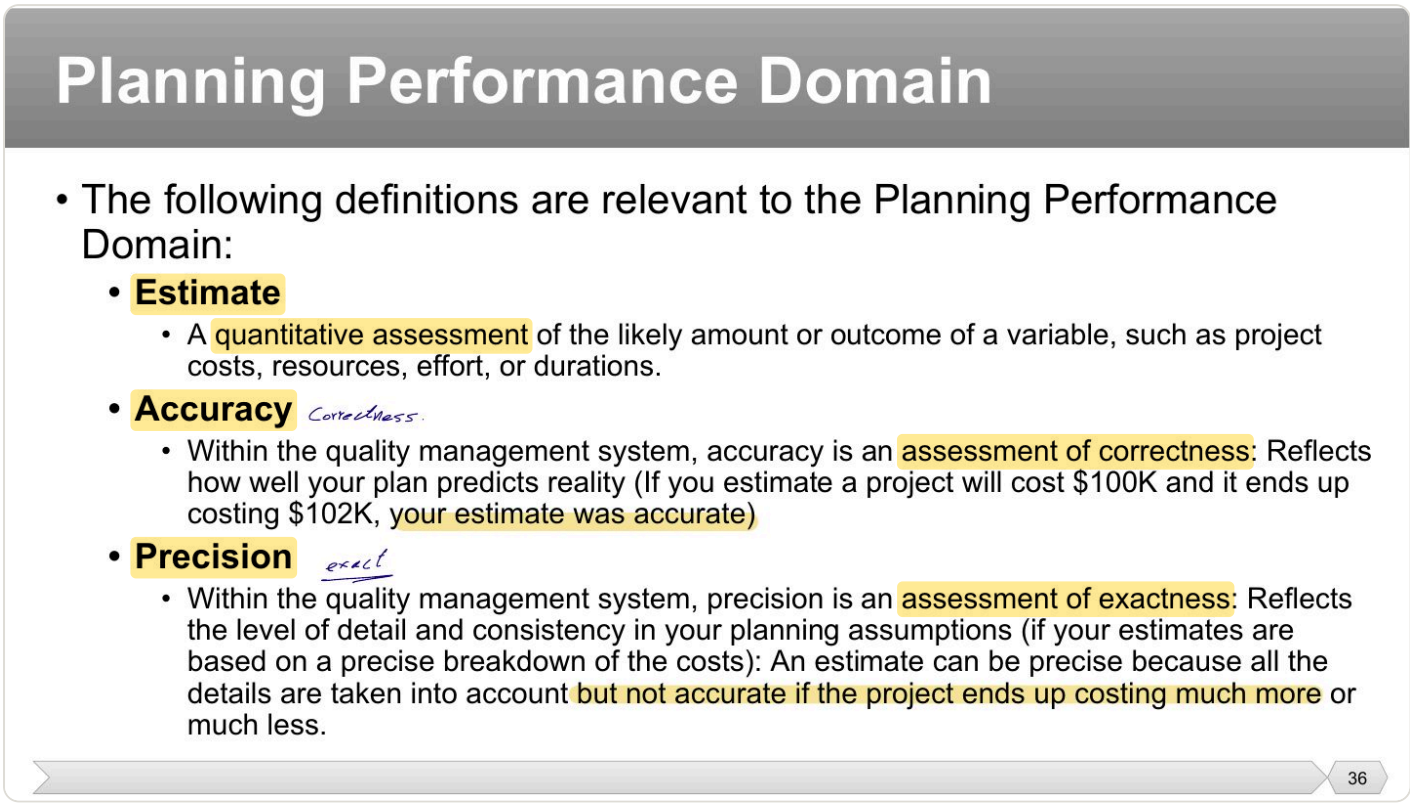


NOTES

1. The **planning** domain covers the initial, ongoing and evolving organization and coordination needed to deliver the deliverables and outcomes.

2. Desired outcomes: the project progresses in an organized, coordinated way; a holistic approach to delivery; evolving information elaborated over time; **planning time appropriate** for the situation (not over-planning); enough information to manage stakeholder expectations; and a process to **adapt plans** as needs change.

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NOTES

1. **Estimate**: a quantitative assessment of the likely amount or outcome of a variable (cost, resources, effort, duration).
2. **Accuracy** = **correctness**: how well the estimate predicts reality. Estimate $100K, actual $102K → accurate.
3. **Precision** = **exactness**: how detailed and consistent the estimate is. A precise estimate (every cost broken down) can still be **inaccurate** if the project ends up costing much more. Accurate = hits the target; precise = tightly grouped.

EXAM ANGLE

Prediction Q1

An estimate built from a detailed, line-by-line cost breakdown ends up 40% over the actual cost. The estimate was:

A. Precise but not accurate
B. Accurate but not precise
C. Both accurate and precise
D. Neither accurate nor precise by definition

Answer: A. Precise but not accurate

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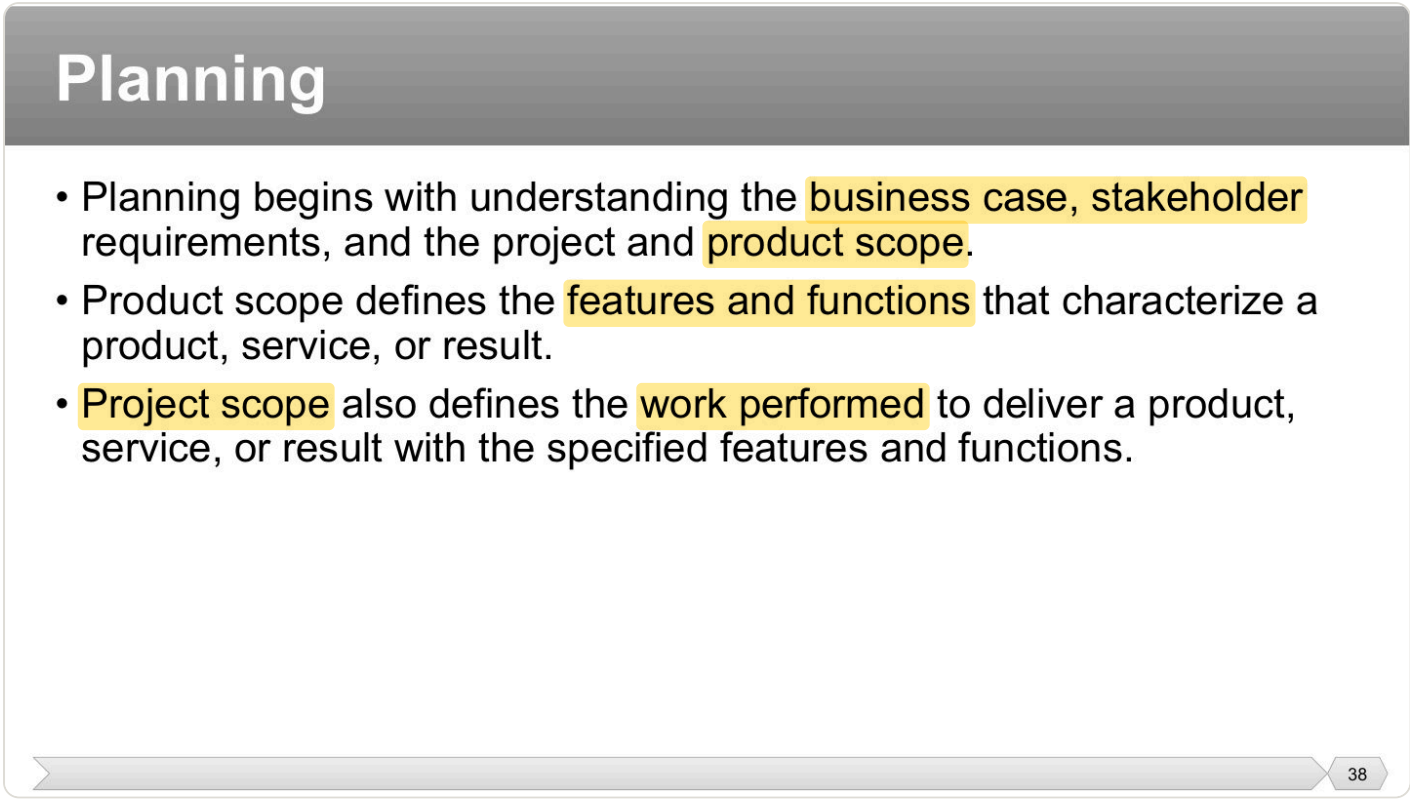
Planning Overview

- The purpose of planning is to proactively (مسبقًا) develop an approach to create the project deliverables.
- High-level planning may begin prior to project authorization.
- The project team progressively elaborates initial project documents, such as a vision statement, project charter, business case, or similar documents to identify or define a coordinated path to achieve the desired outcomes.
- It is becoming more common for initial planning to consider social and environmental impacts in addition to the financial. Such as a product life cycle assessment which evaluates the potential environmental impacts of a product, process, or system (impacts of materials and processes with regards to sustainability, toxicity, and the environment).

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NOTES

1. The purpose of planning is to **proactively** develop an approach to create the deliverables: deciding in advance instead of reacting.
2. High-level planning may start **before the project is authorized** (to build the business case).
3. The team **progressively elaborates** early documents (vision statement, charter, business case): start high-level, add detail as more is learned.
4. Planning increasingly weighs **social and environmental** impacts as well as financial ones, e.g. a product life cycle assessment of materials, toxicity and sustainability.



NOTES

1. Planning starts by understanding the **business case**, the stakeholder requirements, and the project and product scope.
2. **Product scope**: the **features and functions** of the product, service or result (what it has).
3. **Project scope**: the **work** performed to deliver it (what the team must do). Don't swap them; near-identical terms are the instructor's favourite trap.

EXAM ANGLE

Past exam · Final 251 · Q1.3 Q1

Describe the significance of a project scope statement in project management and elaborate on how it aids in mitigating project risks and ensuring successful project outcomes.
Answer: The scope statement defines what the project will and won't deliver: objectives, deliverables, acceptance criteria, exclusions, assumptions and constraints. It reduces risk because everyone agrees on the boundaries up front, it's the baseline for change control (so scope creep is caught and evaluated rather than absorbed), and it's the basis for the WBS, estimates and schedule. That leads to realistic plans and a result the stakeholders will accept.

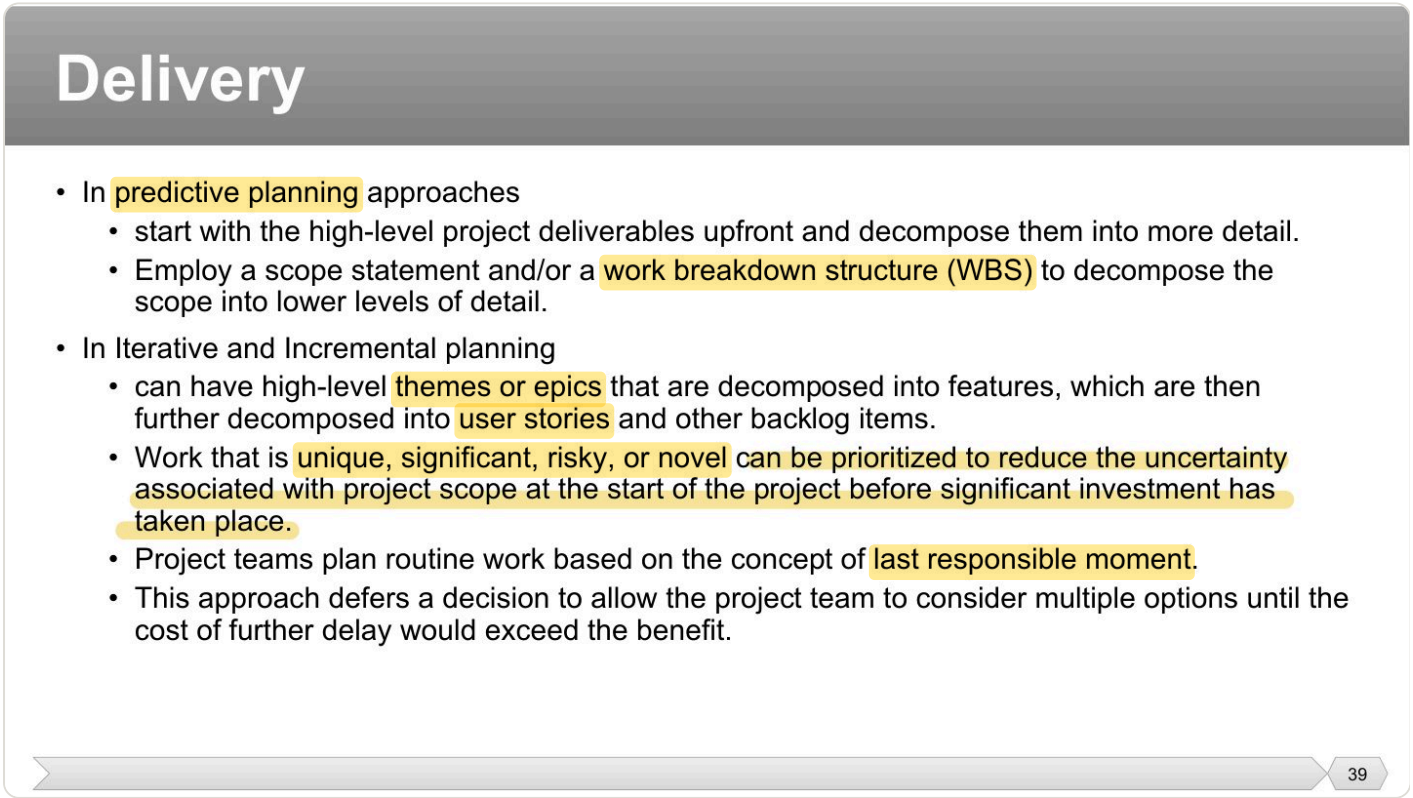
Prediction Q2

"The work performed to deliver a product, service, or result with the specified features and functions" defines:

A. Project scope
B. Product scope
C. The business case
D. Cadence

Answer: A. Project scope

SLIDE 39 / 107 **(cid:68)(cid:101)(cid:108)(cid:105)(cid:118)(cid:101)(cid:114)(cid:121)**



NOTES

1 **Predictive** planning starts with high-level deliverables and decomposes them using a **scope statement** and a **work breakdown structure (WBS)**.

2. **Iterative and incremental** planning decomposes **themes or epics** into features, then into **user stories** and other backlog items.
3. Work that is **unique, significant, risky or novel** is done first, to reduce scope uncertainty before big investment.
4. Routine work is planned at the **last responsible moment**: defer a decision to keep options open until the cost of further delay would exceed the benefit.

EXAM ANGLE

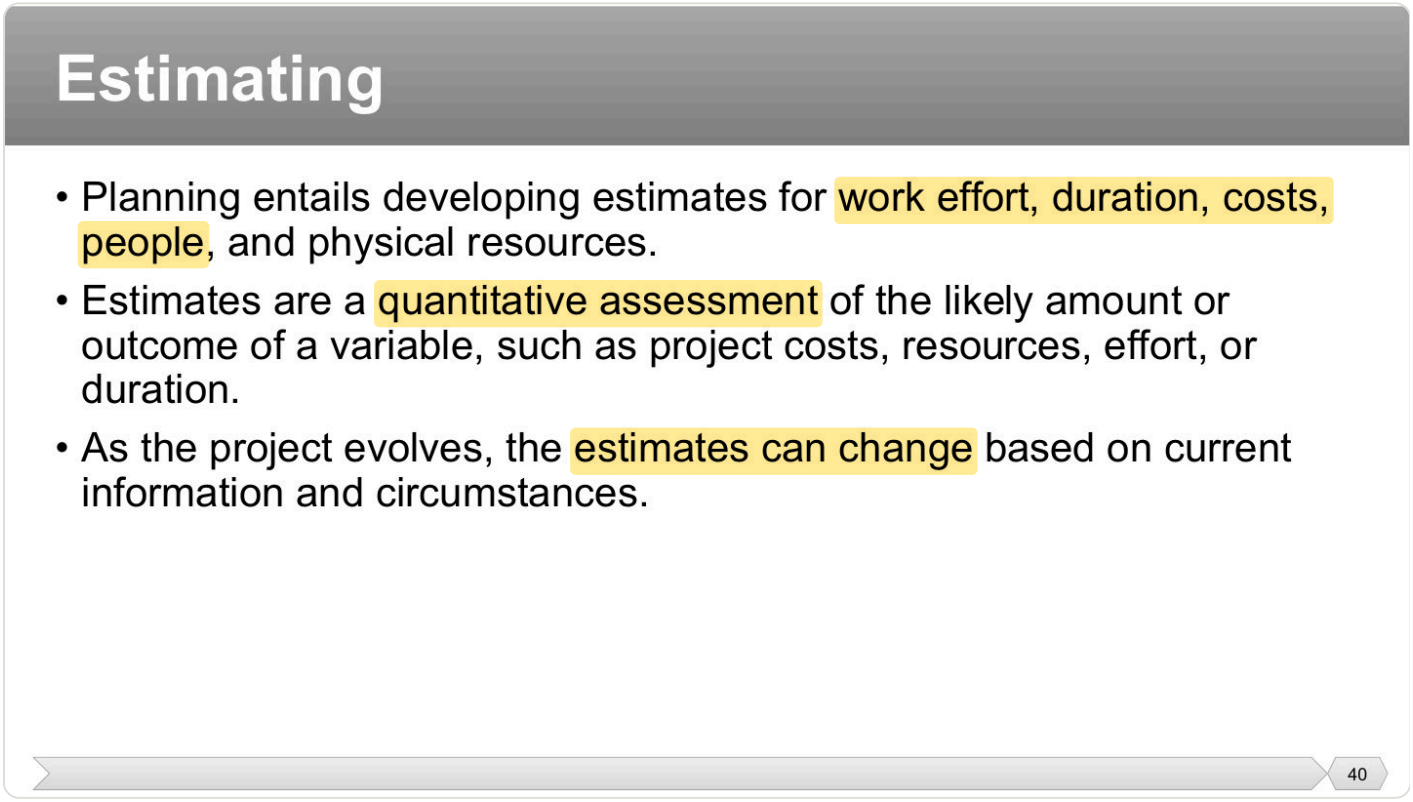
Prediction Q1

Deferring a decision until the cost of further delay would exceed the benefit of waiting is called:

A. The last responsible moment
B. A phase gate
C. Fast-tracking
D. A management reserve

Answer: A. The last responsible moment

SLIDE 40 / 107 **(cid:69)(cid:115)(cid:116)(cid:105)(cid:109)(cid:97)(cid:116)(cid:105)(cid:110)(cid:103)**



NOTES

1. Planning develops estimates for **effort, duration, costs, people and physical resources**.
2. An estimate is a **quantitative assessment** of a likely amount or outcome.

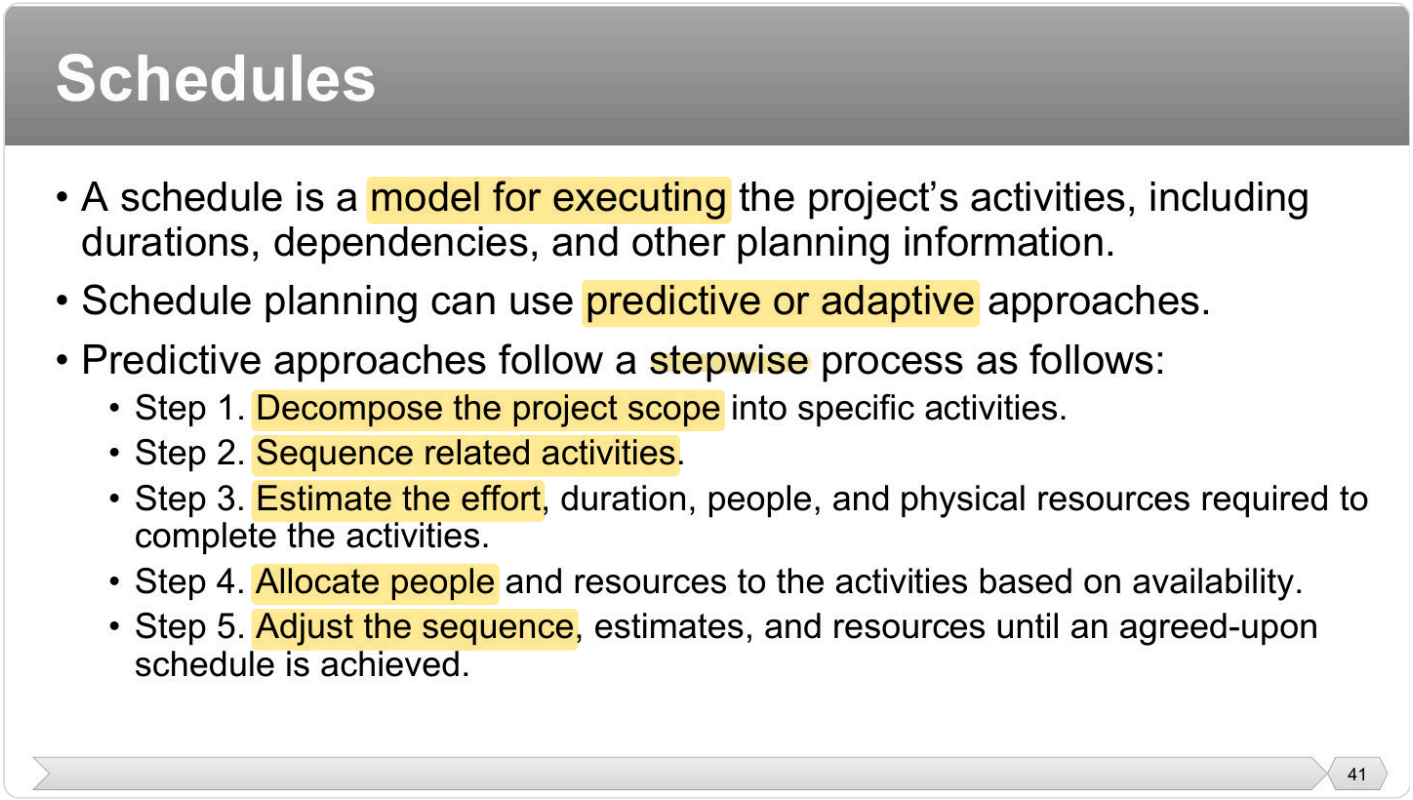
3. Estimates **change** as the project evolves and more is known (the "cone of uncertainty" in Chapter 6).

EXAM ANGLE

Past exam · Final 242 · Q2.2 Q1

To plan accurately for the design, prototyping, and testing phases, you need to estimate the time and effort required. Choose two effort estimation techniques you would apply to this project. Explain how each technique supports better planning and risk control in this scenario.
Answer: Three-point (PERT) estimation: each design, prototyping and test activity gets optimistic, most-likely and pessimistic values, (O + 4M + P) / 6, so the plan reflects the uncertainty of hardware prototyping and the spread shows where contingency is needed. Delphi / expert judgment: hardware and software experts estimate anonymously and converge over rounds, which brings in experience with regulatory testing and removes single-person bias. Comparing the two estimates cross-checks them and exposes risky tasks early. Chapter 6 teaches the techniques.

SLIDE 41 / 107 **(cid:83)(cid:99)(cid:104)(cid:101)(cid:100)(cid:117)(cid:108)(cid:101)(cid:115)**



NOTES

1. A **schedule** is a model for executing the activities: durations, dependencies and other planning information.
2. It can be planned **predictively or adaptively**.
3. The predictive steps: (1) **decompose** scope into activities; (2) **sequence** them; (3) **estimate** effort, duration and resources; (4) **allocate** people and resources by availability; (5) **adjust** until the schedule is agreed. Decompose → sequence → estimate → allocate → adjust.

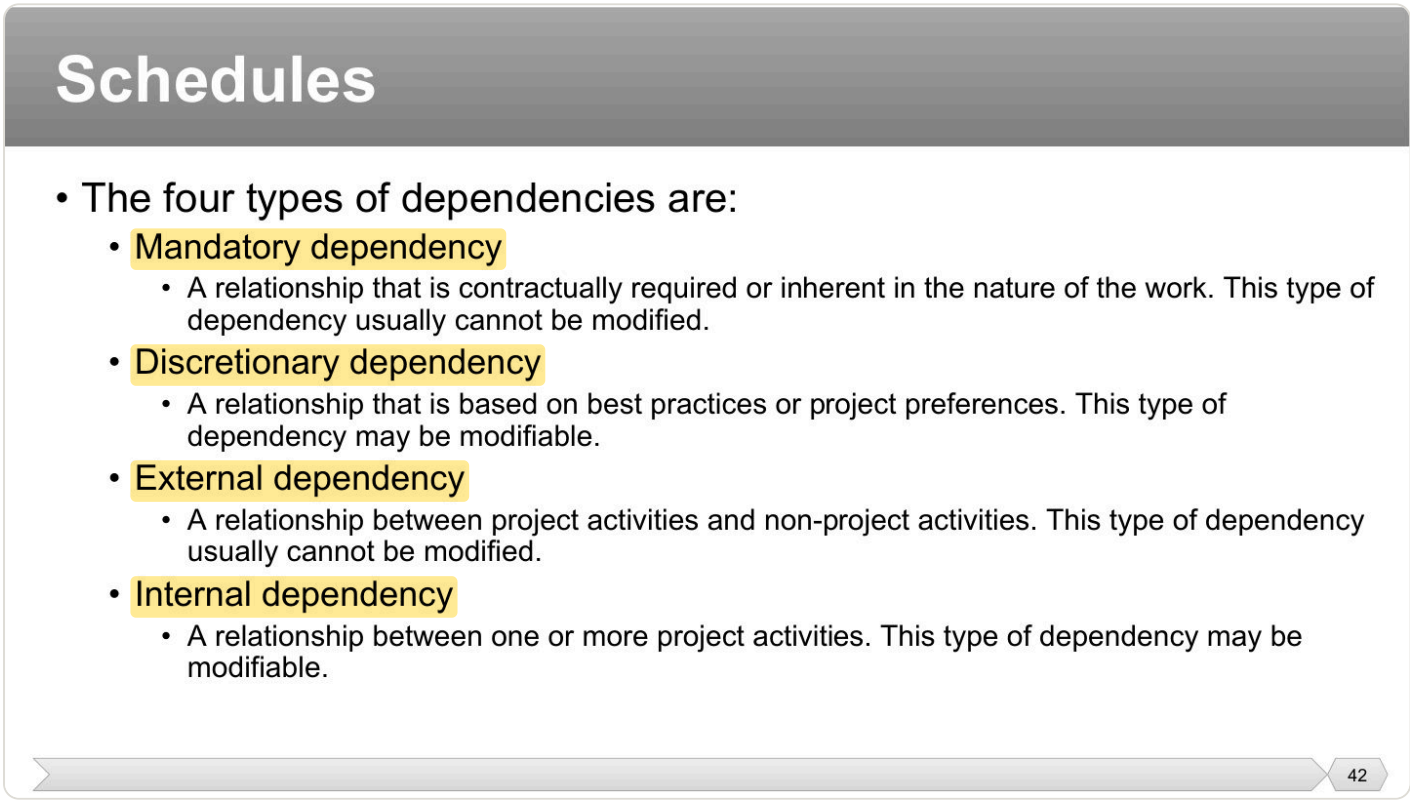
EXAM ANGLE

Past exam · Final 242 · Q2.1 Q1

As the project manager, you must ensure that the 12-month development timeline is realistic and achievable. Identify three project scheduling methods/techniques you would use to plan the sequence and timing of tasks. Briefly explain how each method helps in identifying delays or managing overlapping activities.

Answer: 1) Network diagram + Critical Path Method: shows dependencies and the longest path; any delay on a zero-slack critical activity delays the whole project, while slack shows how much other tasks can slip. 2) Gantt chart: shows each task on a timeline, so overlapping activities, resource clashes and slipping bars (actual vs planned) are easy to spot. 3) PERT: uses optimistic, most-likely and pessimistic durations ((O + 4M + P) / 6), useful for uncertain design, prototyping and regulatory work, so likely delays are built in. Chapter 6.2 covers these techniques.

SLIDE 42 / 107 **(cid:83)(cid:99)(cid:104)(cid:101)(cid:100)(cid:117)(cid:108)(cid:101)(cid:115)**



NOTES

1. **Mandatory**: required by contract or by the nature of the work (you can't test code before it is written). Usually **cannot** be modified.
2. **Discretionary**: based on best practice or preference ("finish the design review before coding"). **May** be modified, which is what fast-tracking relaxes.
3. **External**: between a project activity and a **non-project** one (waiting for a government permit or a vendor's delivery). Usually **cannot** be modified.
4. **Internal**: between project activities, under the team's control. **May** be modified.

EXAM ANGLE

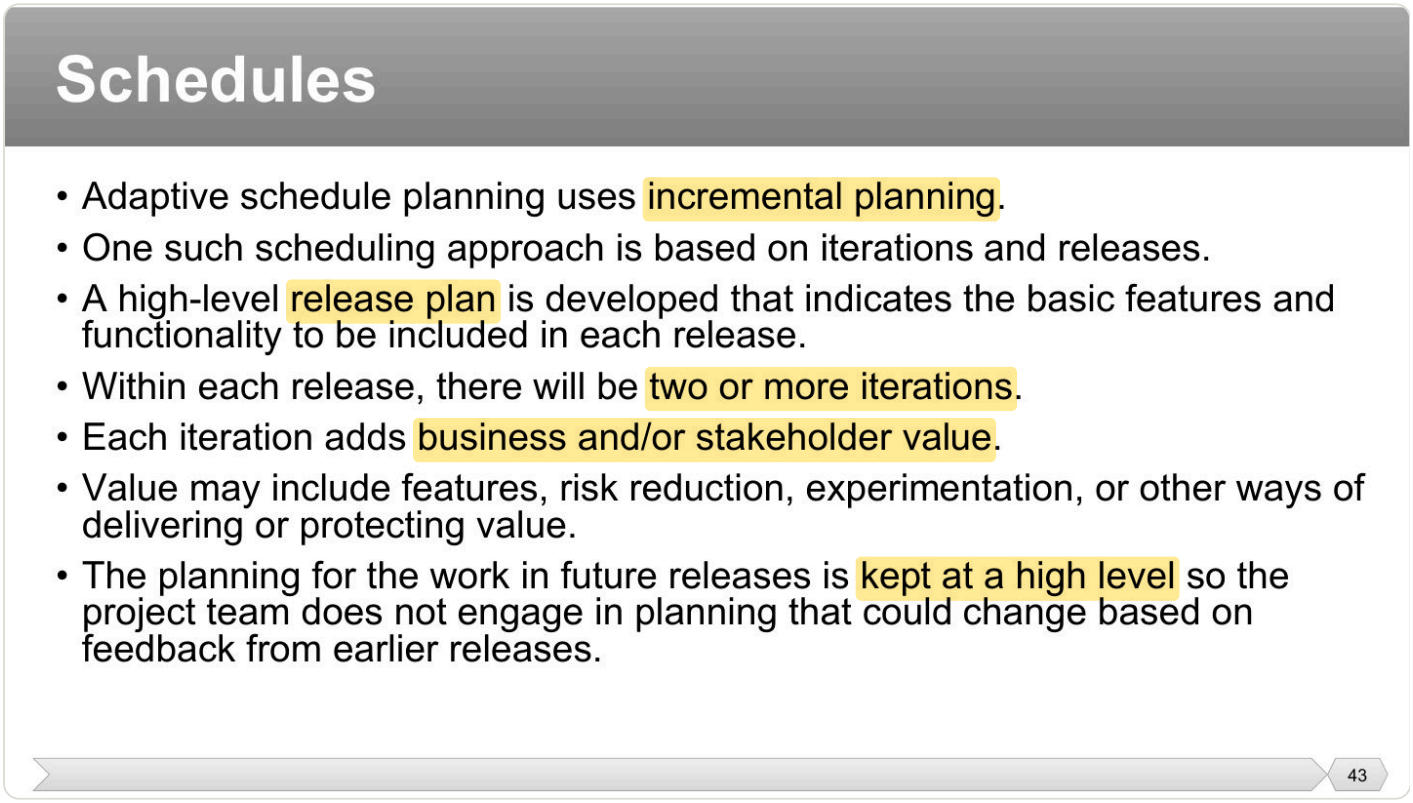
Prediction Q1

Waiting for a government permit before construction can begin is which type of dependency?

A. External
B. Discretionary
C. Internal
D. Neither, it is a constraint

Answer: A. External

SLIDE 43 / 107 **(cid:83)(cid:99)(cid:104)(cid:101)(cid:100)(cid:117)(cid:108)(cid:101)(cid:115)**

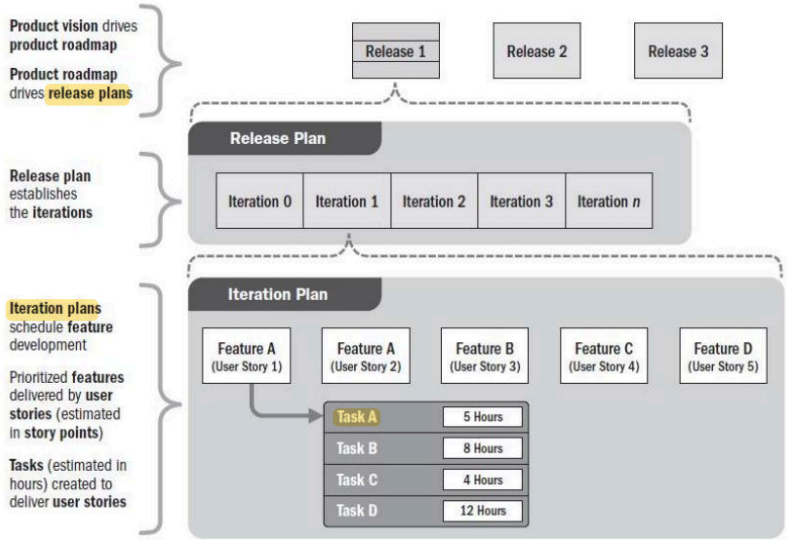


NOTES

1. **Adaptive** scheduling uses **incremental planning**.
2. A high-level **release plan** says which features go in each release.
3. Each release contains **two or more iterations**...
4. ...and each iteration adds **business or stakeholder value** (features, risk reduction, experiments).
5. Future releases are **kept high level** on purpose, because feedback from earlier releases will change them.

 Release and Iteration Plan

Schedules (Release and Iteration Plan)



NOTES

1. Reading the diagram as zoom levels: the product **vision** drives the roadmap, the **roadmap** drives the release plans (Releases 1–3), and each **release plan** sets its iterations (0…n).
2. Each **iteration plan** schedules features delivered as **user stories**, estimated in story points.
3. Stories are broken into **tasks** estimated in hours (Task A 5 h, B 8 h…), the most detailed level, planned only for the current iteration.

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Schedules

- Adaptive approaches often use timeboxes.
- The work in each timebox is based on a prioritized backlog.
- The project team determines the amount of work they can do in each timebox, estimates the work, and self-manages to accomplish the work.
- At the end of the timebox, the project team demonstrates the work completed.
- At that point, the backlog and estimates of work available to be done may be updated or reprioritized for the next timebox.

NOTES

1. Adaptive approaches use **timeboxes**: fixed time, flexible scope (a sprint).
2. The work in each timebox comes from a **prioritized backlog**.
3. The team decides how much it can do, estimates it, and **self-manages** to complete it.
4. At the end the team **demonstrates** the completed work...
5. ...and the backlog and estimates are updated and **reprioritized** for the next timebox.

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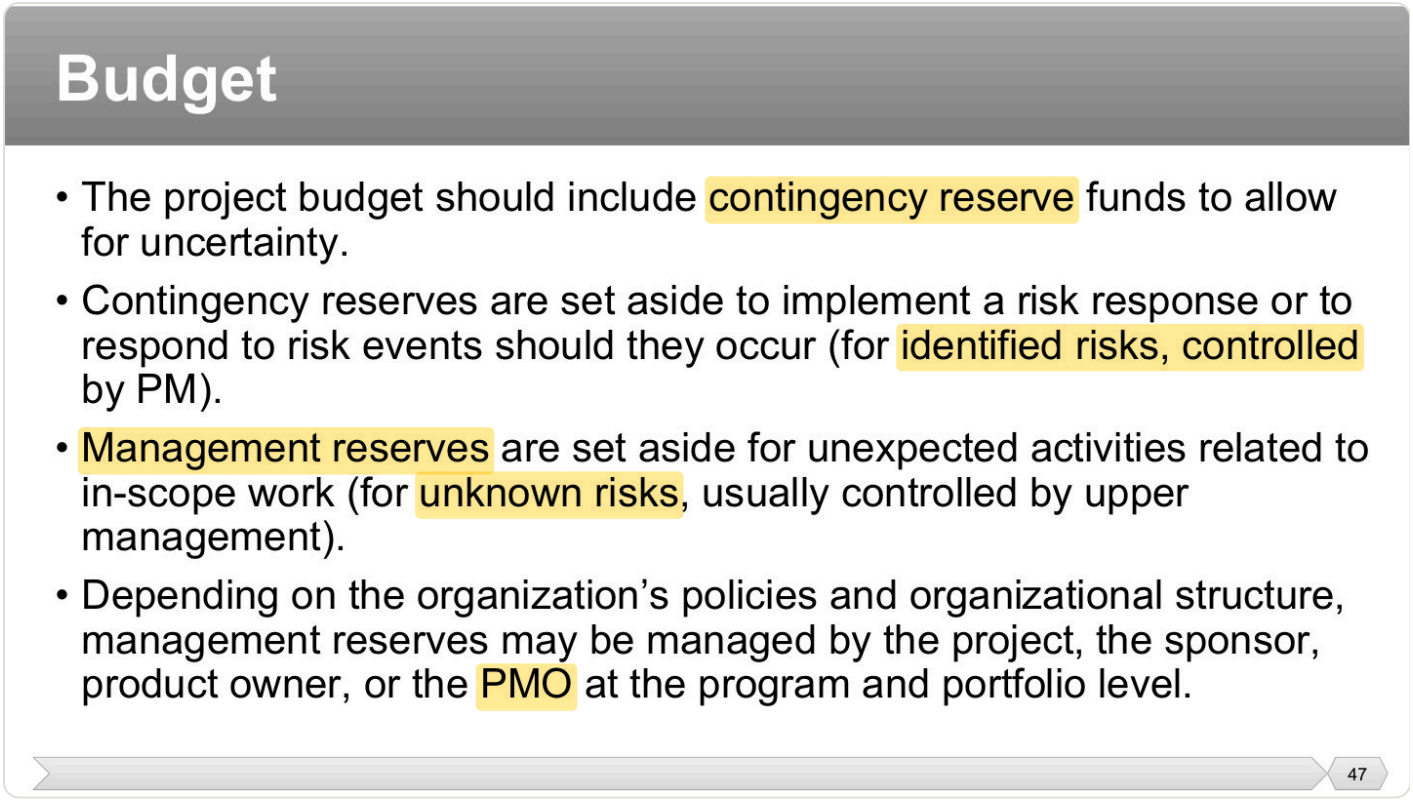
Budget

- The project budget evolves from the agreed estimates for the project.
- Cost estimates are then aggregated to develop the cost baseline.
- The **cost baseline** (time-phased budget) is often allocated across the project schedule to reflect when the costs will be incurred.
- This practice allows project managers to balance the funds approved in a specific budget period with the scheduled work.
- If there are funding limitations for a budget period, the work may need to be rescheduled to meet those limitations.

NOTES

1. The **budget** grows out of the agreed estimates.
2. Cost estimates are aggregated into the **cost baseline**...
3. ...which is a **time-phased budget**: spread across the schedule to show when costs will be incurred.
4. That lets the PM **balance the funds** approved per budget period against the scheduled work.
5. If a period's funding is limited, work may have to be **rescheduled** to fit.

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NOTES

1. The budget should include **contingency reserve** for uncertainty.
2. **Contingency reserve**: set aside to implement a risk response or respond to **identified** risks if they occur. Controlled by the **PM**. ("Known unknowns".)
3. **Management reserve**: set aside for **unknown**, unexpected in-scope work. Usually controlled by **upper management**. ("Unknown unknowns".)
4. Depending on the organization, management reserve may be held by the project, the sponsor, the product owner or the **PMO**.

EXAM ANGLE

Prediction Q1

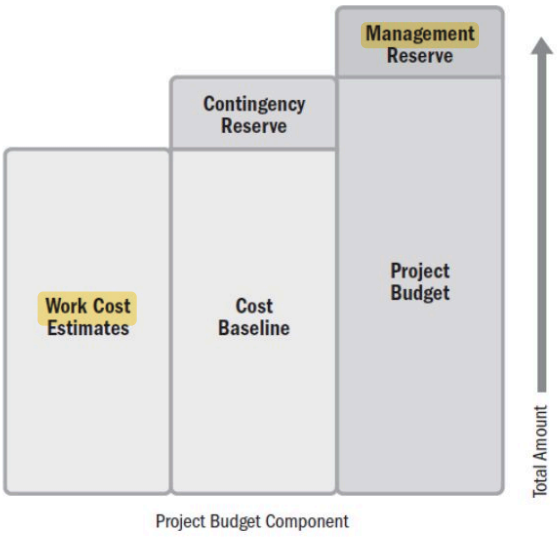
A reserve for unknown, unplanned in-scope work, usually controlled by upper management, is the:

A. Management reserve
B. Contingency reserve
C. Cost baseline
D. Work package budget

Answer: A. Management reserve

 Budget Build Up

Budget Build Up



NOTES

1. Build the budget bottom-up: **work cost estimates** come first.
2. Add the **contingency reserve** → the **cost baseline** (what the PM manages against).
3. Add the **management reserve** → the total **project budget**. So contingency is inside the baseline; management reserve is outside it.

EXAM ANGLE

Prediction Q1

Which amount is included in the cost baseline but not in the management reserve?

A. Contingency reserve
B. Management reserve
C. Both reserves
D. Neither reserve

Answer: A. Contingency reserve

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Physical Resources

- Physical resources apply to any resource that is not a person.
- It can include materials, equipment, software, testing environments, licenses, and so forth.
- Planning for physical resources entails estimating, as well as supply chain, logistics, and management.
- Projects with significant physical resources, such as engineering and construction projects, will need to plan for procurement activities to acquire the resources.
- Planning for physical resources includes taking into account lead time for delivery, movement, storage, and disposition of materials, as well as a means to track material inventory from arrival on site to delivery of an integrated product.

NOTES

1. **Physical resources** are any resource that is **not a person**...
2. ...materials, equipment, software, **testing environments**, licenses.
3. Planning them includes estimating plus **supply chain**, logistics and management.
4. Projects with large physical needs (engineering, construction) must plan **procurement** to acquire them.
5. It also covers **lead time** for delivery, movement, storage, disposal, and tracking inventory from arrival to delivery.

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Procurement

- Procurement can happen at any time during a project. However, upfront planning helps to set expectations that ensure the procurement process is performed smoothly.
- Once the high-level scope is known, project teams conduct a make-or-buy analysis.
- This includes identifying those deliverables and services that will be developed in-house, and those that will be purchased from external sources.
- This information impacts the project team and the schedule.

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NOTES

1. **Procurement** can happen at any time, but planning it up front sets expectations so it runs smoothly.
2. Once the high-level scope is known, the team runs a **make-or-buy analysis**...
3. ...deciding what to develop **in-house** and what to **purchase**. The answer affects the team's composition and the schedule.

EXAM ANGLE

Prediction Q1

Deciding which deliverables to develop in-house and which to purchase is called:

A. Make-or-buy analysis
B. A bidder conference
C. A phase gate
D. Resource leveling

Answer: A. Make-or-buy analysis

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Metrics

- There is a natural linkage between planning, delivering, and measuring work.
- That linkage is metrics.
- Establishing metrics includes setting the thresholds that indicate whether work performance is as expected, trending positively or negatively away from expected performance, or unacceptable.
- Deciding what to measure and "how often" is best informed by the phrase "only measure what matters."

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NOTES

1. Planning, delivering and measuring work are naturally linked...
2. ...and the link is **metrics**.
3. Setting metrics includes **thresholds** that show whether performance is as expected, trending positively or negatively, or unacceptable.
4. Decide what to measure and how often by the rule **"only measure what matters"**: measuring costs effort.

 Checking Outcomes: Planning

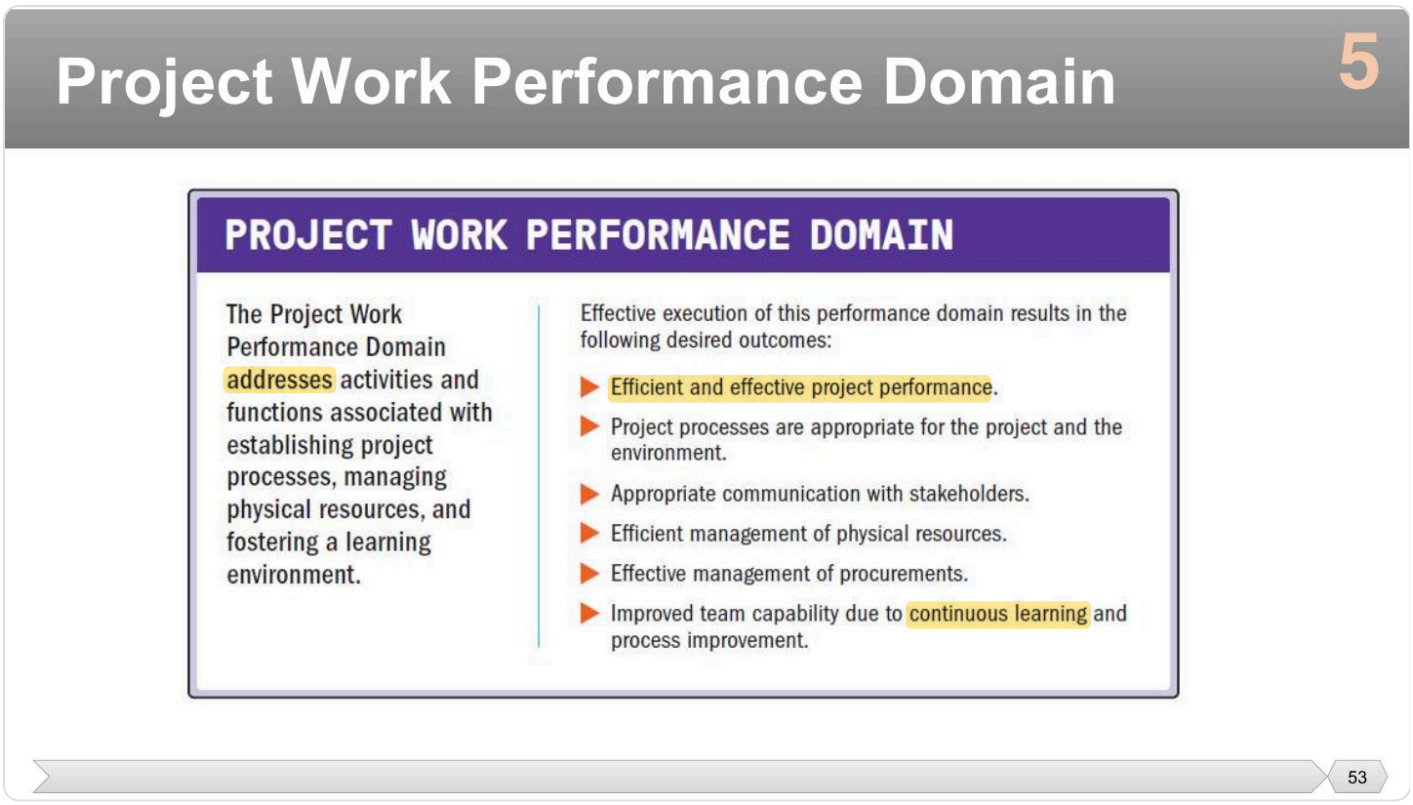
Checking Outcomes—Planning Performance Domain

Outcome	Check
The project progresses in an organized, coordinated, and deliberate manner.	A performance review of project results against the project baselines and other measurement metrics demonstrates that the project is progressing as planned. Performance variances are within thresholds.
There is a holistic approach to delivering the project outcomes.	The delivery schedule, funding, resource availability, procurements, etc., demonstrate that the project is planned in a holistic manner with no gaps or areas of misalignment.
Evolving information is elaborated to produce the deliverables and outcomes for which the project was undertaken.	Initial information about deliverables and requirements compared to current information demonstrates appropriate elaboration. Current information compared to the business case indicates the project will produce the deliverables and outcomes it was undertaken to deliver.
Time spent planning is appropriate for the situation.	Project plans and documents demonstrate that the level of planning is appropriate for the project.
Planning information is sufficient to manage stakeholder expectations.	The communications management plan and stakeholder information indicate that the communications are sufficient to manage stakeholder expectations.
There is a process for the adaptation of plans throughout the project, based on emerging and changing needs or conditions.	Projects using a backlog show the adaptation of plans throughout the project. Projects using a change control process have change logs and documentation from change control board meetings that demonstrate the change control process is being applied.

NOTES

1 The “checking outcomes” table: the left column is what success looks like for the domain, the right column is the evidence that it happened. You won’t be asked to reproduce it, but it is a good structure for a written answer about the domain. Planning: the project progresses in an **organized** way (checked against baselines), a **holistic** approach (schedule, budget and resources integrated), plans **elaborated over time**, planning effort **appropriate** to the situation, and plans that **adapt**.

 Project Work Performance Domain

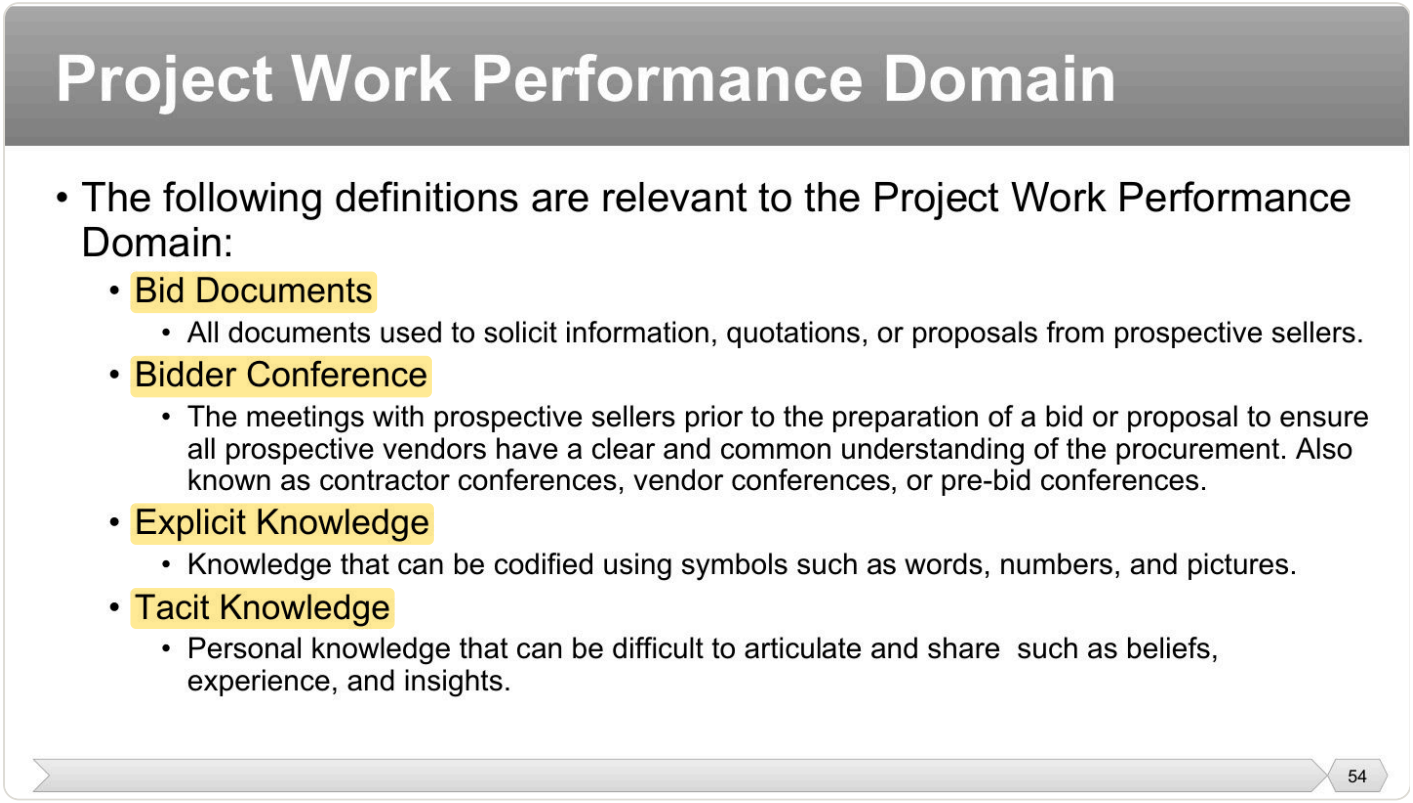


NOTES

1. The **project work** domain covers establishing processes, managing physical resources and fostering a learning environment.

2. Outcomes: efficient and effective performance, appropriate processes, good stakeholder communication, efficient management of physical resources and procurement, and improved capability through **continuous learning**.

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NOTES

1. **Bid documents**: all documents used to request information, quotations or proposals from prospective sellers.
2. **Bidder conference**: meetings with prospective sellers **before** they prepare bids, so all have the same understanding of the procurement (also called contractor, vendor or pre-bid conference).
3. **Explicit knowledge**: can be codified in words, numbers and pictures (documents, wikis).
4. **Tacit knowledge**: personal and hard to articulate (beliefs, experience, insights), e.g. a senior developer's instinct for which module will break. It is shared through mentoring, pairing and conversation.

EXAM ANGLE

Prediction Q1

An experienced developer's instinct for which modules usually break under load is an example of:

A. Tacit knowledge
B. Explicit knowledge
C. A bid document
D. A metric

Answer: A. Tacit knowledge

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Project Work Performance Domain

- Project work keeps the project team focused and project activities running smoothly.
- This includes but is not limited to:
  - Managing the flow of existing work, new work, and changes to work;
  - Keeping the project team focused;
  - Establishing efficient project systems and processes;
  - Communicating with stakeholders;
  - Managing material, equipment, supplies, and logistics;
  - Working with contracting professionals and vendors to plan and manage procurements and contracts;
  - Monitoring changes that can affect the project; and
  - Enabling project learning and knowledge transfer.

NOTES

1. Project work keeps the team **focused** and activities running smoothly.
2. It includes managing the **flow of work and changes**, keeping the team focused, efficient systems and processes, and communicating with stakeholders.
3. It includes managing materials and logistics and working with contracting professionals and vendors on **procurements and contracts**, and monitoring changes that affect the project.
4. And it enables **learning and knowledge transfer** across the team.

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Project Processes

- The project manager and the project team establish and periodically review the processes the project team is using to conduct the work.
- This can take the form of reviewing task boards to determine if there are bottlenecks in the process, if work is flowing at the expected rate, and if there are any impediments that are blocking progress.

NOTES

1. The PM and team establish and **periodically review** the processes they use.
2. For example, review **task boards** for **bottlenecks** (work piling up in one column), whether work flows at the expected rate, and **impediments** blocking progress.

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Project Processes

- Process tailoring can be used to optimize the process for the needs of the project.
- In general, large projects have more process compared to small projects, and critical projects have more process than less significant projects.
- Tailoring takes into consideration the demands of the environment.
- Ways of optimizing the processes for the environment include:
  - Lean production methods.
  - Retrospectives or lessons learned.
  - Where is the next best funding spent?

NOTES

1. **Process tailoring** optimizes the process for the project.
2. **Large** and **critical** projects get more process; small or less significant ones get less.
3. Ways to optimize: **lean** methods (remove waste), **retrospectives** and lessons learned, and asking where the next unit of funding is best spent.

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Project Processes

- Reviewing processes can entail determining if processes are efficient, or if there is waste in the process that can be eliminated.
- Time spent tracking conformance to process is time the project team cannot spend on delivering the outcomes for which the project was commissioned.
- Therefore, project teams utilize just enough time reviewing process conformance to maximize the benefits delivered from the review while still satisfying the governance needs of process.

NOTES

1. Process reviews look for inefficiency and **waste** that can be eliminated.
2. **Time spent checking conformance** to process is time not spent delivering outcomes.
3. So teams spend **just enough** time on process reviews to get their benefit and satisfy governance.

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Balancing Competing Constraints

- Successfully leading a project includes understanding the constraints associated with the work.
- Constraints can take the form of fixed delivery dates, compliance to regulatory codes, a predetermined budget, quality policies, and so forth. The constraints may shift and change throughout the project.
- A new stakeholder requirement may entail expanding the schedule and budget.
- A reduction in budget may entail relaxing a quality requirement or reducing scope.

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NOTES

1. Leading a project means understanding its **constraints**.
2. They include **fixed delivery dates**, regulatory compliance, a **predetermined budget** and quality policies.
3. Constraints **shift** during the project.
4. A **new stakeholder requirement** may expand the schedule and budget...
5. ...and a **budget cut** may relax a quality requirement or reduce scope. They are interrelated: change one, and the others move.

EXAM ANGLE

Past exam · Final 242 · Q1.1 Q1

Define the term ‘project constraints’ and explain how they influence project planning and execution. Provide at least three examples of common constraints and discuss their interrelationship.
Answer: Project constraints are the limits a project must be delivered within. The classic three are scope, time and cost, often with quality, resources and risk. They drive planning because every schedule, budget and scope decision has to fit inside them, and during execution the PM keeps trading them off. They are interrelated (the triple constraint): cutting the schedule usually needs more cost or less scope, and adding scope needs more time or money; quality sits in the middle and suffers if they are squeezed.

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Balancing Competing Constraints

- Balancing these shifting constraints, while maintaining stakeholder satisfaction, is an ongoing project activity.
- At times, it may include meeting with the customer, sponsor, or product owner to present alternatives and implications.
- Other times, the decisions and potential variances may be within the project team's authority to make trade-offs to deliver the end result.

NOTES

1. Balancing shifting constraints while keeping stakeholders satisfied is an **ongoing** activity.
2. Big trade-offs go to the customer, sponsor or product owner, with **alternatives and their implications** presented.
3. Smaller trade-offs and variances may be within the **team's own authority**.

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Working with Procurement

- Many projects involve some form of contracting or procurement.
- Procurement can cover everything from material, capital equipment, and supplies to solutions, labor, and services.
- In most organizations, project managers do not have contracting authority.
- Rather, they work with contracting officers or other people with expertise in contracts, laws, and regulations.

NOTES

1. Many projects involve **contracting or procurement**...
2. ...of anything from **materials and equipment** to solutions, labor and services.
3. In most organizations PMs **do not have contracting authority** (they can't sign contracts)...
4. ...so they work with **contracting officers** and experts in contracts, laws and regulations.

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Working with Procurement

- Organizations usually have rigorous policies and procedures associated with procurements. The policies identify who has authority to enter into a contract, the limits of authority, and the processes and procedures that should be followed.
- Prior to conducting a procurement, the project manager and technically qualified project team members work with contracting professionals to develop the request for proposals (RFP), statement of work (SOW), terms and conditions, and other necessary documents to go out to bid.

NOTES

1. Organizations have **rigorous procurement policies** that define who can sign contracts, the limits of that authority, and the process to follow.
2. Before procuring, the PM and technical team members work with contracting professionals to write the **request for proposals (RFP)**, the **statement of work (SOW)** (what work the seller must do), terms and conditions, and other bid documents.

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The Bid Process

- The bid process includes developing and publicizing bid documents, bidder conferences, and selecting a bidder.
- Bid documents can include:
  - Request for information (Example: A company wants to build a new software platform but isn't sure what technologies or vendors are out there. They issue an RFI to learn about different capabilities and approaches.)
  - Request for proposal (Example: After reviewing RFIs, the company decides on a specific type of software and issues an RFP to invite vendors to bid on building it.)
  - Request for quote (Example: A company needs 500 laptops with specific specs. They issue an RFQ to multiple vendors asking for their best price and delivery timeline.)

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NOTES

1. The **bid process**: develop and publish bid documents, hold bidder conferences, select a bidder.
2. **RFI** (request for information): "what's out there?", used to learn about technologies and vendors before deciding.
3. **RFP** (request for proposal): "how would you build it?", used once the type of solution is chosen, to invite vendors to propose and bid.
4. **RFQ** (request for quote): "what's your price?", used for a known, specified item (500 laptops with exact specs).

EXAM ANGLE

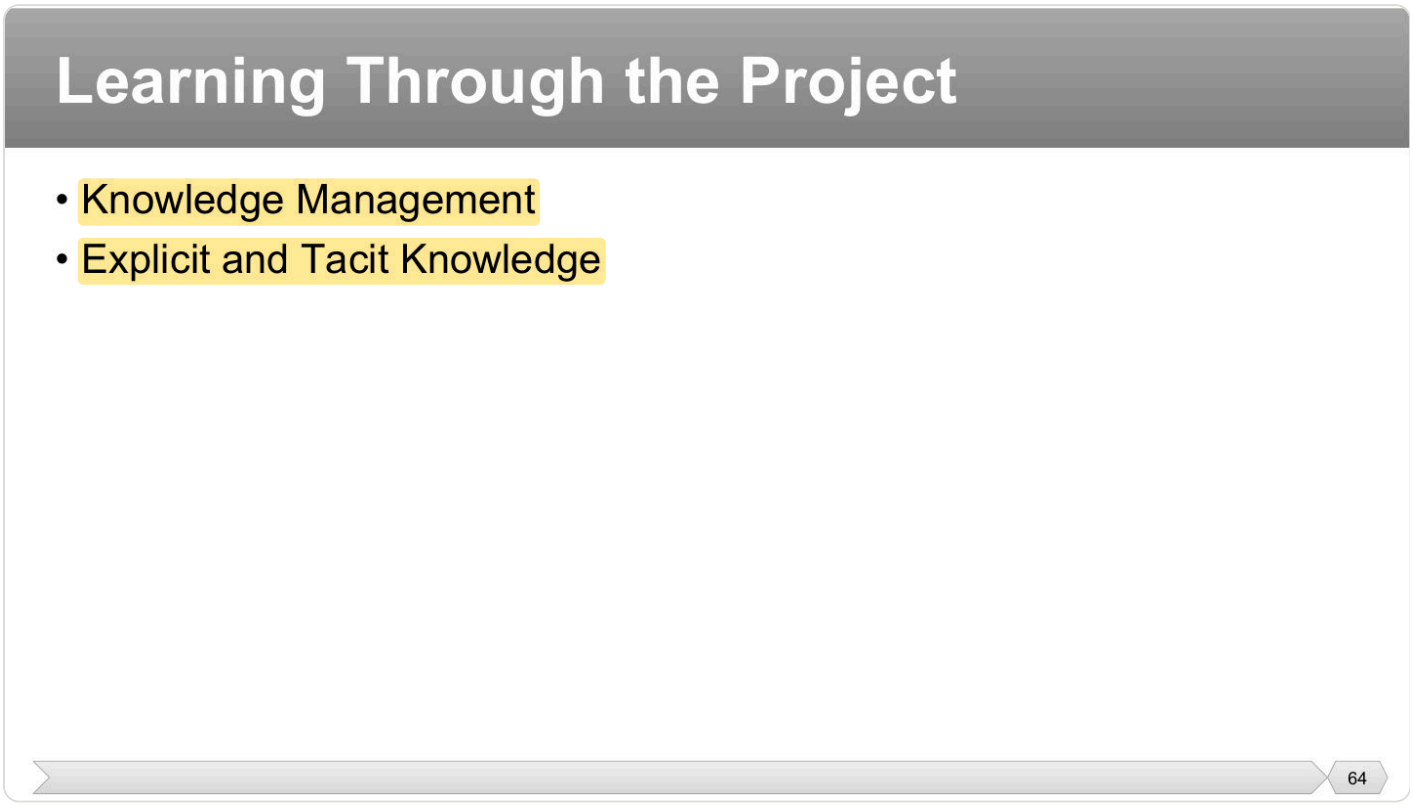
Prediction Q1

A company needs 500 laptops with exact specifications and asks several vendors for their best price and delivery time. Which bid document does it issue?

A. Request for quote (RFQ)
B. Request for information (RFI)
C. Request for proposal (RFP)
D. Statement of work (SOW)

Answer: A. Request for quote (RFQ)

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NOTES

1. Learning through the project is **knowledge management**: capturing and sharing what the team learns so it isn't lost when people leave.

2. It covers both **explicit** knowledge (documents, lessons-learned registers) and **tacit** knowledge (shared through mentoring, pairing, communities of practice). The diagram shows explicit knowledge flowing through information management and tacit knowledge through people.

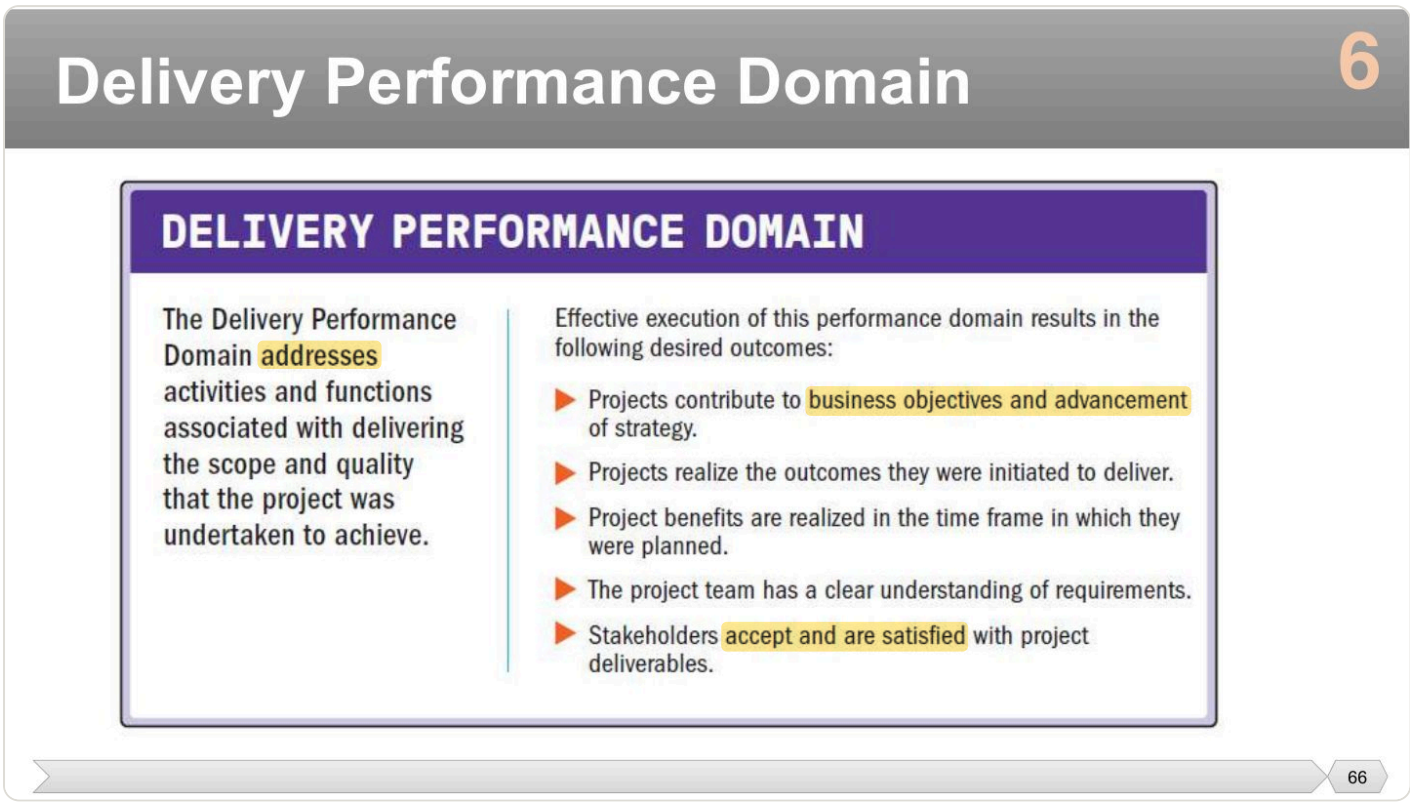
Checking Outcomes: Project Work

Outcome	Check
Efficient and effective project performance	Status reports show that project work is efficient and effective.
Project processes that are appropriate for the project and the environment	Evidence shows that the project processes have been tailored to meet the needs of the project and the environment. Process audits and quality assurance activities show that the processes are relevant and being used effectively.
Appropriate communication and engagement with stakeholders	The project communications management plan and communication artifacts demonstrate that the planned communications are being delivered to stakeholders. There are few ad hoc requests for information or misunderstandings that might indicate engagement and communication activities are not effective.
Efficient management of physical resources	The amount of material used, scrap discarded, and amount of rework indicate that resources are being used efficiently.
Effective management of procurements	A procurement audit demonstrates that appropriate processes utilized were sufficient for the procurement and that the contractor is performing to plan.
Effective handling of change	Projects using a predictive approach have a change log that demonstrates changes are being evaluated holistically with consideration for scope, schedule, budget, resource, stakeholder, and risk impacts. Projects using an adaptive approach have a backlog that shows the rate of accomplishing scope and the rate of adding new scope.
Improved capability due to continuous learning and process improvement	Team status reports show fewer errors and rework with an increase in velocity.

NOTES

1 The “checking outcomes” table: the left column is what success looks like for the domain, the right column is the evidence that it happened. You won’t be asked to reproduce it, but it is a good structure for a written answer about the domain. Project work: efficient performance (status reports show the work on track), appropriate processes, good communication, efficient physical resources and procurement, and improved capability through learning.

 Delivery Performance Domain



NOTES

1. The **delivery** domain covers delivering the scope and quality the project was undertaken to achieve.
2. Outcomes: projects contribute to business objectives and strategy, realize the outcomes they were initiated for, realize benefits in the planned time, the team clearly understands the requirements, and **stakeholders accept and are satisfied** with the deliverables.

SLIDE 67 / 107 **(cid:68)(cid:101)(cid:108)(cid:105)(cid:118)(cid:101)(cid:114)(cid:121)(cid:32)(cid:80)(cid:101)(cid:114)(cid:102)(cid:111)(cid:114)(cid:109)(cid:97)(cid:110)(cid:99)(cid:101)(cid:32)(cid:68)(cid:111)(cid:109)(cid:97)(cid:105)(cid:110)**

Delivery Performance Domain

- The following definitions are relevant to the Delivery Performance Domain:
  - Definition of Done (DoD)
    - A checklist of all the criteria required to be met so that a deliverable can be considered ready for customer use.
  - Quality
    - The degree to which a set of inherent characteristics fulfills requirements.
  - Cost of Quality (COQ)
    - All costs incurred over the life of the product by investment in preventing nonconformance to requirements, appraisal (تقييم) of the product or service for conformance to requirements, and failure to meet requirements.

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NOTES

1. **Definition of done (DoD)**: a checklist of all criteria a deliverable must meet to be considered **ready for customer use** (coded, reviewed, tested, documented).
2. **Quality**: the degree to which a set of **inherent characteristics fulfills requirements**.
3. **Cost of quality (COQ)**: all costs over the product's life from **prevention** (training, standards), **appraisal** (testing, inspection) and **failure** (rework, warranty, lost customers). Chapter 7 develops it.

EXAM ANGLE

Prediction Q1

Which three kinds of cost make up the cost of quality?

A. Prevention, appraisal, and failure
B. Design, build, and test
C. Fixed, variable, and sunk
D. Labor, materials, and overhead

Answer: A. Prevention, appraisal, and failure

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Delivery of Value

h

- Projects that use a development approach that supports **releasing deliverables throughout the project life cycle** can start delivering value to the business, customer, or other stakeholders during the project.
- Business value often continues to be captured long after the initial project has ended.
- A business case document often provides the business justification and a projection of anticipated business value from a project.
- The format of this business case varies based on the development approach and life cycle selected.

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NOTES

1. Approaches that **release** deliverables throughout the life cycle start delivering value **during** the project.
2. Business value often keeps being captured **long after** the project ends.
3. A **business case** gives the justification and projected value; its format depends on the approach.

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Quality

- Delivery is more than just scope and requirements.
- Scope and requirements focus on what needs to be delivered.
- Quality focuses on the performance levels that are required to be met.
- Quality requirements may be reflected in the completion criteria, definition of done, statement of work, or requirements documentation.

NOTES

1. Delivery is **more than scope and requirements**.
2. Scope and requirements say **what** must be delivered.
3. Quality says **how well**: the **performance levels** that must be met (response time, reliability, accuracy).
4. Quality requirements appear in completion criteria, the **definition of done**, the statement of work or requirements documents.

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Quality

- Much of the costs associated with quality are born by the sponsoring organization and are reflected in policies, procedures, and work processes.
- For example, organizational policies that govern how work is performed and procedures that prescribe work processes are often part of the organization's quality policy.
- The cost of overhead, training, and process audit are born by the organization, though they are employed by the project.
- Inherent in projects is balancing the quality needs of the processes and products with the costs associated with meeting those needs.
- The cost of quality methodology is used to find the appropriate balance for investing in quality prevention and appraisal to avoid defect or product failures.

NOTES

1. Much of the cost of quality is borne by the **sponsoring organization** through its policies, procedures and processes.
2. Policies that govern how work is done are part of the organization's **quality policy**.
3. Overhead, **training and process audits** are paid by the organization though used by the project.
4. Projects **balance** the quality needs of processes and products against the cost of meeting them.
5. The **cost of quality** method finds that balance: invest in prevention and appraisal to avoid more expensive defects and failures.

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Suboptimal Outcomes

- All projects attempt to deliver outcomes, though some may fail to do so or may produce suboptimal outcomes.
- The potential for suboptimal outcomes exists in every project.
- In the case of a fully experimental project, the organization is attempting to achieve a breakthrough, such as the creation of a completely new technology, for example.
- This requires deliberate investment in an uncertain outcome.
- Companies that produce new medicines or compounds may experience several failures before finding a successful formula.
- Some projects may fail to deliver outcomes because the market opportunity has passed or competitors were first to market with their offering.
- Effective project management can minimize negative outcomes, but such possibilities are part of the uncertainty of attempting to produce a unique deliverable.

NOTES

1. Some projects fail or produce **suboptimal outcomes**, and the possibility exists in every project.
2. A **fully experimental** project (a new technology or medicine) deliberately invests in an uncertain outcome; several failures may come before success.
3. Others fail because the **market opportunity passed** or a competitor got there first.
4. Good project management **minimizes** negative outcomes but can't eliminate them: uncertainty is part of producing something unique.

 Checking Outcomes: Delivery

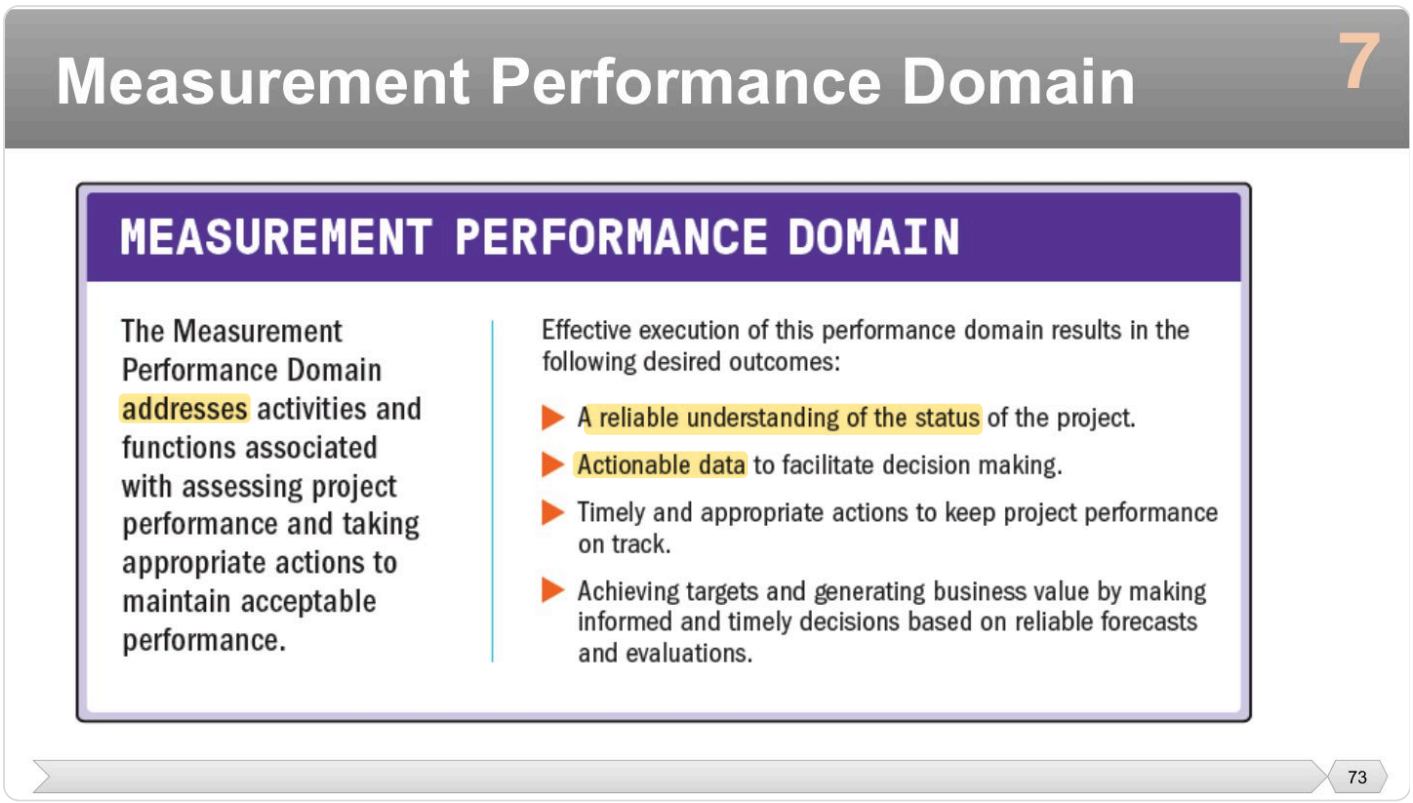
Checking Outcomes—Delivery Performance Domain

Outcome	Check
Projects contribute to business objectives and advancement of strategy	The business plan and the organization's strategic plan, along with the project authorizing documents, demonstrate that the project deliverables and business objectives are aligned.
Projects realize the outcomes they were initiated to deliver	The business case and underlying data indicate the project is still on track to realize the intended outcomes.
Project benefits are realized in the time frame in which they were planned	The benefits realization plan, business case, and/or schedule indicate that the financial metrics and scheduled deliveries are being achieved as planned.
The project team has a clear understanding of requirements	In predictive development, little change in the initial requirements reflects understanding. In projects where requirements are evolving, a clear understanding of requirements may not take place until well into the project.
Stakeholders accept and are satisfied with project deliverables	Interviews, observation, and end user feedback indicate stakeholder satisfaction with deliverables. Levels of complaints and returns can also be used to indicate satisfaction.

NOTES

1 The "checking outcomes" table: the left column is what success looks like for the domain, the right column is the evidence that it happened. You won't be asked to reproduce it, but it is a good structure for a written answer about the domain. Delivery: contribution to strategy and business objectives (checked against the business case), outcomes realized, benefits realized on time (benefits realization plan), clear understanding of requirements, and stakeholder acceptance.

 Measurement Performance Domain



NOTES

1. The **measurement** domain covers assessing performance and taking action to keep it acceptable.
2. Outcomes: a **reliable understanding of status**, **actionable data** for decisions, timely actions to stay on track, and business value from informed, timely decisions based on reliable forecasts.

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Measurement Performance Domain

- The following definitions are relevant to the Measurement Performance Domain:
  - Metric
    - A description of a project or product attribute and how to measure it (ex: work progress %).
  - Baseline
    - The approved version of a measure used as a basis for comparison to actual results (cost of a project, duration).
  - Dashboard
    - A set of charts and graphs showing progress or performance against important measures of the project.

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NOTES

1. **Metric**: a description of a project or product attribute and **how to measure** it (e.g. percent of work complete).
2. **Baseline**: the **approved version** of a measure used to compare actual results against (the approved budget or schedule).
3. **Dashboard**: a set of charts and graphs showing progress or performance against the important measures.

EXAM ANGLE

Prediction Q1

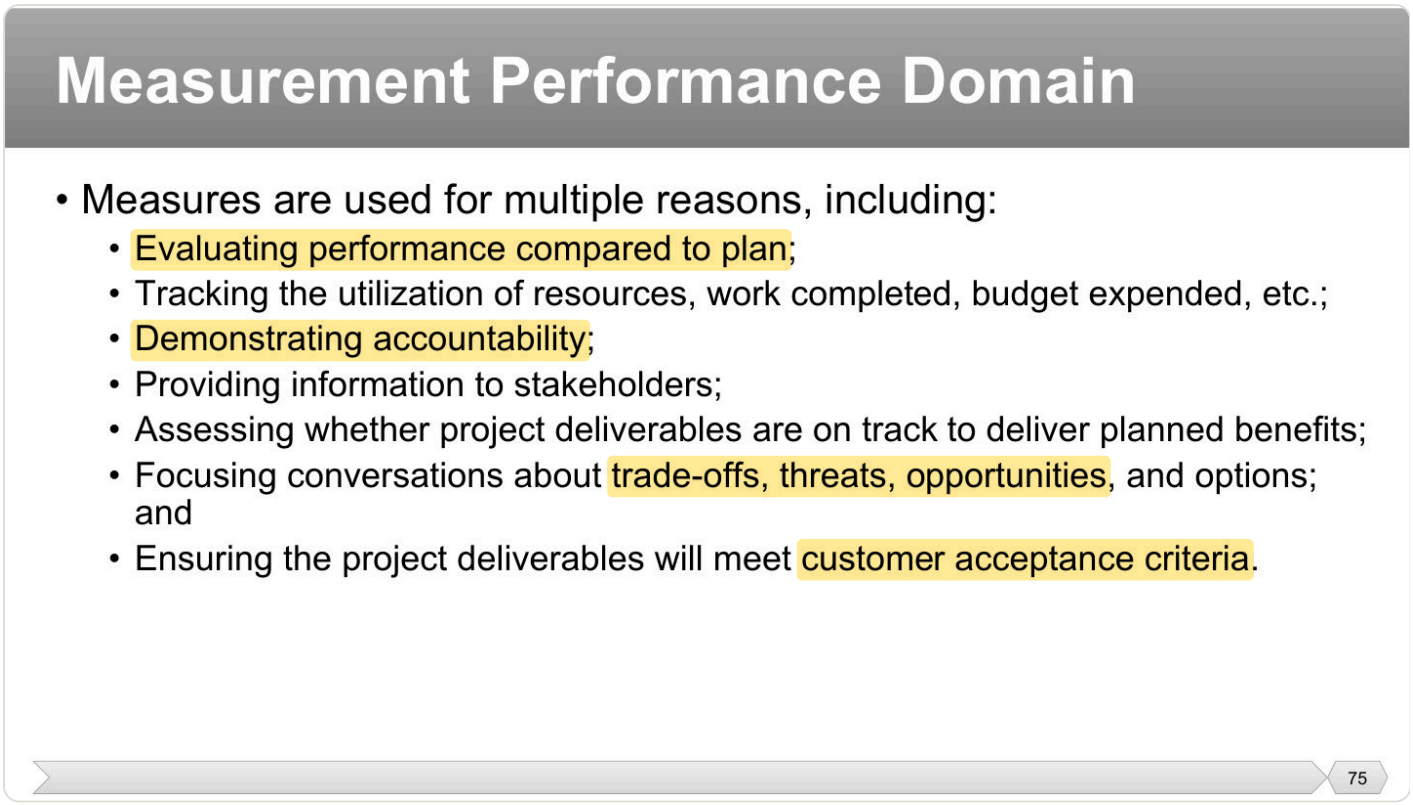
The approved version of a measure that actual results are compared against is called a:

A. Baseline
B. Metric
C. Dashboard
D. Forecast

Answer: A. Baseline

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NOTES

1. Why measure: to **evaluate performance against the plan** and track resource use, work done and budget spent.
2. To **demonstrate accountability** and inform stakeholders.
3. To check whether deliverables will produce the planned benefits, and to **focus conversations** about trade-offs, threats, opportunities and options.
4. To make sure deliverables will meet **customer acceptance criteria**.

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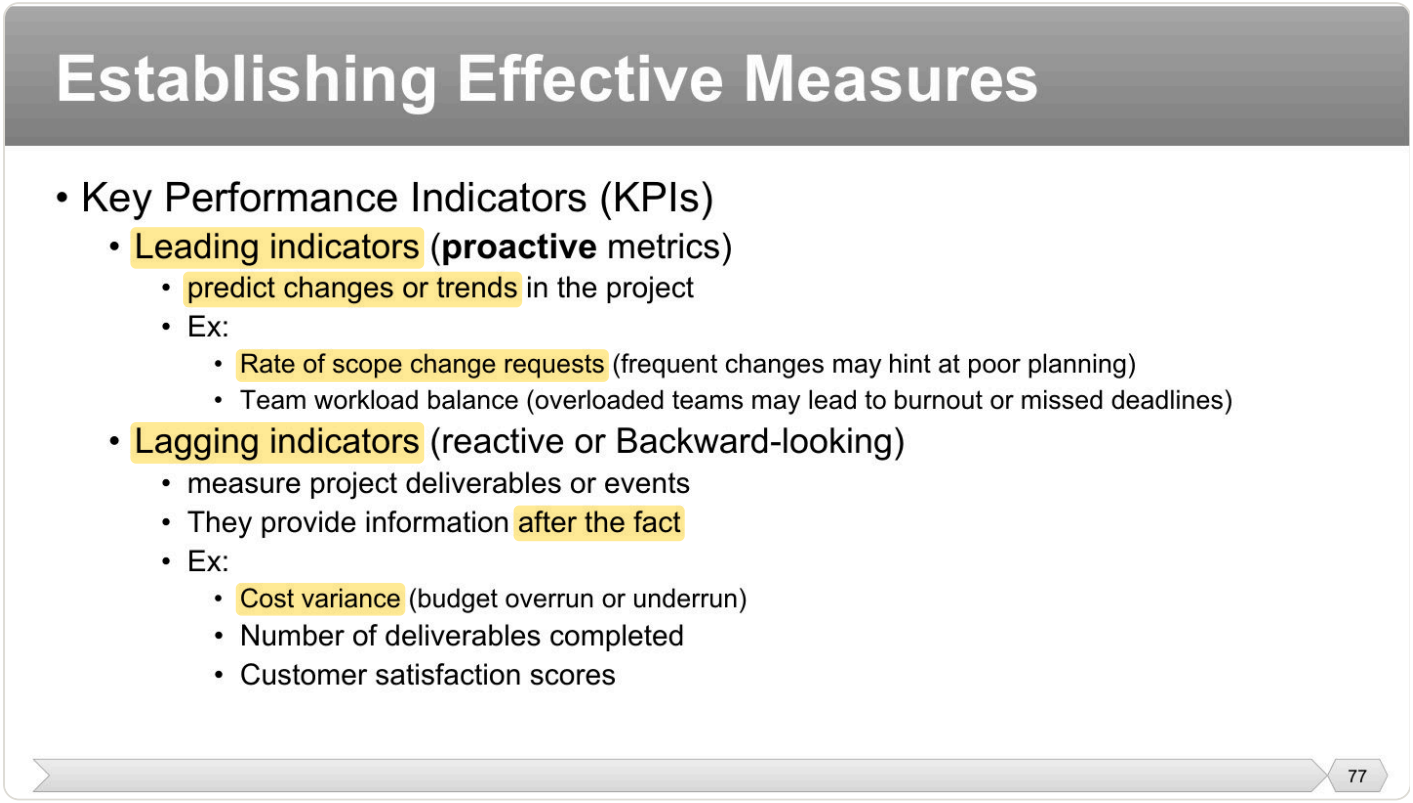
Measurement Performance Domain

- The value of measurements is not in the collection and dissemination of the data, but rather in the conversations about how to use the data to take appropriate action.
- It helps to ensure the right things are measured and reported to stakeholders.
- Effective measures allow for tracking, evaluating, and reporting information that can communicate project status, help improve project performance, and reduce the likelihood of performance deterioration.

NOTES

1. The value of measurement is **not in collecting data** but in the **conversations** about how to act on it.
2. It helps make sure the **right things** are measured and reported.
3. Good measures let the team track, evaluate and report status, improve performance and **reduce the chance of performance deteriorating** unnoticed.

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NOTES

1. **Key performance indicators (KPIs)** come in two kinds. **Leading** indicators are proactive: they **predict** changes or trends while there is still time to act.
2. Examples: the **rate of scope change requests** (many changes hint at poor planning) and **team workload balance** (overload predicts burnout and missed dates).
3. **Lagging** indicators are reactive: they measure deliverables or events **after the fact**.
4. Examples: **cost variance**, number of deliverables completed, customer satisfaction scores. Leading looks ahead; lagging looks back.

EXAM ANGLE

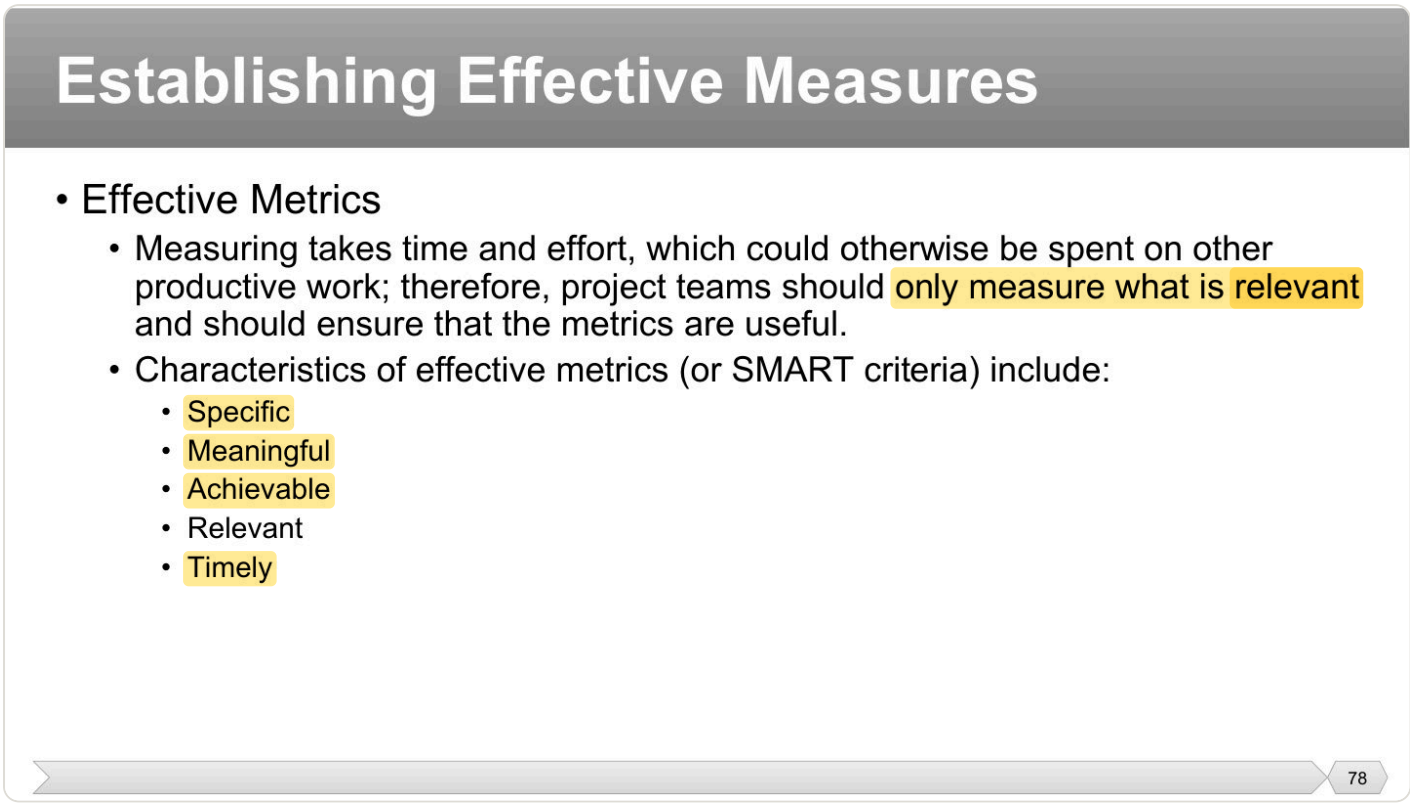
Prediction Q1

Which of these is a leading indicator?

A. Team workload balance
B. Cost variance
C. Customer satisfaction score
D. Number of deliverables completed

Answer: A. Team workload balance

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NOTES

1. Measuring takes time that could go to productive work, so teams **measure only what is relevant** and useful.

2. Effective metrics are **SMART**: Specific, **Meaningful**, Achievable, Relevant, Timely. Note that in this slide the M is “meaningful”, not the usual “measurable”: a classic trap.

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What to Measure

- What is measured, the parameters, and the measurement method depend on the project objectives, the intended outcomes, and the environment in which the project takes place.
- Common categories of metrics include:
  - Deliverable metrics,
  - Delivery,
  - Baseline performance,
  - Resources,
  - Business value,
  - Stakeholders, and
  - Forecasts.

NOTES

1. What to measure depends on the **objectives, intended outcomes and environment**.
2. Seven common categories: **deliverable** metrics, **delivery**, **baseline performance**, **resources**, **business value**, **stakeholders** and **forecasts**. Slides 80–81 give examples of each.

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What to Measure

- Deliverable metrics
  - Information on errors or defects, Measures of performance, Technical performance measures
- Delivery
  - Work in progress, Lead time (total elapsed time between the identification of a need and the delivery of the result), Cycle time, Process efficiency
- Baseline performance
  - Start and finish dates, Effort and duration, Schedule variance

NOTES

1. **Deliverable metrics**: errors or defects, performance measures, technical performance measures.
2. **Delivery**: work in progress, **lead time** (from identifying the need to delivering the result: the customer's whole wait), **cycle time** (from starting the work to finishing it), process efficiency.
3. **Baseline performance**: start and finish dates, effort and duration, schedule variance.

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What to Measure

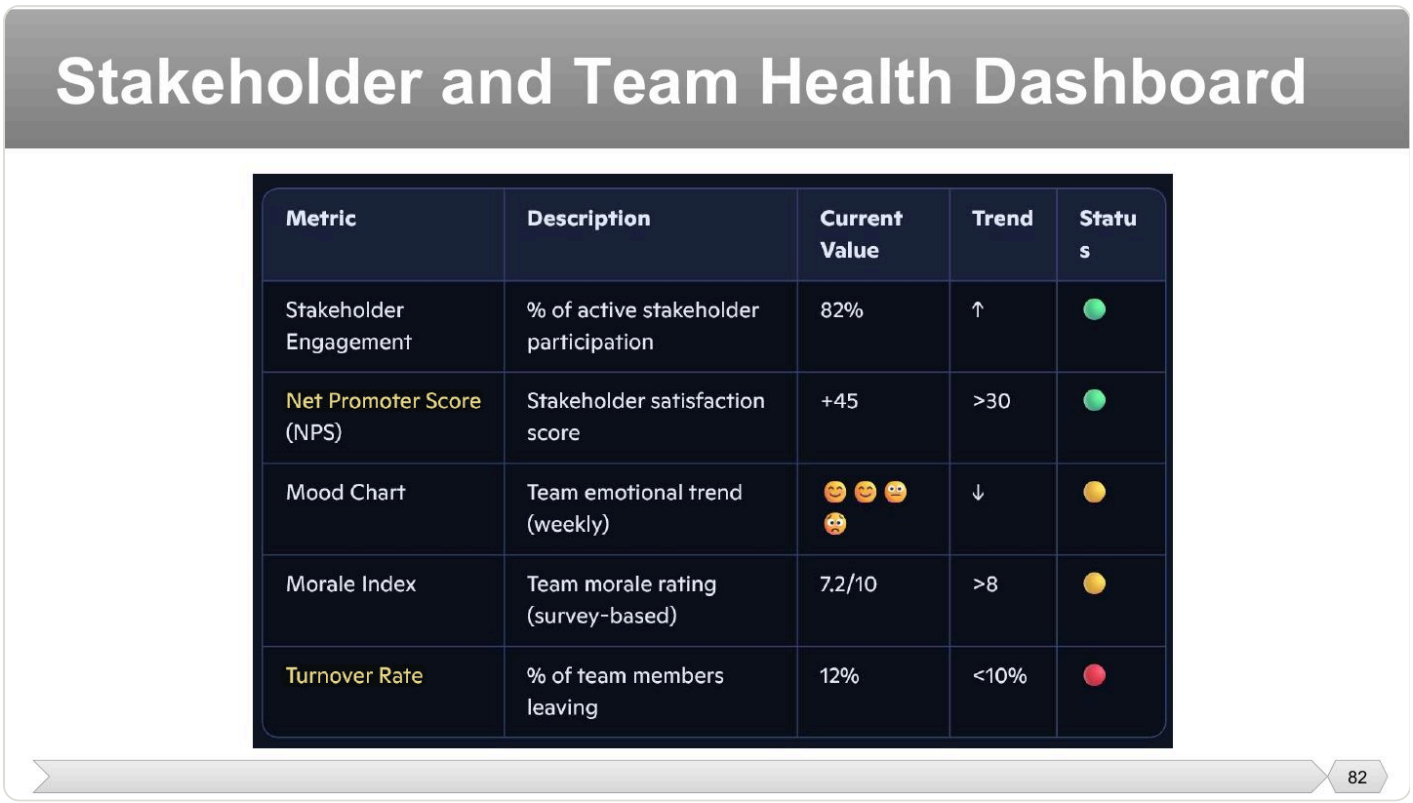
- Resources
  - Planned resource utilization compared to actual resource utilization, Planned resource cost compared to actual resource cost
- Business value
  - Cost-benefit ratio, Return on investment (ROI), Net present value (NPV)
- Stakeholders
  - Net Promoter Score (NPS), Mood chart, Morale, Turnover
- Forecasts
  - Estimate to complete (ETC), Estimate at completion (EAC), Variance at completion (VAC)

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NOTES

1. **Resources**: planned vs. actual resource use and cost.
2. **Business value**: cost-benefit ratio, **return on investment (ROI)**, net present value (NPV).
3. **Stakeholders**: **Net Promoter Score (NPS)**, mood charts, morale, turnover.
4. **Forecasts**: estimate to complete (ETC: what is left), **estimate at completion (EAC**: the expected total), variance at completion (VAC = budget – EAC).

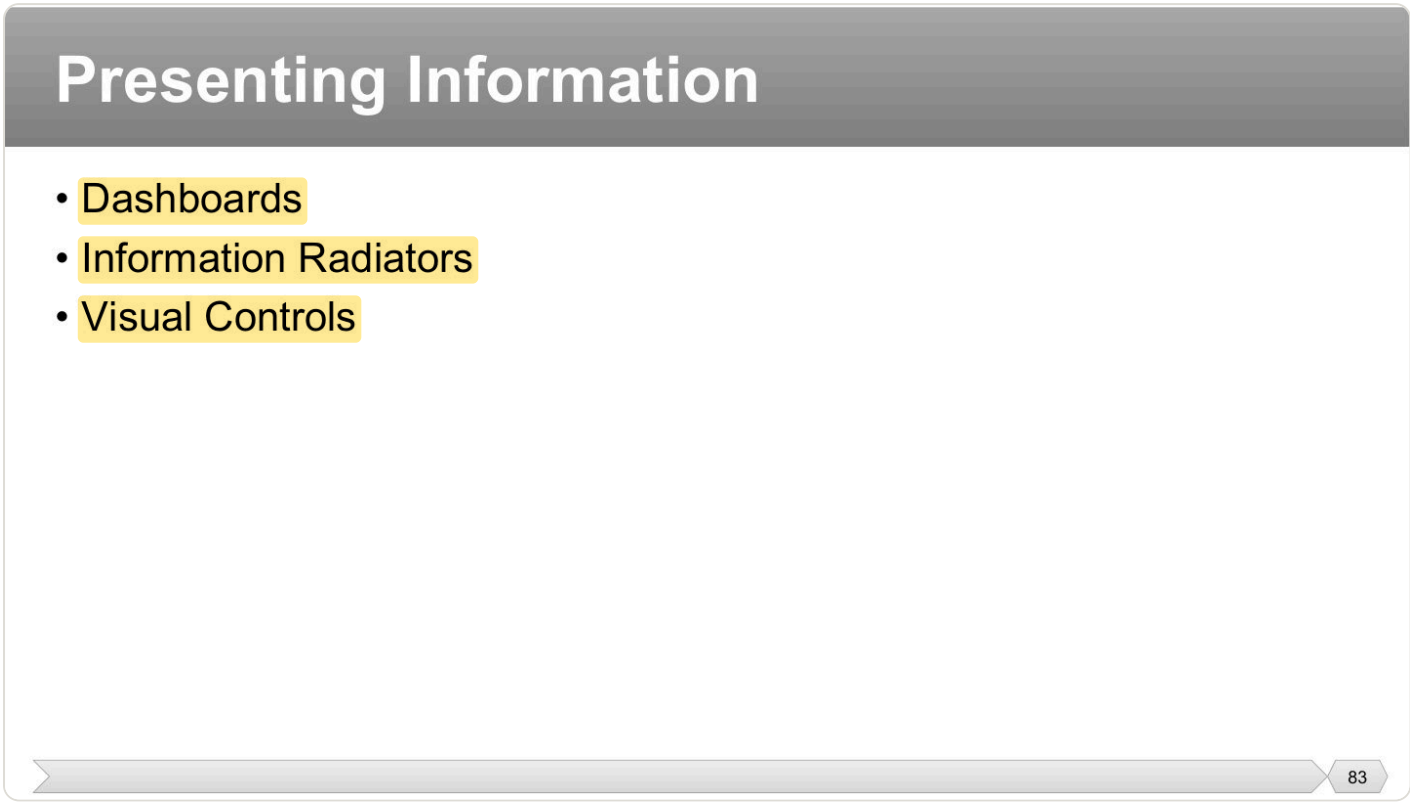
 Stakeholder and Team Health Dashboard



NOTES

1. An example dashboard for the stakeholder and team metrics of slide 81. Each row has a current value, a trend arrow, a target and a red/amber/green status. **Stakeholder engagement**: 82% active participation, rising, green.
2. **NPS** +45 against a target of >30 (green). Mood chart and morale (7.2/10 vs. >8) are amber.
3. **Turnover** 12% against a target of <10% is red: the item needing action. A dashboard's job is exactly this, pointing attention at what is off track.

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NOTES

1. **Dashboards**: summary displays of the key measures, usually electronic (slide 84).
2. **Information radiators**: large, visible displays placed where everyone sees them without asking (slide 85).
3. **Visual controls**: task boards, burn charts and other charts that show the flow of work (slide 86).

SLIDE 84 / 107 **Dashboards**

Presenting Information

- Dashboards

Organization Project Name						
Project Name and High-Level Description						
Exec Sponsor:			PM:			
Start Date:		End Date:		Report Period:		
Status:	Schedule	Resources	Budget			
Key Activities	Recent Accomplishments	Upcoming Key Deliverables	Status			
Activity #1			Concern			
Activity #2			On Track			
Activity #3			Issue			
On Track	Complete	Concern	Issue	On Hold	Canceled	Not Started

84

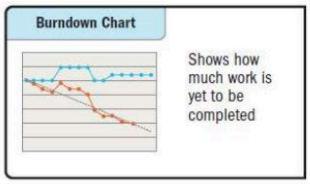
NOTES

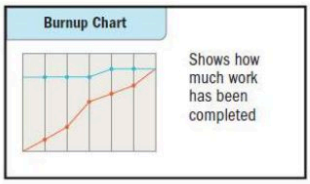
1 A project **dashboard** template: project name, sponsor, PM, dates, overall status for schedule, resources and budget, then key activities with recent accomplishments, upcoming deliverables and a coloured status (concern, on track, issue). The legend at the bottom defines the colour codes.

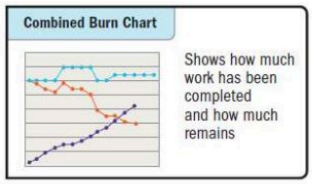
SLIDE 85 / 107 **Information Radiators**

Presenting Information

- Information Radiators







Reference	Risk Description	Date	Likelihood	Impact	Risk Rating	Response	Owner
1	The main supplier cannot deliver on time because of other commercial commitments	03/21	Likely	High	High	Include financial penalties in contract; build contingency into the schedule; monitor contractor performance	Annie
2	The lead time for the leased line exceeds 90 days	03/21	Unlikely	Medium	Medium	Order leased line earlier than necessary; incur additional rental fees	Jim
3	Release of the new system is delayed because user acceptance testing commences after the planned start	03/21	Very likely	High	High	Employ temporary staff to free up resources for testing; revise project schedule	Mark
4	There is insufficient capacity to create additional database instances for data migration and testing	04/18	Very unlikely	Medium	Low	Prioritize projects; temporarily remove alternative development instance	Jim

Risk Log

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NOTES

1. An **information radiator** is a big visible display, such as a burnup or burndown chart or a risk log, placed where the team works, so status is visible without asking anyone. (Also called a "big visible chart".)

EXAM ANGLE

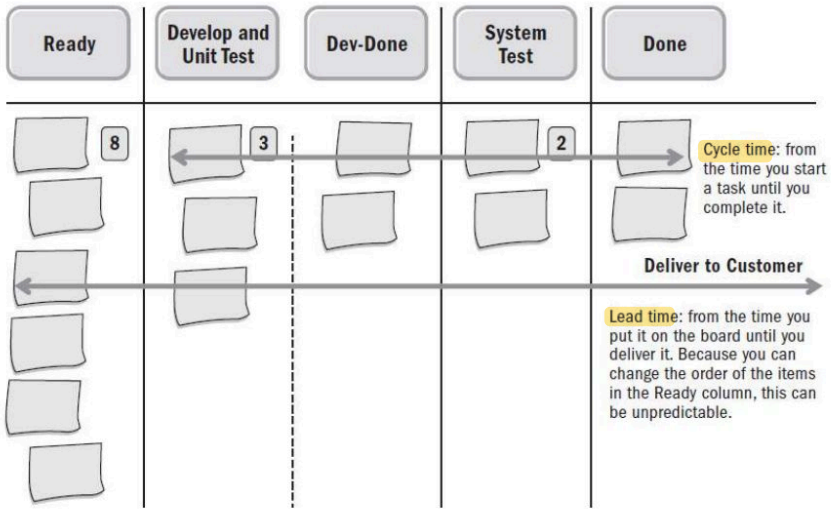
Past exam · Final 242 · Q6.2 Q1

How can a burn chart help a project manager track progress and identify if the team is ahead or behind schedule?
Answer: A burndown chart plots the work remaining (e.g. story points) against time, next to an ideal line from the total down to zero at the end of the sprint. If the actual line is above the ideal line the team is behind; below it, ahead. A burnup chart shows completed work rising toward the scope line and also reveals scope changes. Either one gives an early warning, so the PM can act (remove blockers, re-prioritize, adjust scope) before the deadline.

SLIDE 86 / 107 **Visual Controls**

Presenting Information

- Visual Controls
  - Task boards
  - Burn charts
  - Other types of charts



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NOTES

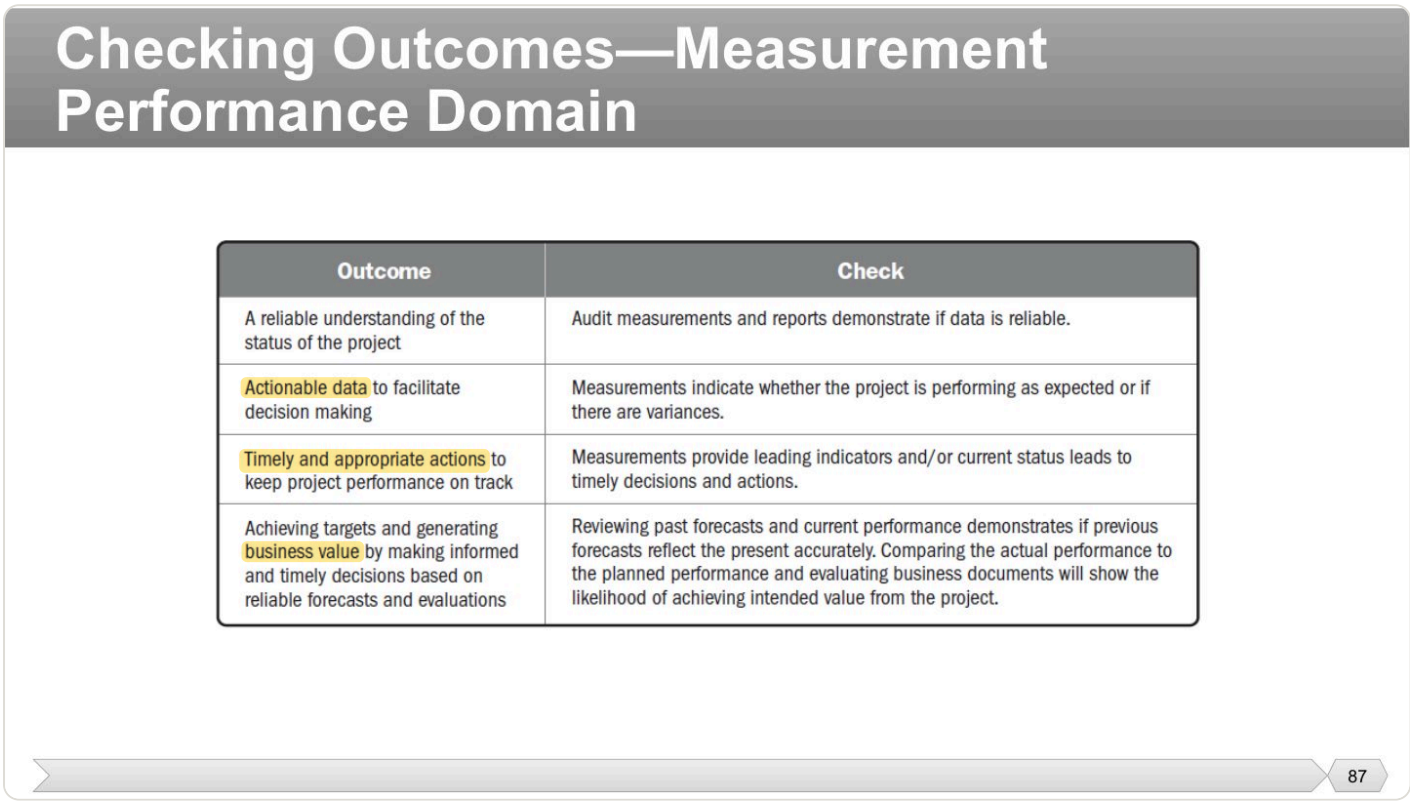
1. **Task boards** make the flow of work visible: columns Ready → Develop and unit test → Dev-done → System test → Done, with work-in-progress limits (the boxed numbers).
2. **Burn charts** show work remaining (burndown) or completed (burnup) over time.
3. On the board: **cycle time** runs from when work starts on an item until it is complete; **lead time** runs from when the item is put on the board until it is delivered. Lead time is longer, because items wait in Ready and their order can change.

EXAM ANGLE

Past exam · Final 251 · Q6.1 Q1

What does a burndown chart measure, and how are its two axes labeled in Agile project management?
Answer: It measures the work remaining against time, showing whether the team will finish on schedule. X-axis: time (days of the sprint, or sprints/iterations). Y-axis: work remaining (story points, hours or tasks).

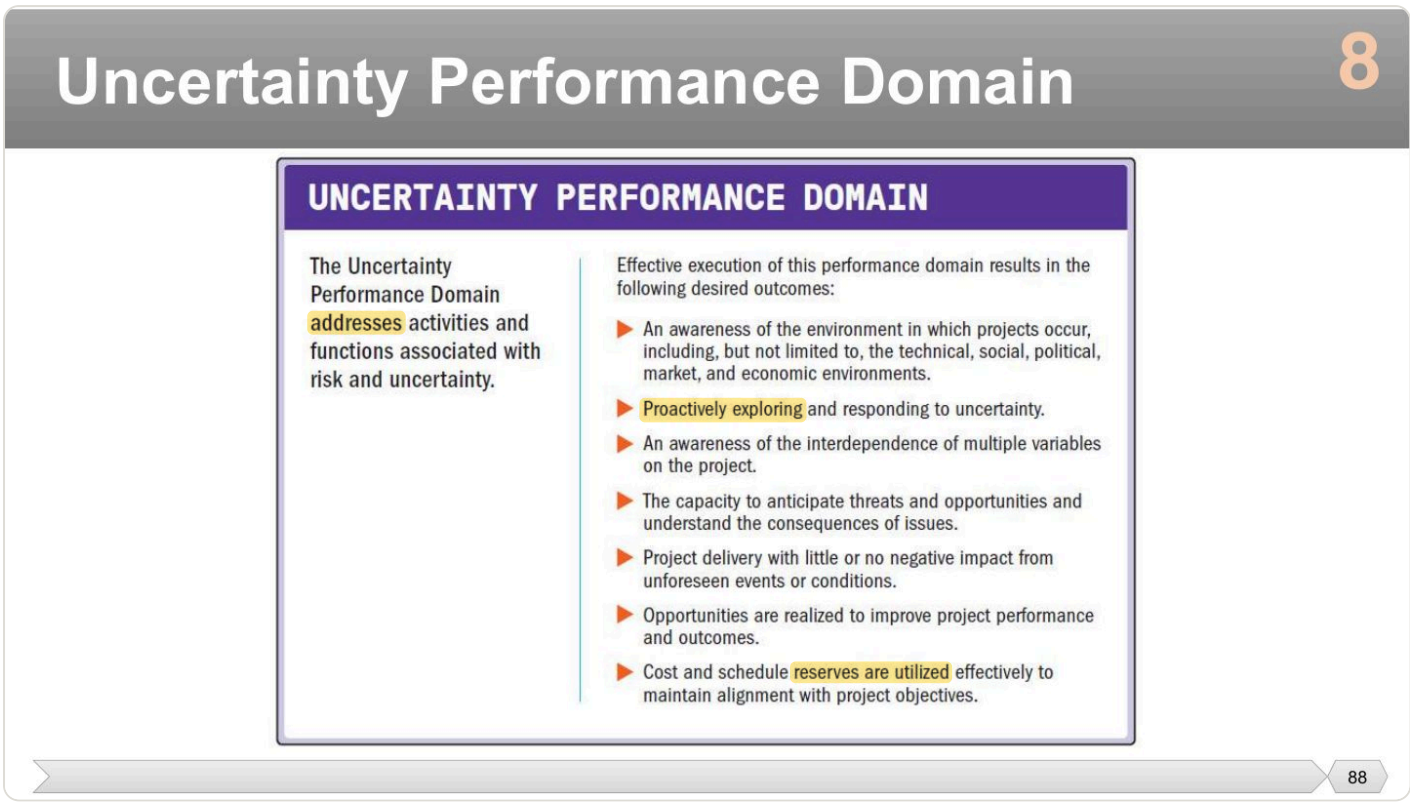
SLIDE 87 / 107 **Checking Outcomes: Measurement**



NOTES

1. The “checking outcomes” table: the left column is what success looks like for the domain, the right column is the evidence that it happened. You won’t be asked to reproduce it, but it is a good structure for a written answer about the domain. Measurement: a reliable understanding of status (audits of measures), actionable data (data shows whether to act), timely action (leading indicators prevent surprises) and business value from informed decisions.

 Uncertainty Performance Domain



NOTES

1. The **uncertainty** domain covers activities related to risk and uncertainty.
2. Outcomes: awareness of the environment (technical, social, political, market, economic), **proactive** exploration of and response to uncertainty, awareness of how variables depend on each other, anticipating threats and opportunities, delivery with little impact from surprises, and opportunities realized.
3. And **cost and schedule reserves used effectively** to stay aligned with the objectives.

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Uncertainty Performance Domain

- The following definitions are relevant to the Uncertainty Performance Domain:
  - Uncertainty
    - A lack of understanding and awareness of issues, events, paths to follow, or solutions to pursue.
  - Ambiguity
    - A state of being unclear, having difficulty in identifying the cause of events, or having multiple options from which to choose.
  - Complexity
    - A characteristic of a program or project or its environment that is difficult to manage due to human behavior, system behavior, and ambiguity.
  - Volatility
    - The possibility for rapid and unpredictable change.
  - Risk
    - An uncertain event or condition that, if it occurs, has a positive or negative effect on one or more project objectives.

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NOTES

1. **Uncertainty**: a lack of understanding or awareness of issues, events, paths or solutions.
2. **Ambiguity**: being unclear, having difficulty identifying the cause of events, or having **multiple options** to choose from.
3. **Complexity**: a characteristic that makes a project difficult to manage because of human behavior, system behavior and ambiguity.
4. **Volatility**: the possibility of **rapid and unpredictable change**.
5. **Risk**: an uncertain event or condition that, if it occurs, has a **positive or negative** effect on objectives.

EXAM ANGLE

Prediction Q1

"The possibility for rapid and unpredictable change" defines:

A. Volatility
B. Ambiguity
C. Complexity
D. Risk

Answer: A. Volatility

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Uncertainty Performance Domain

- Uncertainty in the broadest sense is a state of not knowing or unpredictability.
- There are many nuances to uncertainty, such as:
  - Risk associated with not knowing future events,
  - Ambiguity associated with not being aware of current or future conditions, and
  - Complexity associated with dynamic systems having unpredictable outcomes.

NOTES

1. **Uncertainty** in the broadest sense is a state of **not knowing** or unpredictability.
2. It has nuances: **risk** (not knowing future events), **ambiguity** (not being aware of current or future conditions) and **complexity** (dynamic systems with unpredictable outcomes).

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Uncertainty Performance Domain

- Successfully navigating uncertainty begins with understanding the larger environment within which the project is operating.
- Aspects of the environment that contribute to project uncertainty include, but are not limited to:
  - Economic factors such as volatility in prices, availability of resources, ability to borrow funds, and inflation/deflation;
  - Technical considerations such as new or emerging technology, complexity associated with systems, and interfaces;
  - Legal or legislative constraints or requirements;
  - Physical environment as it pertains to safety, weather, and working conditions;
  - Ambiguity associated with current or future conditions;
  - Social and market influences shaped by opinion and media; and,
  - Political influences, either external or internal to the organization.

NOTES

1. Navigating uncertainty starts with understanding the **larger environment**.
2. Sources: **economic** (price volatility, resource availability, borrowing, inflation), **technical** (new technology, system complexity, interfaces), **legal**, physical (safety, weather, working conditions), ambiguity, and social and market influences shaped by opinion and media.
3. **Political** influences, inside or outside the organization. (A PESTLE-style checklist.)

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Ambiguity

- There are two categories of ambiguity:
  - Conceptual ambiguity
    - The lack of effective understanding—occurs when people use similar terms or arguments in different ways.
    - For example, the statement, “The schedule was reported on track last week,” is not clear.
    - It isn’t clear whether the schedule was on track last week or whether it was reported on last week.
  - Situational ambiguity
    - Surfaces when more than one outcome is possible.
    - Having multiple options to solve a problem is a form of situational ambiguity.

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NOTES

1. **Conceptual ambiguity**: a lack of shared understanding, when people use the same terms differently. “The schedule was reported on track last week”: was it on track last week, or was it reported last week? Disagreeing about what “done” means is the same problem.
2. **Situational ambiguity**: **more than one outcome** is possible; having several options to solve a problem is a form of it.

EXAM ANGLE

Prediction Q1

Team members argue in a status meeting because they mean different things by “done”. Which kind of uncertainty is this?

A. Conceptual ambiguity
B. Situational ambiguity
C. Volatility
D. A threat

Answer: A. Conceptual ambiguity

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Complexity

- Complexity is a characteristic of a program, project, or its environment, which is difficult to manage due to human behavior, system behavior, or ambiguity.
- Complexity exists when there are many interconnected influences that behave and interact in diverse ways.

NOTES

1. **Complexity** makes a project or its environment **difficult to manage** because of human behavior, system behavior or ambiguity.
2. It exists when there are **many interconnected influences** that behave and interact in diverse ways.

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Complexity

- In complex environments, it is not uncommon to see an aggregation of individual elements leading to unforeseen or unintended outcomes.
- The effect of complexity is that there is no way of making accurate predictions about the likelihood of any potential outcome or even of knowing what outcomes might emerge.
- There are numerous ways to work with complexity; some of them are systems-based, some entail reframing, and others are based on process.

NOTES

1. In complex environments, individual elements combine into **unforeseen or unintended outcomes**.
2. The effect: there is **no way to predict** the likelihood of outcomes accurately, or even which outcomes might emerge.
3. Three families of ways to work with complexity: **systems-based**, **reframing** and **process-based** (next three slides).

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Systems-Based Complexity

- Examples of working with complexity that is systems based include:
  - Decoupling
    - Disconnecting parts of the system to both simplify the system and reduce the number of connected variables.
    - Determining how a piece of a system works on its own reduces the overall size of the problem.
  - Simulation
    - There may be similar though unrelated scenarios that can be used to simulate components of a system.
    - A project to build a new airport that includes an area with shopping and restaurants can learn about consumer buying habits by seeking out analogous information on shopping malls and entertainment establishments.

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NOTES

1. **Decoupling** (systems-based): disconnect parts of the system to simplify it and reduce connected variables. Understanding how one piece works on its own shrinks the problem.
2. **Simulation**: use similar but unrelated scenarios to model parts of the system. A new airport with shops and restaurants can study **shopping malls** to learn about consumer buying habits.

EXAM ANGLE

Prediction Q1

A team designing a new airport's retail area studies shopping malls to predict consumer buying habits. Which complexity technique is this?

A. Simulation
B. Decoupling
C. Fail safe
D. Balance

Answer: A. Simulation

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Reframing Complexity

- Examples of working with complexity that entail reframing are:
  - Diversity
    - Complex systems require viewing the system from diverse perspectives.
    - This can include brainstorming with the project team to open up divergent ways of seeing the system.
    - It can also include Delphi-like processes to move from divergent to convergent thinking.
  - Balance
    - Balancing the type of data used rather than only using forecasting data or data that report on the past or lagging indicators provides a broader perspective.
    - This can include using elements whose variations are likely to counteract each other's potential negative effects.

NOTES

1. **Diversity** (reframing): view the system from diverse perspectives, e.g. brainstorming to open divergent ideas, then Delphi-like processes to converge.
2. **Balance**: balance the type of data used (forecasts and lagging indicators, not just one) for a broader view, and use elements whose variations **counteract** each other's negative effects.

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Process-Based Complexity

- Examples of working with complexity that is process based include:
  - Iterate
    - Build iteratively or incrementally.
    - Add features one at a time. After each iteration, identify what worked, what did not work, customer reaction, and what the project team learned.
  - Engage
    - Build in opportunities to get stakeholder engagement.
    - This reduces the number of assumptions and builds learning and engagement into the process.
  - Fail safe
    - For elements of a system that are critical, build in redundancy or elements that can provide a graceful degradation of functionality in the event of a critical component failure.

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NOTES

1. **Iterate** (process-based): build iteratively or incrementally, one feature at a time, and after each iteration learn what worked, what didn't and how customers reacted.
2. **Engage**: build in stakeholder engagement, which **reduces assumptions** and builds learning into the process.
3. **Fail safe**: for critical elements, build in **redundancy** or **graceful degradation** so a component failure doesn't take the whole system down (a backup server).

EXAM ANGLE

Prediction Q1

Adding a backup server so a critical system keeps working in reduced form if its main server fails is which complexity technique?

A. Fail safe
B. Iterate
C. Engage
D. Decoupling

Answer: A. Fail safe

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Volatility

- Volatility exists in an environment that is subject to rapid and unpredictable change.
- Volatility can occur when there are ongoing fluctuations in available skill sets or materials.
- Volatility usually impacts cost and schedule.

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NOTES

1. **Volatility** exists where change is **rapid and unpredictable**.
2. It often comes from **fluctuations** in available skills or materials (a key specialist leaves; chip prices spike).
3. It usually hits **cost and schedule**.

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Volatility

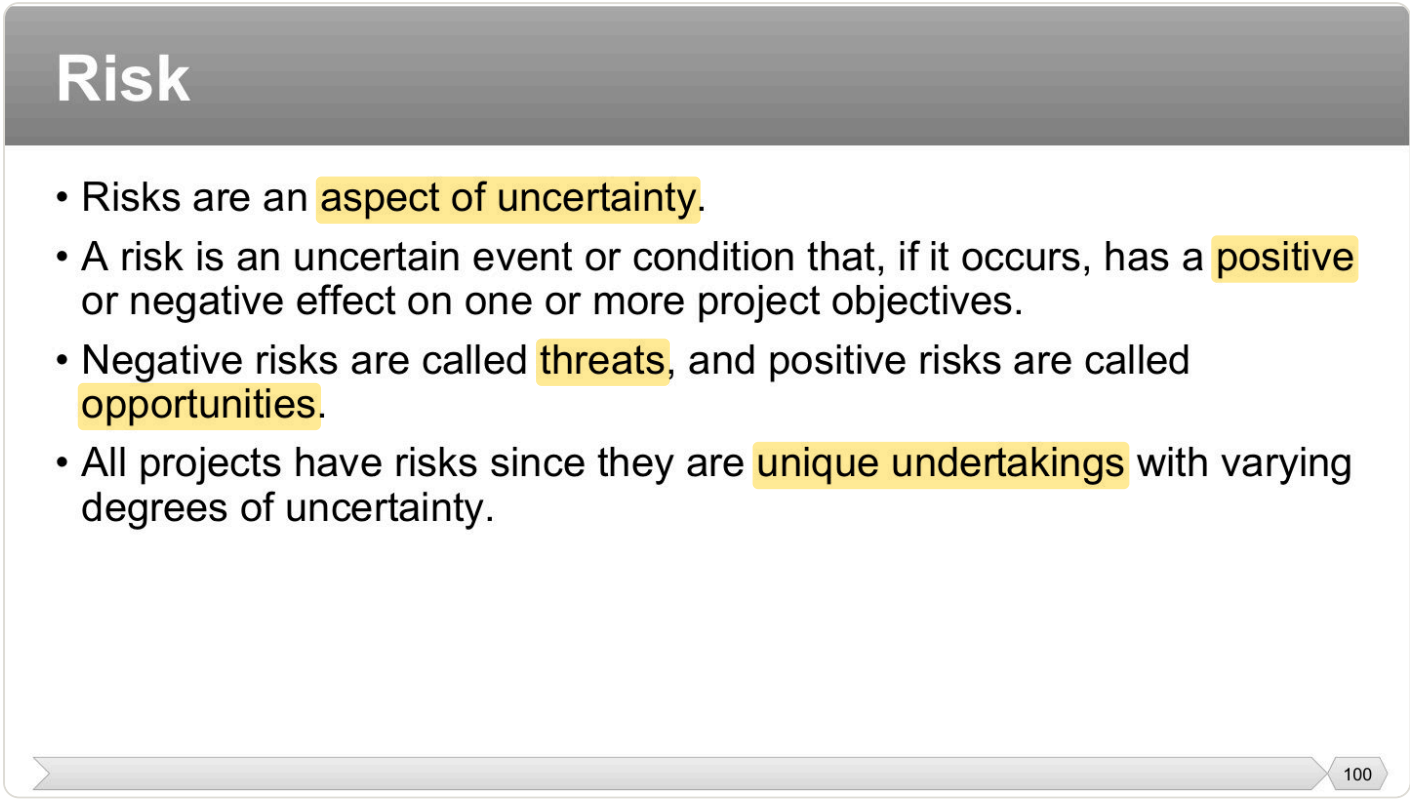
- Alternatives analysis and use of cost or schedule reserve address volatility.
  - Alternatives analysis
    - Finding and evaluating alternatives, such as looking at different ways to meet an objective, such as using a different mix of skills, resequencing work, or outsourcing work
  - Reserve
    - Cost reserve can be used to cover budget overruns due to price volatility.

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NOTES

1. Volatility is addressed with **alternatives analysis** and **reserves**.
2. **Alternatives analysis**: find and evaluate different ways to meet the objective: a different **mix of skills**, resequencing work, or **outsourcing**.
3. **Reserve**: a **cost reserve** covers budget overruns caused by price volatility (and a schedule reserve covers time).

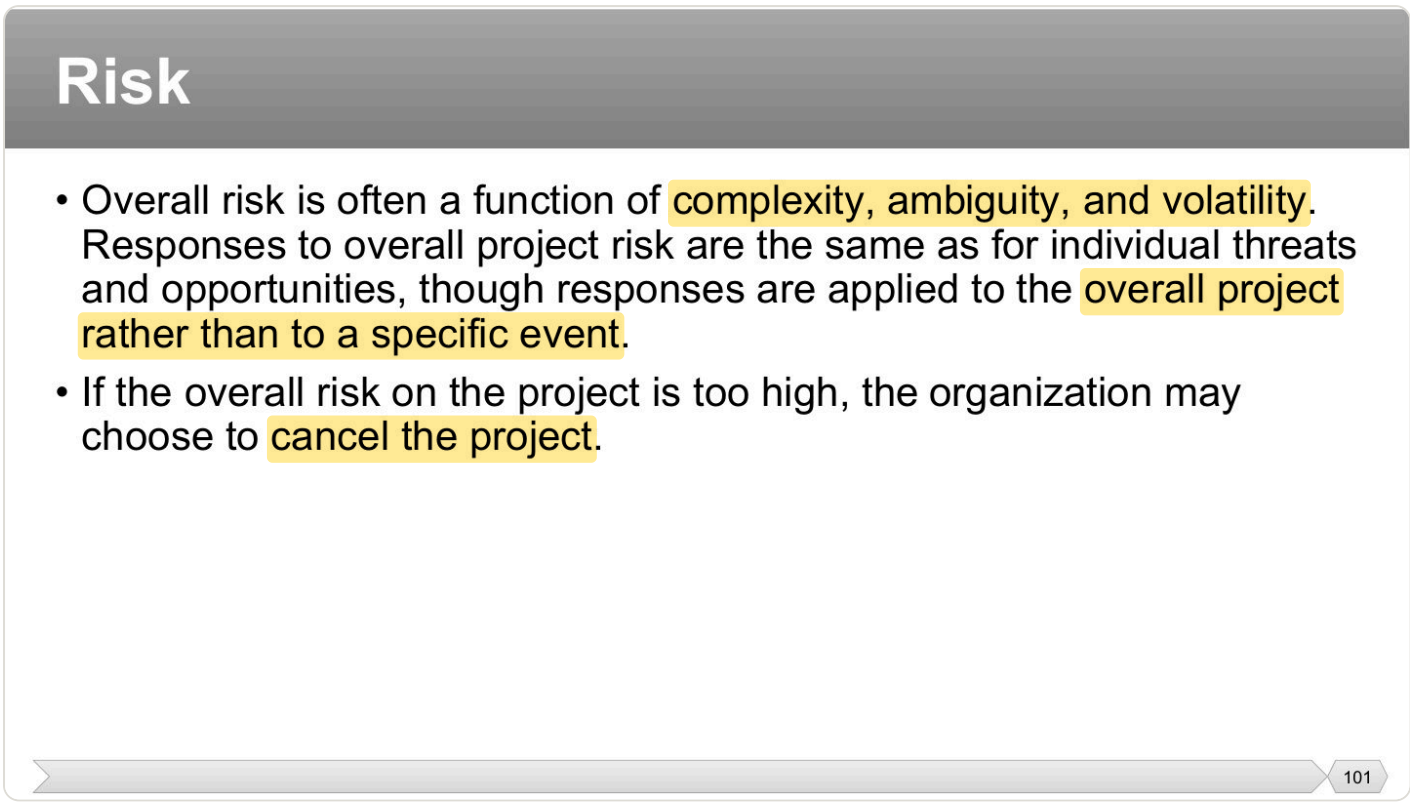
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NOTES

1. **Risk** is one aspect of uncertainty.
2. It is an uncertain event or condition with a **positive or negative** effect on objectives if it occurs.
3. Negative risks are **threats**; positive risks are **opportunities**.
4. Every project has risk because projects are **unique** undertakings.

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NOTES

1. **Overall project risk** is often a function of **complexity, ambiguity and volatility**.
2. Responses to overall risk use the same strategies as individual threats and opportunities, applied to the **project as a whole**.
3. If overall risk is too high, the organization may **cancel** the project: a legitimate outcome, not a failure of risk management.

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Risk (Threat)

- A threat is an event or condition that, if it occurs, has a negative impact on one or more objectives.
- Five alternative strategies may be considered for dealing with threats, as follows:
  - Avoid
  - Escalate
  - Transfer
  - Mitigate
  - Accept



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NOTES

1. A **threat** is an event or condition with a **negative** impact on objectives if it occurs.
2. **Avoid**: eliminate the threat or protect the project from it (change the plan, drop the risky feature).
3. **Escalate**: the threat is outside the PM's authority or scope, so pass it to the right level (program, portfolio, sponsor).
4. **Transfer**: shift ownership and impact to a third party (insurance, a fixed-price contract).
5. **Mitigate**: reduce the probability or impact (extra testing, a prototype).
6. **Accept**: acknowledge it without action, perhaps with a contingency reserve.

EXAM ANGLE

Prediction Q1

Buying insurance against a data-centre fire is which threat response strategy?

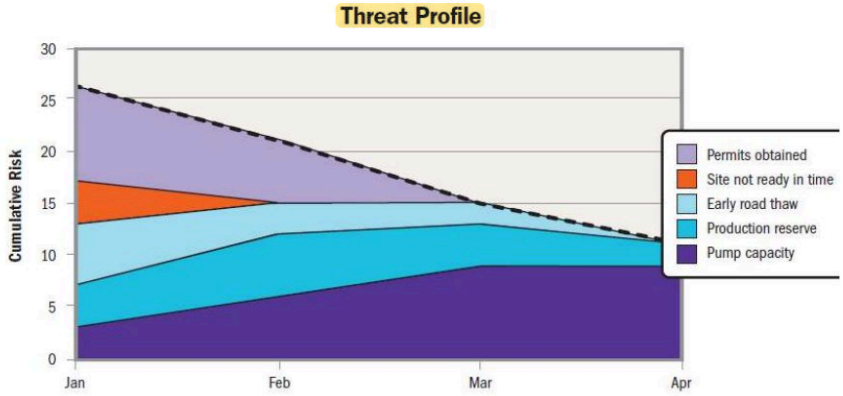
A. Transfer
B. Avoid
C. Mitigate
D. Exploit

Answer: A. Transfer

 Risk Reduction over Time

Risk Reduction over Time

		Jan			Feb			Mar			Apr		
ID	Short Risk Name	Impact	Probability	Severity	Impact	Probability	Severity	Impact	Probability	Severity	Impact	Probability	Severity
1	Permits obtained	3	3	9	3	2	6	3	0	0	3	0	0
2	Site not ready in time	2	2	4	2	0	0	2	0	0	2	0	0
3	Early road thaw	3	2	6	3	1	3	2	1	2	3	0	0
4	Production reserve	2	2	4	2	3	6	2	2	4	2	1	2
5	Pump capacity	3	1	3	3	2	6	3	3	9	3	3	9



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NOTES

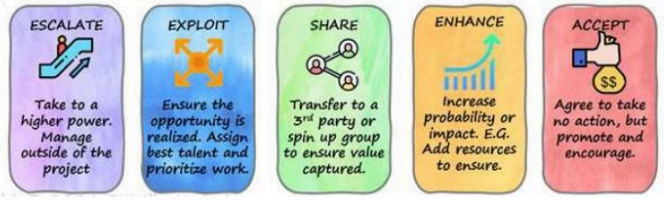
1. The table scores five risks each month (impact × probability = severity), and the **threat profile** chart stacks their severities over time. The total (dashed line) falls from about 26 in January to about 11 in April.

2. Some risks disappear as they are dealt with (**permits obtained** drops to 0 once permits are secured; “site not ready in time” too), while “pump capacity” stays. A falling profile is the evidence that risk responses are working.

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Risk (Opportunities)

- An opportunity is an event or condition that, if it occurs, has a positive impact on one or more project objectives.
- An example of an opportunity could be a time and materials-based subcontractor who finishes work early, resulting in lower costs and schedule savings.
- Five alternative strategies may be considered for dealing with opportunities, as follows:
  - Exploit
  - Escalate
  - Share
  - Enhance
  - Accept



NOTES

1. An **opportunity** is an event or condition with a **positive** impact on objectives if it occurs.
2. Example: a time-and-materials subcontractor **finishes early**, saving cost and schedule.
3. **Exploit**: act to make sure the opportunity **happens** (raise its probability to 100%).
4. **Share**: give part of the ownership to a third party better able to capture the benefit (a partnership).
5. **Enhance**: increase its **probability or impact** without guaranteeing it. **Escalate** and **accept** apply to opportunities too.

EXAM ANGLE

Past exam · Midterm 261 · MCQ — Q1

A team identifies a chance to finish early and takes action to ensure that the opportunity occurs. Which strategy is this?

A. Exploit
B. Transfer
C. Enhance
D. Accept

Answer: A. Exploit — Exploit makes sure the opportunity happens. Enhance only increases its chance; transfer is a threat strategy.

Prediction — Q2

A firm partners with another company so that both can capture a positive risk neither could capture alone. Which opportunity strategy is this?

A. Share
B. Exploit
C. Enhance
D. Transfer

Answer: A. Share

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Risk (Management and Contingency Reserve)

- Reserve is an amount of time or budget set aside to account for handling risks.
- Contingency reserve is set aside to address identified risks should they occur.
- Management reserve is a budget category used for unknown events such as unplanned, in-scope work.

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NOTES

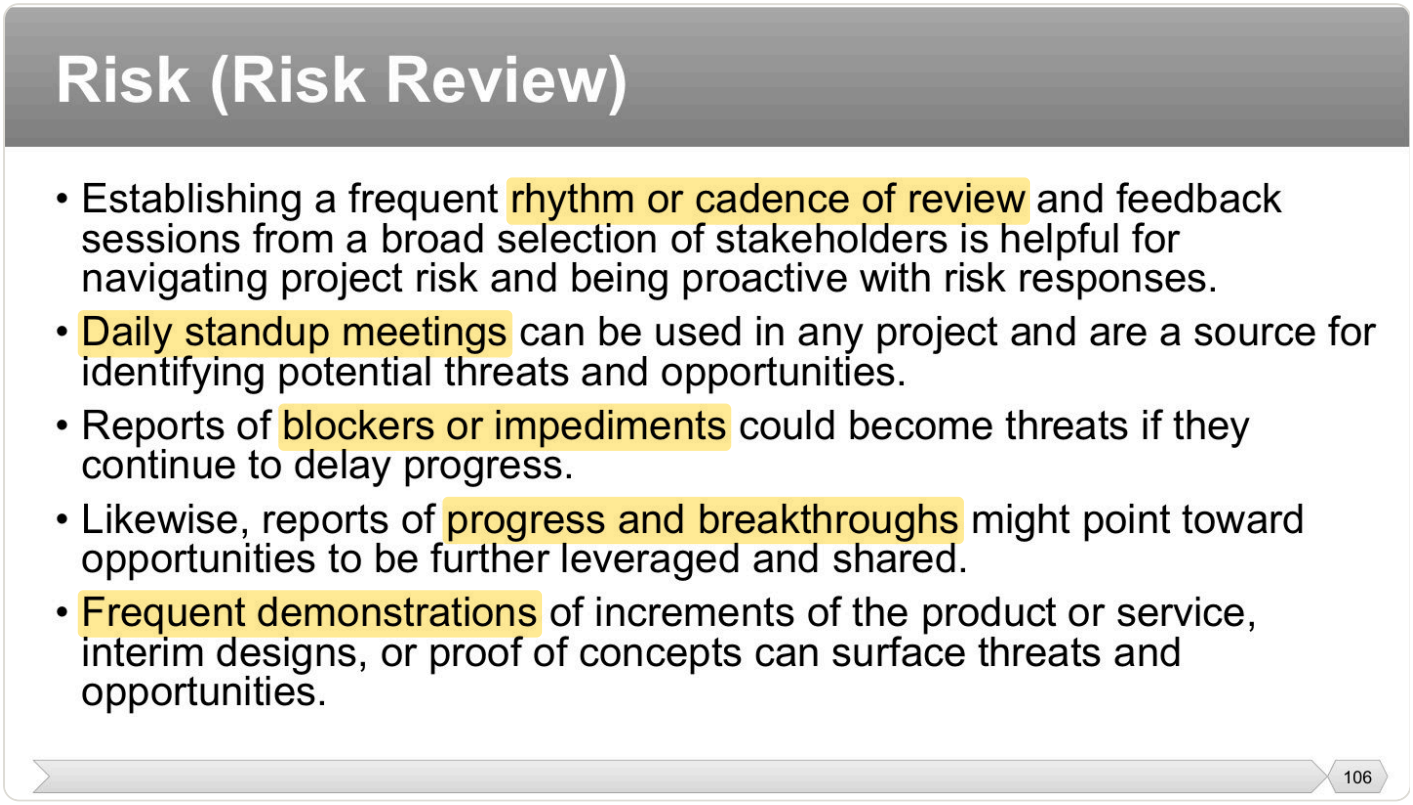
1. A **reserve** is time or budget set aside for handling risks.
2. **Contingency reserve**: for **identified** risks, should they occur.
3. **Management reserve**: for **unknown** events such as unplanned, in-scope work (slide 47).

EXAM ANGLE

Prediction Q1

True or false: contingency reserve is set aside for unknown, unplanned events.
Answer: False — Contingency reserve covers identified risks; management reserve covers unknown events.

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NOTES

1. A regular **cadence of review** and feedback with a broad set of stakeholders helps navigate risk proactively.
2. **Daily stand-ups** (usable in any project) surface potential threats and opportunities.
3. Reported **blockers** may become threats if they keep delaying progress...
4. ...while **progress and breakthroughs** may point to opportunities to exploit and share.
5. **Frequent demonstrations** of increments, interim designs or proofs of concept also surface threats and opportunities early.

Checking Outcomes: Uncertainty

Outcome	Check
An awareness of the environment in which projects occur, including, but not limited to, the technical, social, political, market, and economic environments	The team incorporates environmental considerations when evaluating uncertainty, risks, and responses.
Proactively exploring and responding to uncertainty	Risk responses are aligned with the prioritization of project constraints, such as budget, schedule, and performance.
An awareness of the interdependence of multiple variables on the project	Actions to address complexity, ambiguity, and volatility are appropriate for the project.
The capacity to anticipate threats and opportunities and understand the consequences of issues	Systems for identifying, capturing, and responding to risk are appropriately robust.
Project delivery with little or no negative impact from unforeseen events or conditions	Scheduled delivery dates are met, and the budget performance is within the variance threshold.
Realized opportunities to improve project performance and outcomes	Teams use established mechanisms to identify and leverage opportunities.
Cost and schedule reserves used effectively to maintain alignment with project objectives	Teams take steps to proactively prevent threats, thereby limiting use of cost or schedule reserve.

NOTES

1. The “checking outcomes” table: the left column is what success looks like for the domain, the right column is the evidence that it happened. You won’t be asked to reproduce it, but it is a good structure for a written answer about the domain. Uncertainty: awareness of the environment (environmental factors considered in estimates and plans), proactive responses (risk register and responses kept current), awareness of interdependencies, anticipation of threats and opportunities, little impact from surprises, realized opportunities, and effective use of reserves.

Original slides from the SE423 lecture deck, with study notes. Check exact wording against the original deck.
Notes and questions are study aids. Made by Shoug Alomran.